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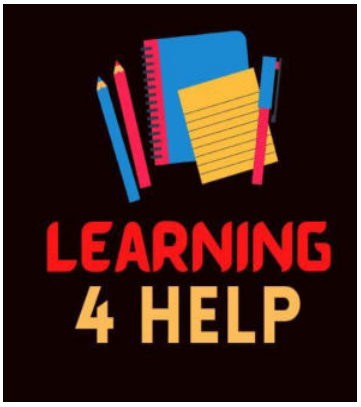
Asslam-o-Alaikum! In this file we'll provide you **MTH601 Important Final term MCQs. Must Prepare before Final term Exams.** I am not responsible for any answer. Verify Answers yourself too. If you found mistake then inform me.

If you found mistake then let us know.

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Or you can send us Gmail: Learning4help@gmail.com

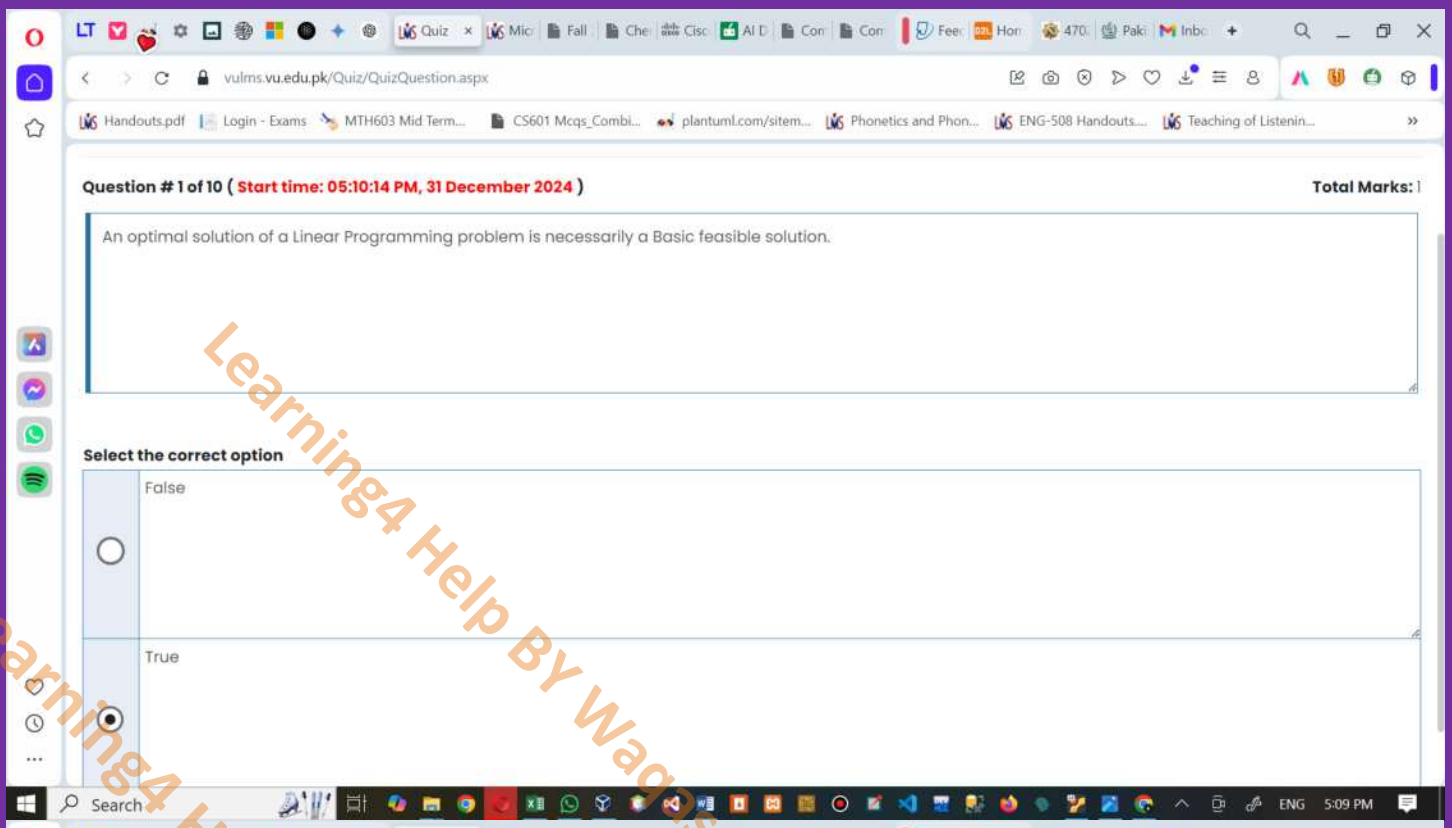


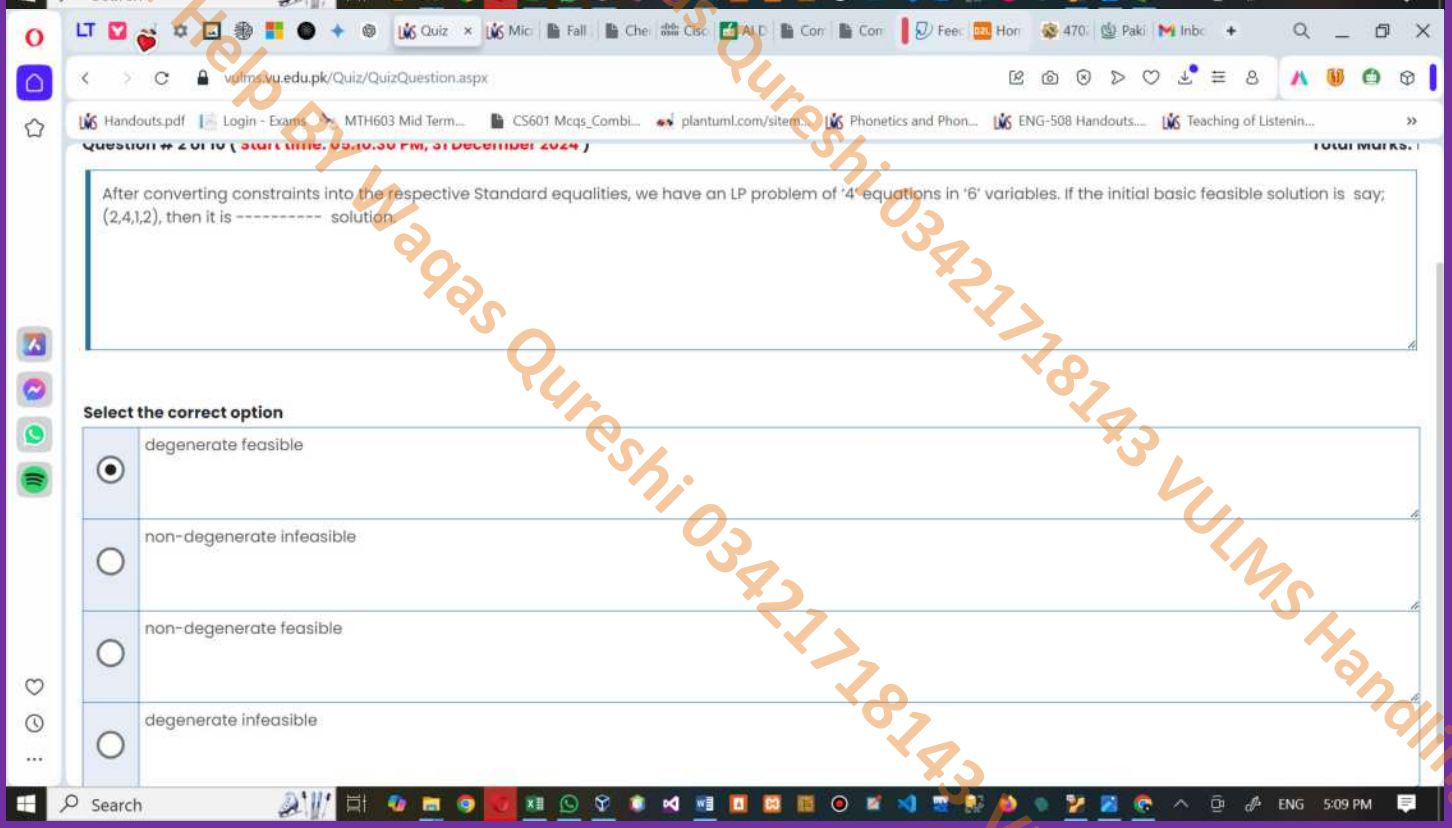


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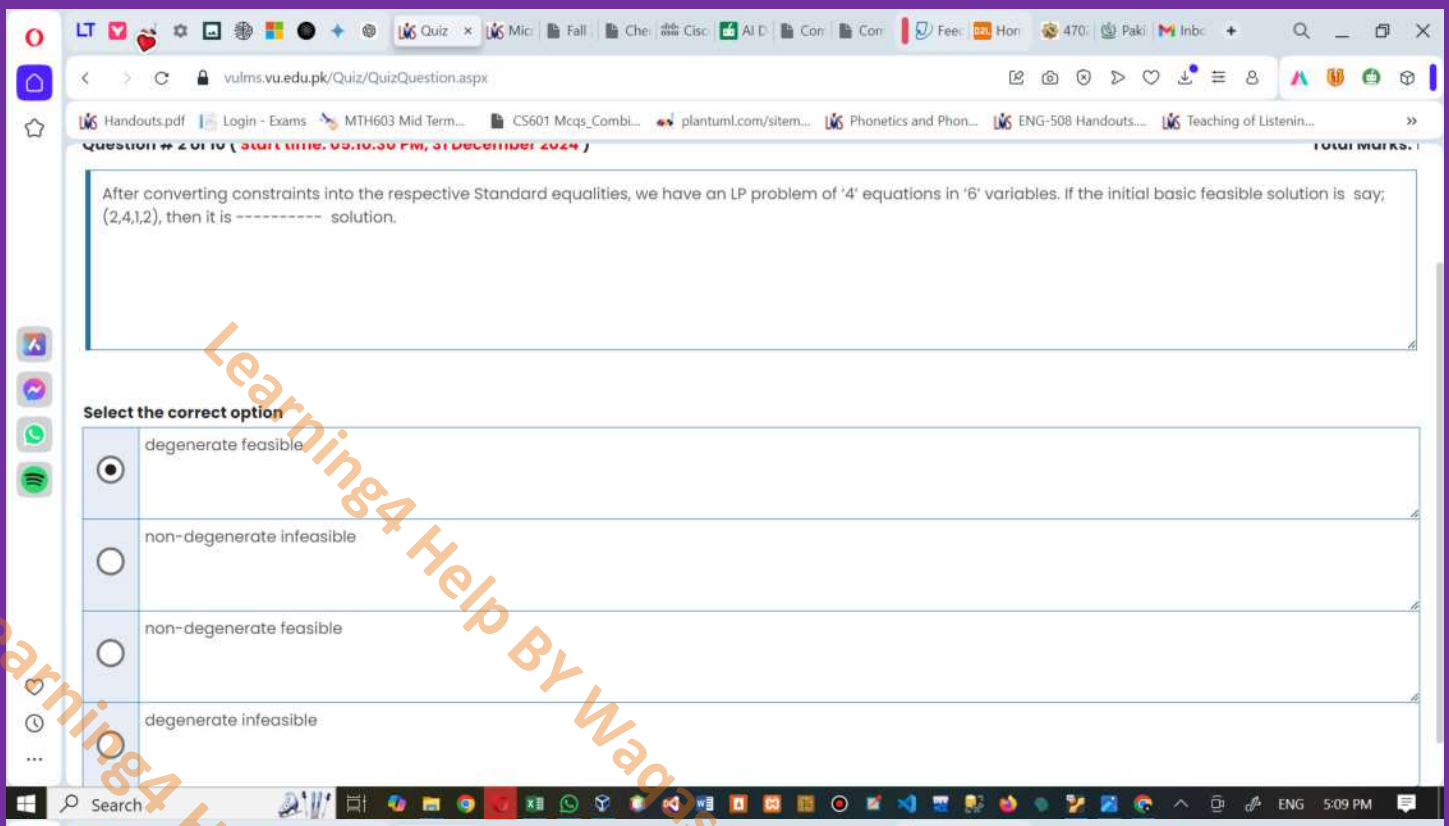
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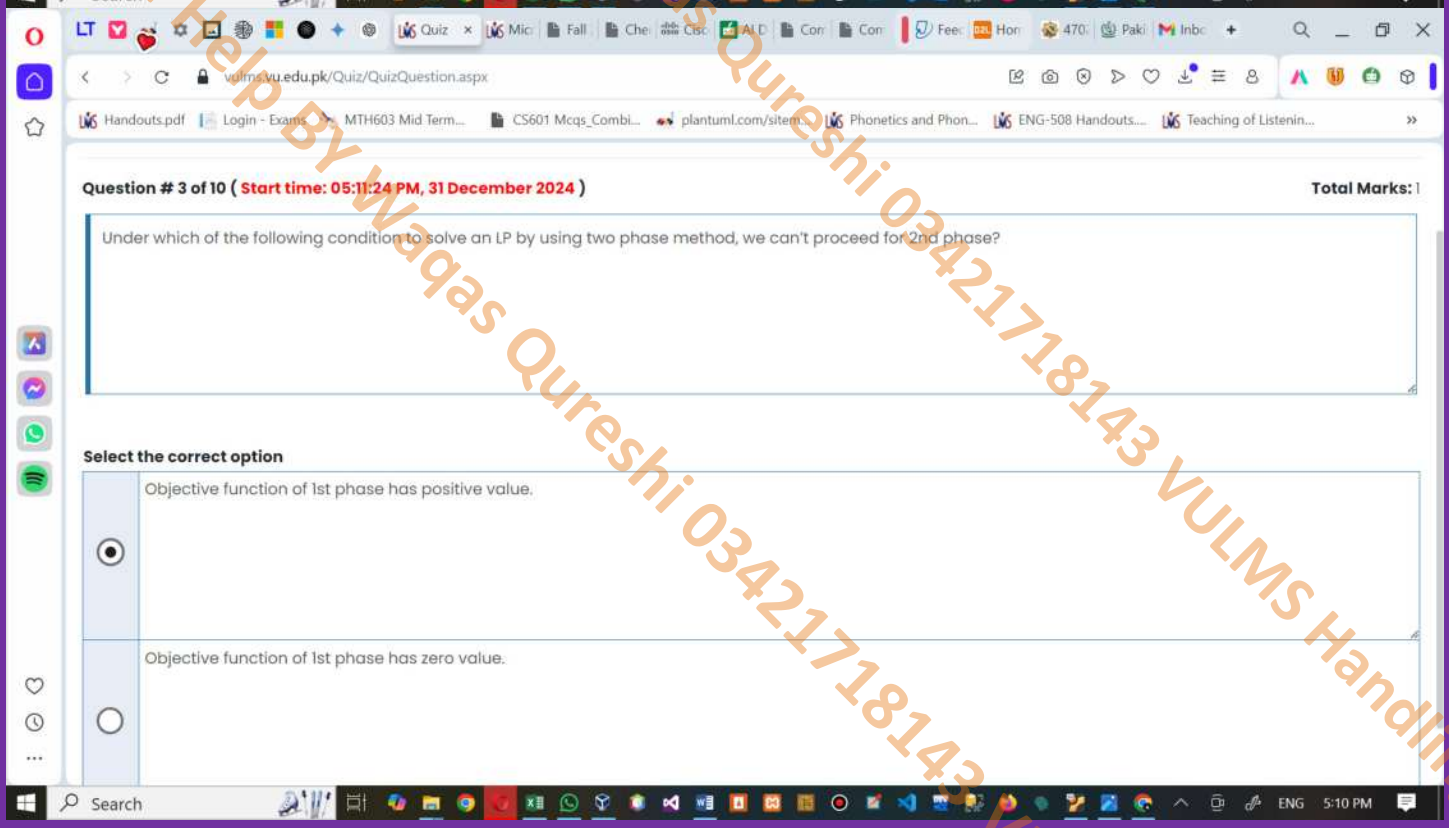
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After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say; (2,4,1,2), then it is ---------- solution.

Select the correct option

- (●) degenerate feasible
- ( ) non-degenerate infeasible
- ( ) non-degenerate feasible
- ( ) degenerate infeasible



Question # 3 of 10 (Start time: 05:11:24 PM, 31 December 2024) **Total Marks:** 1

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?

Select the correct option

- (●) Objective function of 1st phase has positive value.
- ( ) Objective function of 1st phase has zero value.

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If the optimum solution of a Linear Programming problem is finitely unique and then its corresponding objective function must have---------- maximum or minimum.

Select the correct option

- NO
- infinite
- arbitrary
- finite

MTH601 - Operations Research (Quiz 02) **Quiz Start Time: 05:10 PM, 31 December 2024**

Question # 5 of 10 (Start time: 05:13:16 PM, 31 December 2024) **Total Marks:** 1

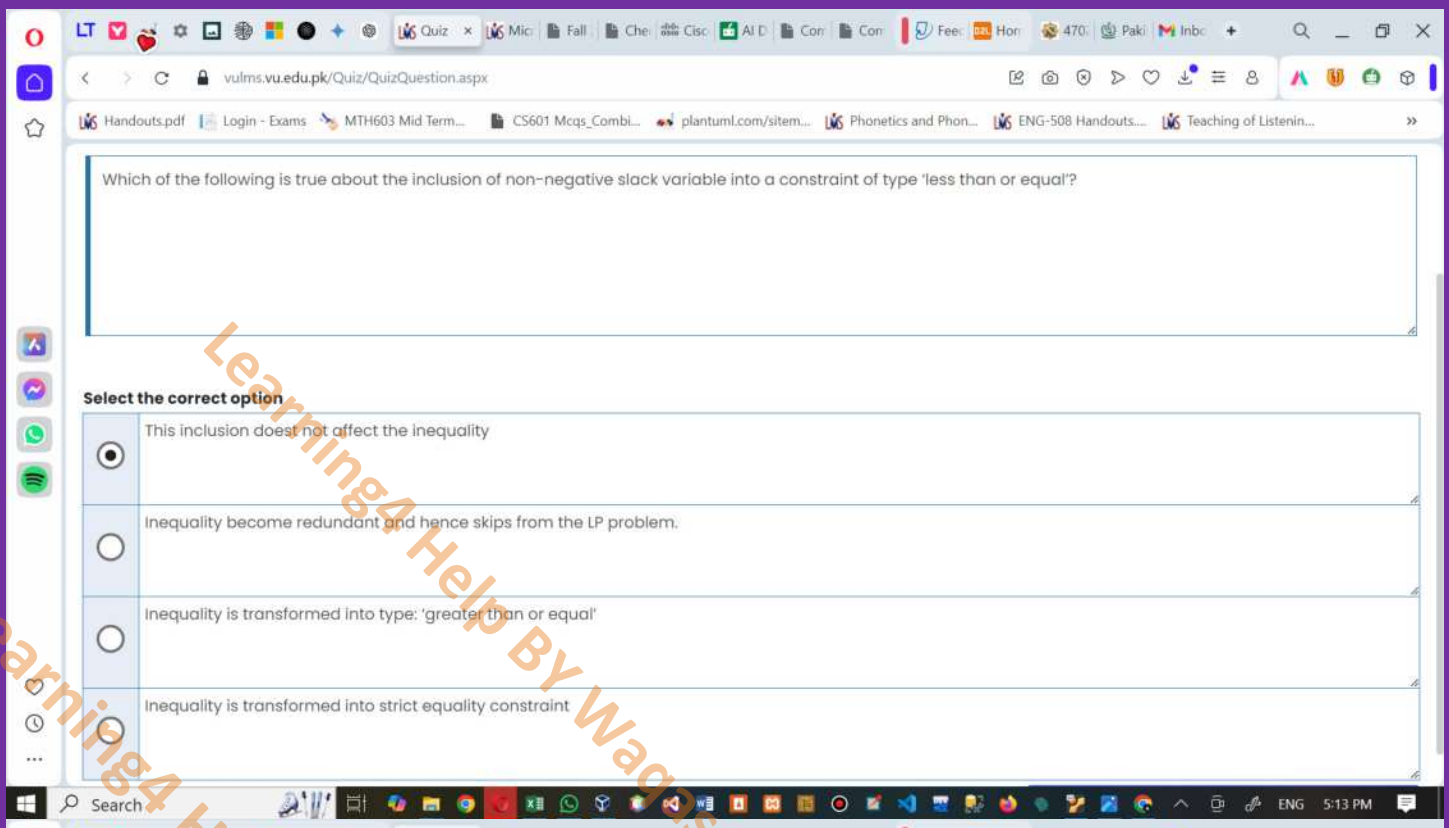
If "$Max\ Z = x - y$, subject to $x > 3, x < 2, x, y > 0$" is solved by two phase method, then which of the following would be the objective function of 1st phase?

Select the correct option

Reload Math Equations

- Max $Z = A$
- Max $Z = x - y + 0s_1 + 0s_2 + MA$
- Min $Z = A$
- Min $Z = x - y + 0s_1 + 0s_2 + MA$

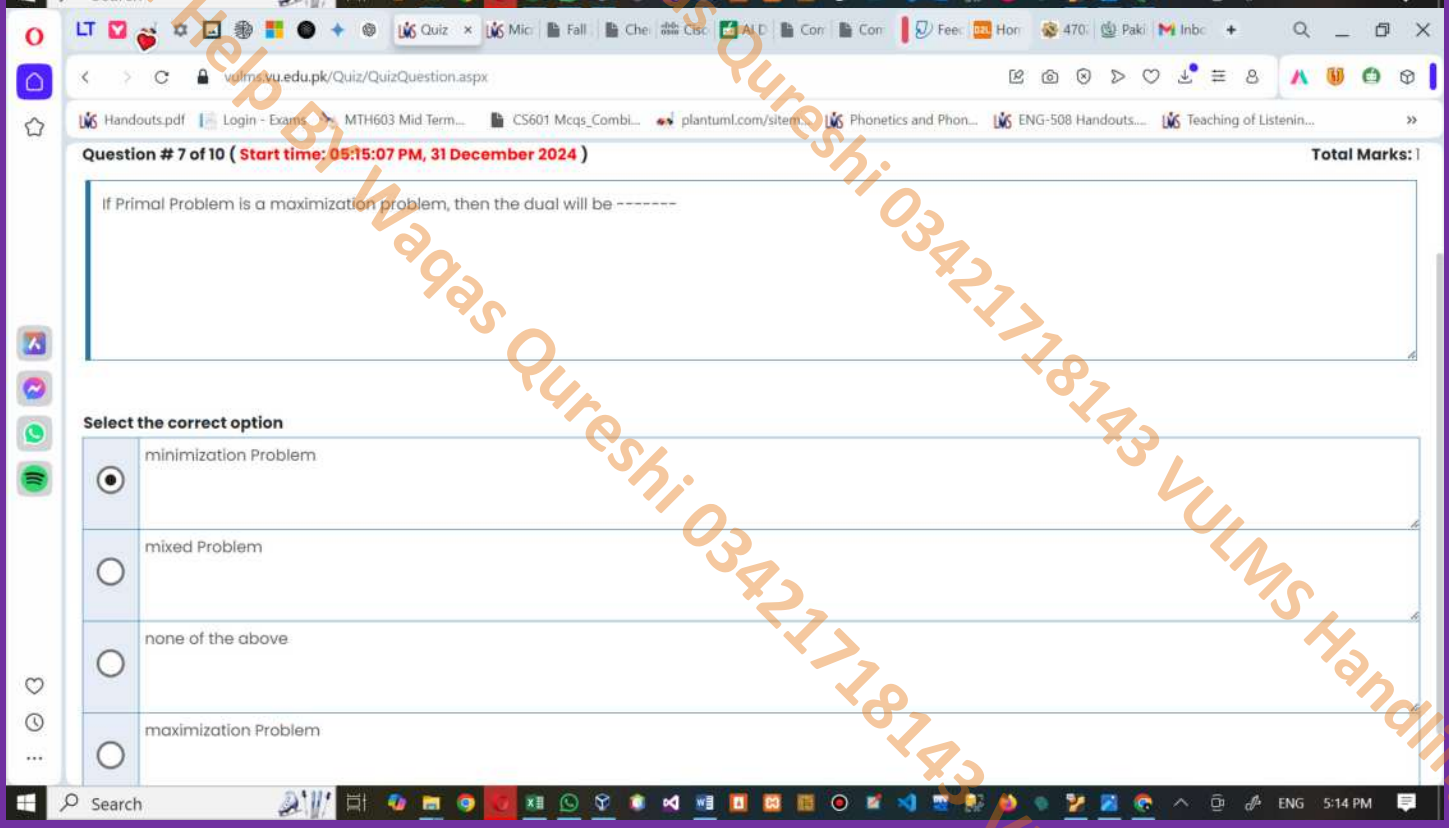
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Which of the following is true about the inclusion of non-negative slack variable into a constraint of type 'less than or equal'?

Select the correct option

- (●) This inclusion doesn't not affect the inequality
- ( ) Inequality become redundant and hence skips from the LP problem.
- ( ) Inequality is transformed into type: 'greater than or equal'
- ( ) Inequality is transformed into strict equality constraint



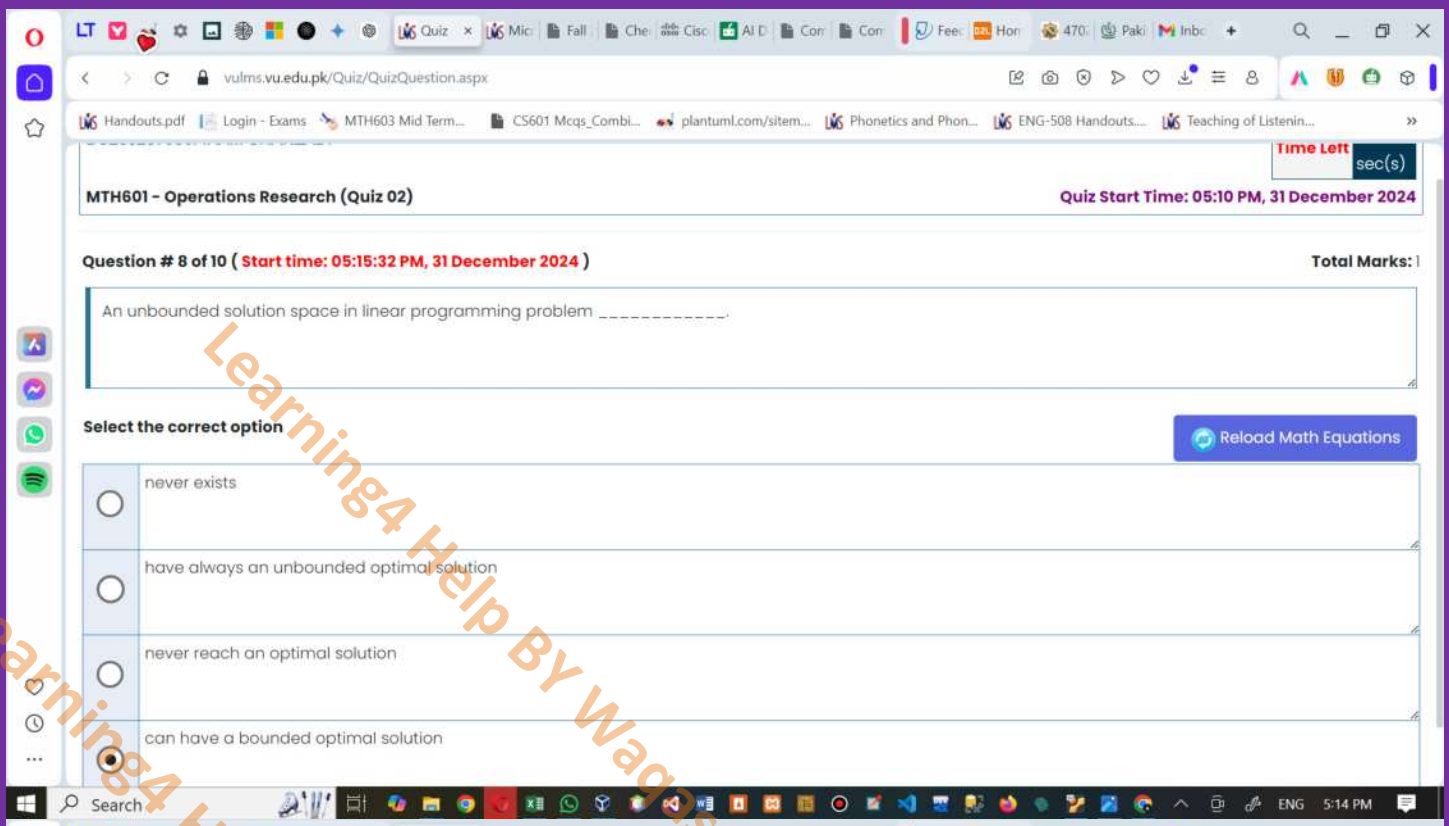
Question # 7 of 10 (Start time: 05:15:07 PM, 31 December 2024) **Total Marks:**

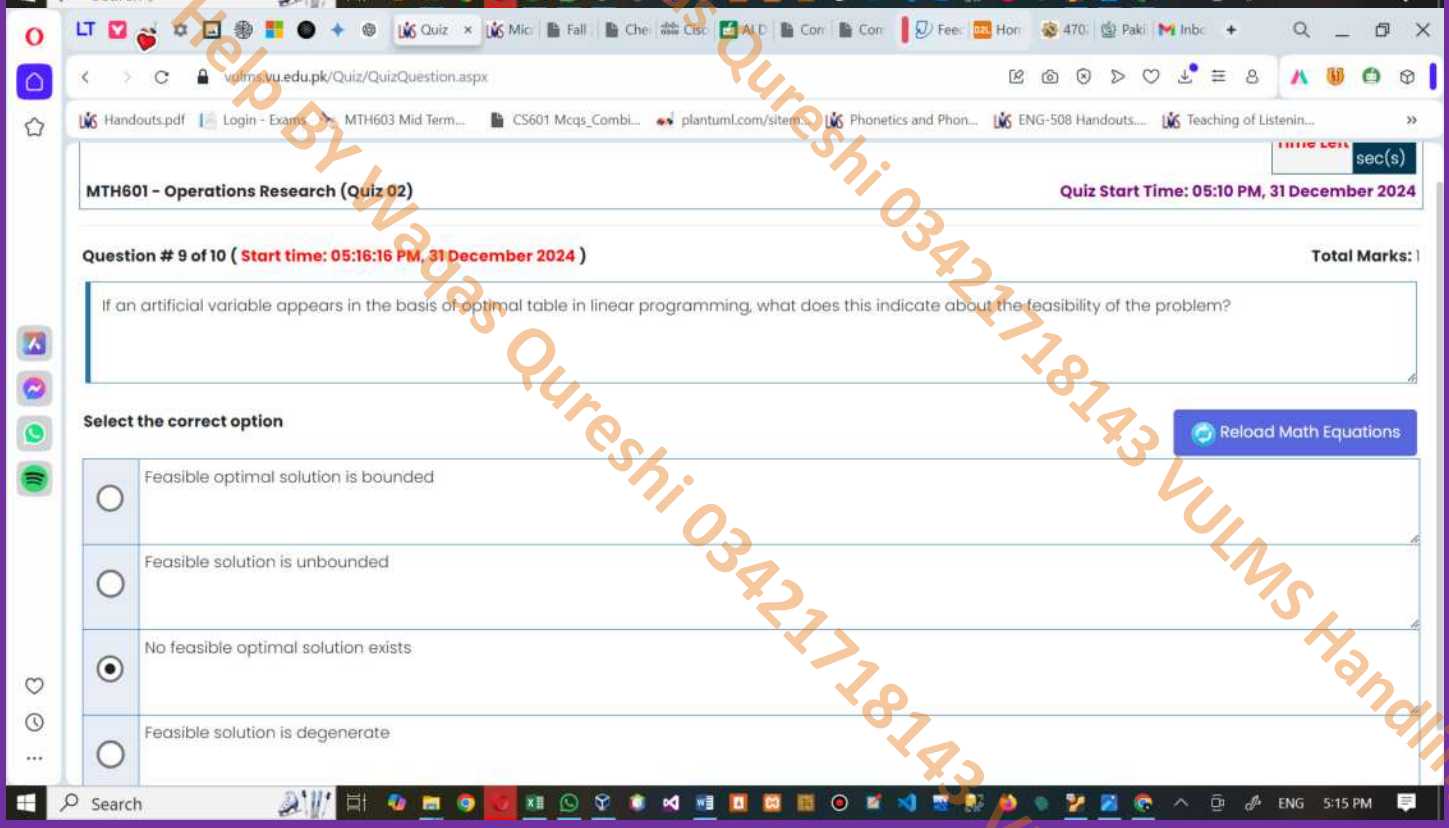
If Primal Problem is a maximization problem, then the dual will be -------

Select the correct option

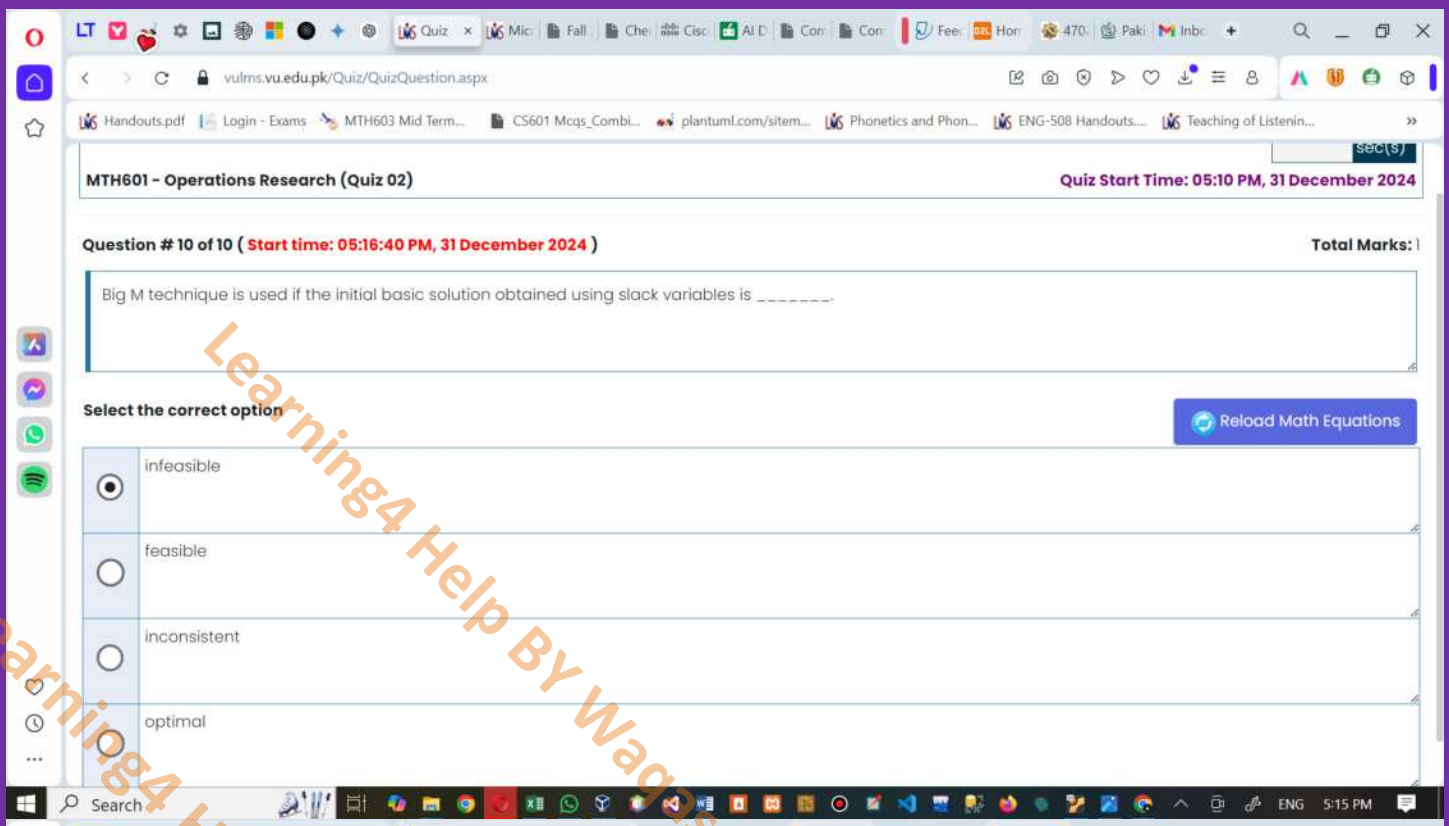
- (●) minimization Problem
- ( ) mixed Problem
- ( ) none of the above
- ( ) maximization Problem

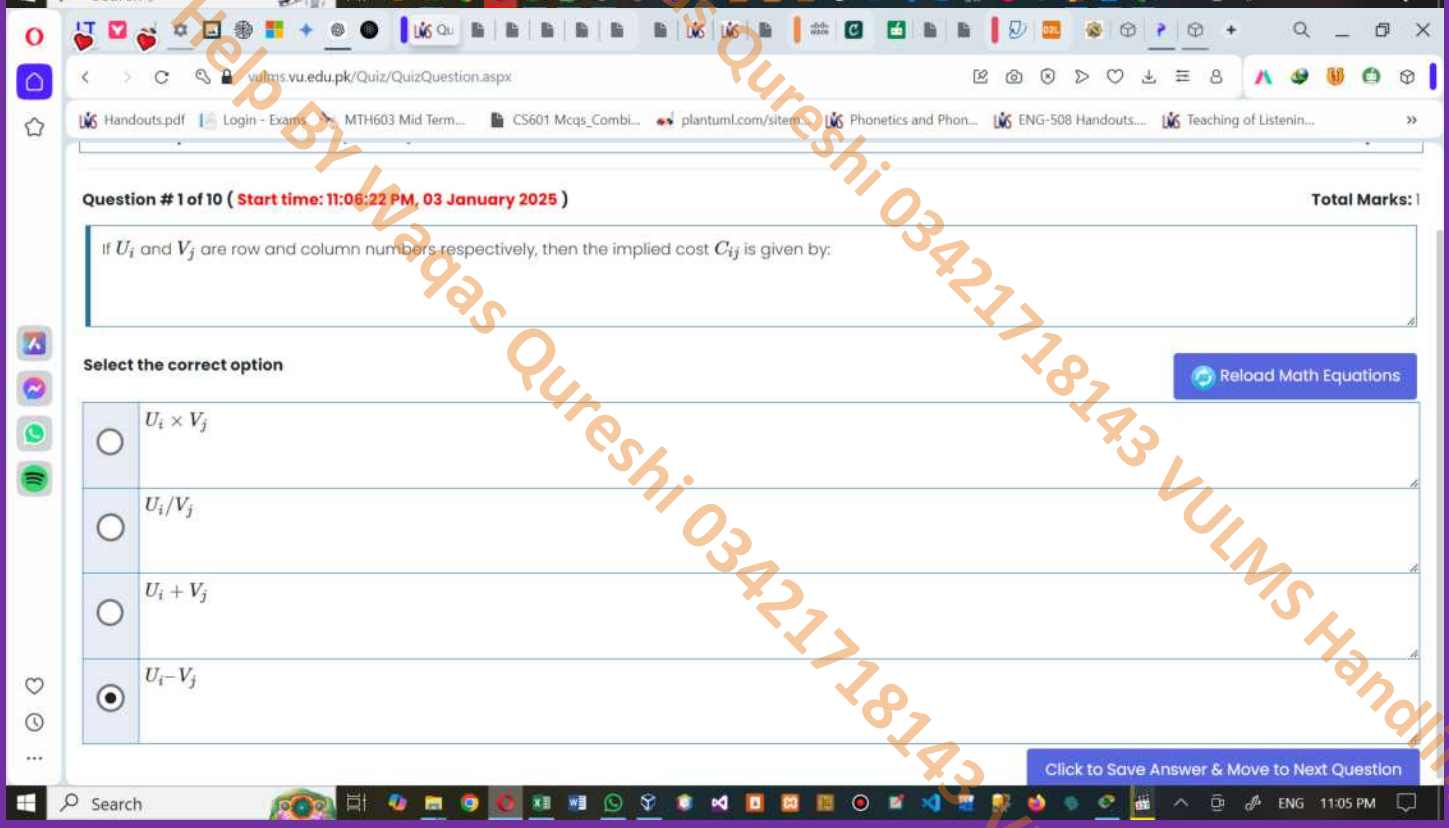
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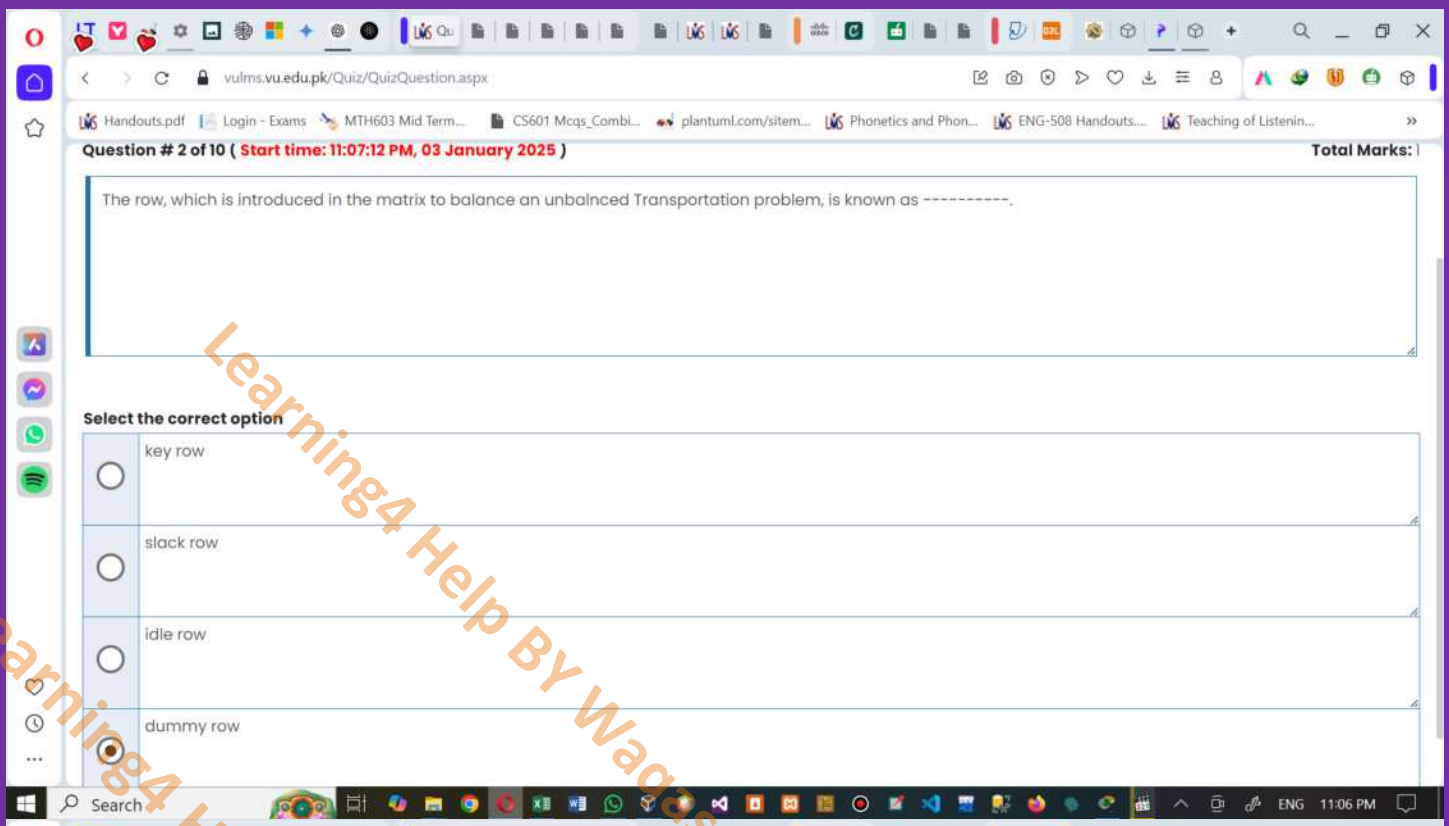


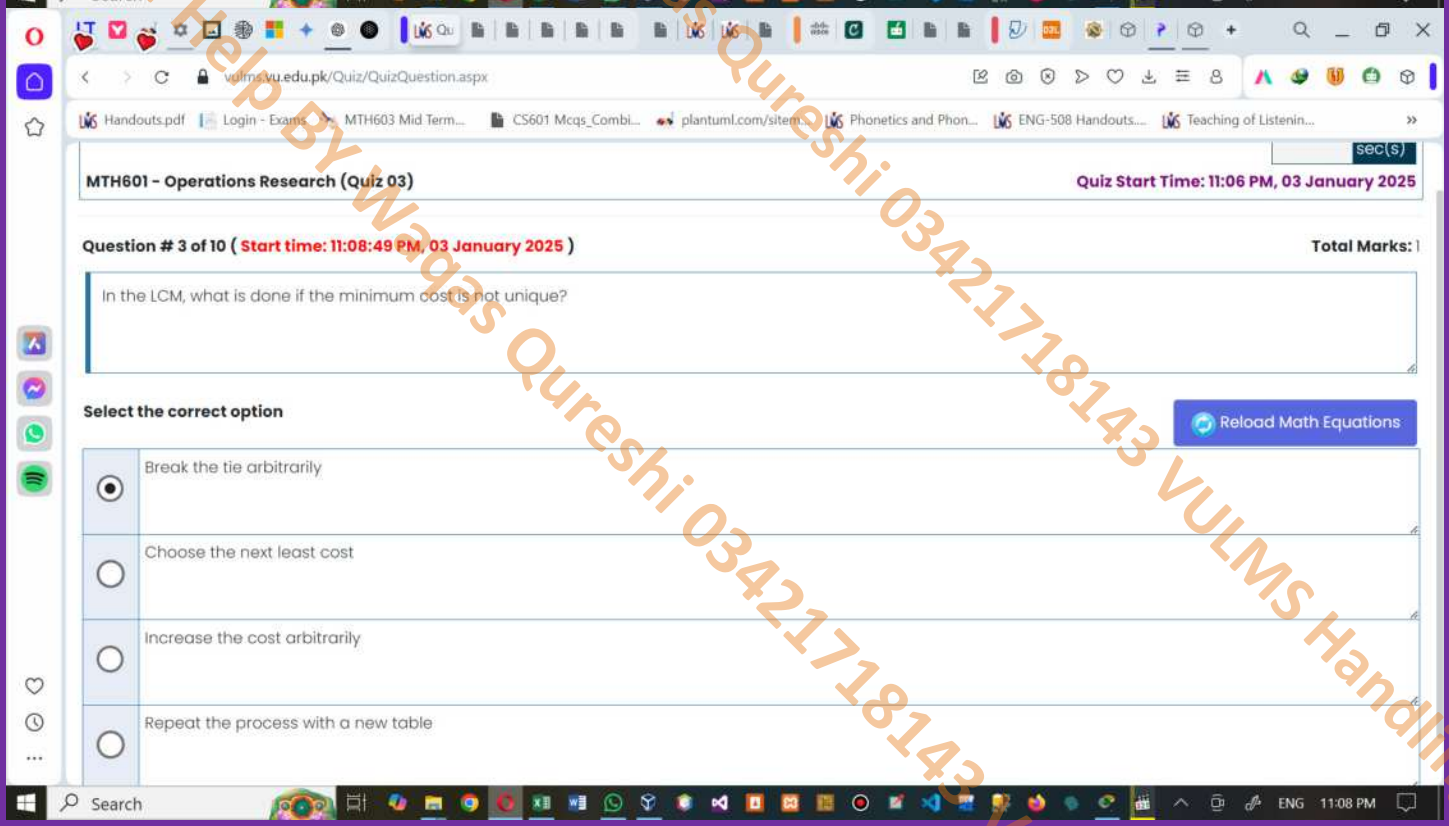
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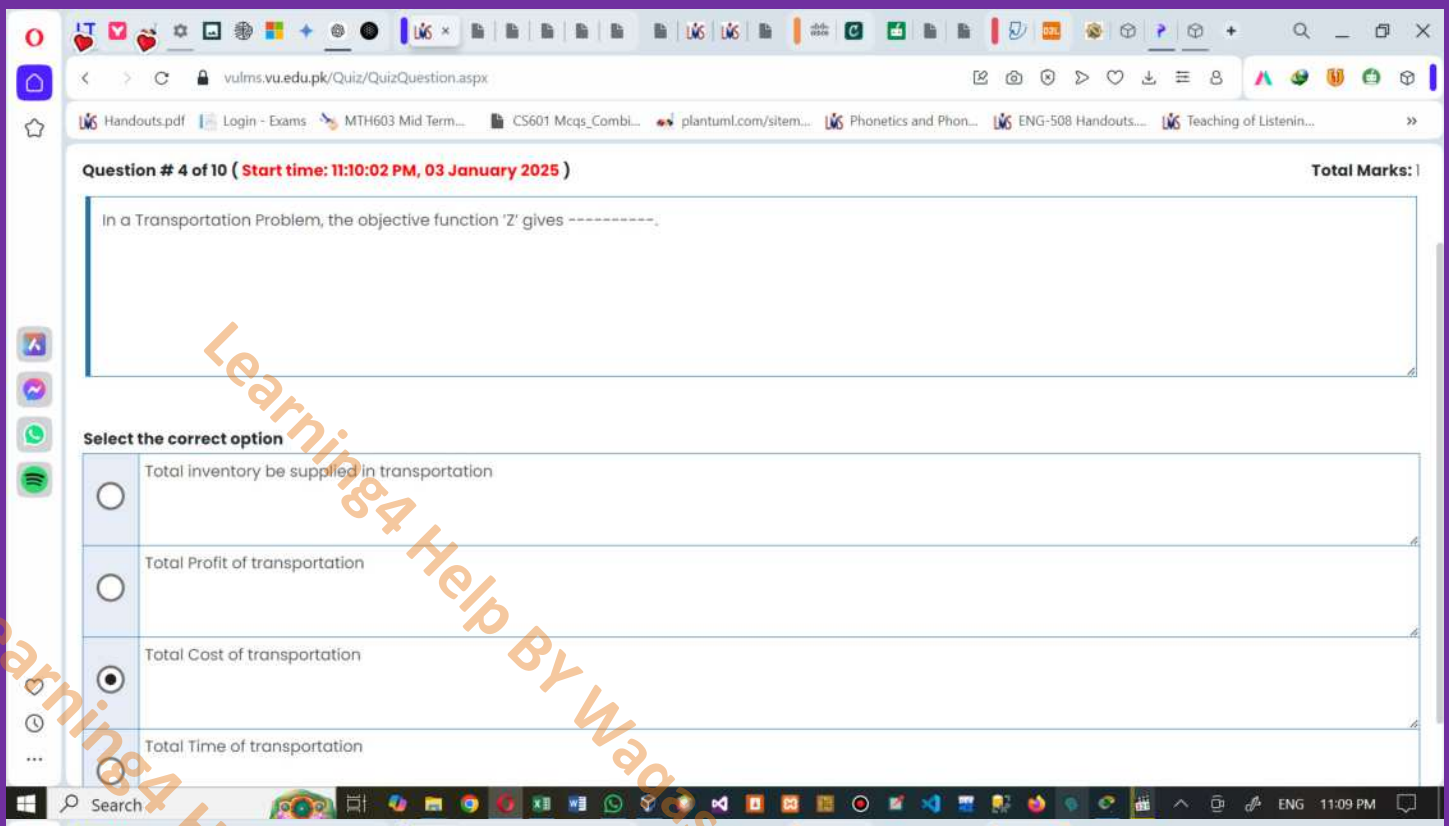


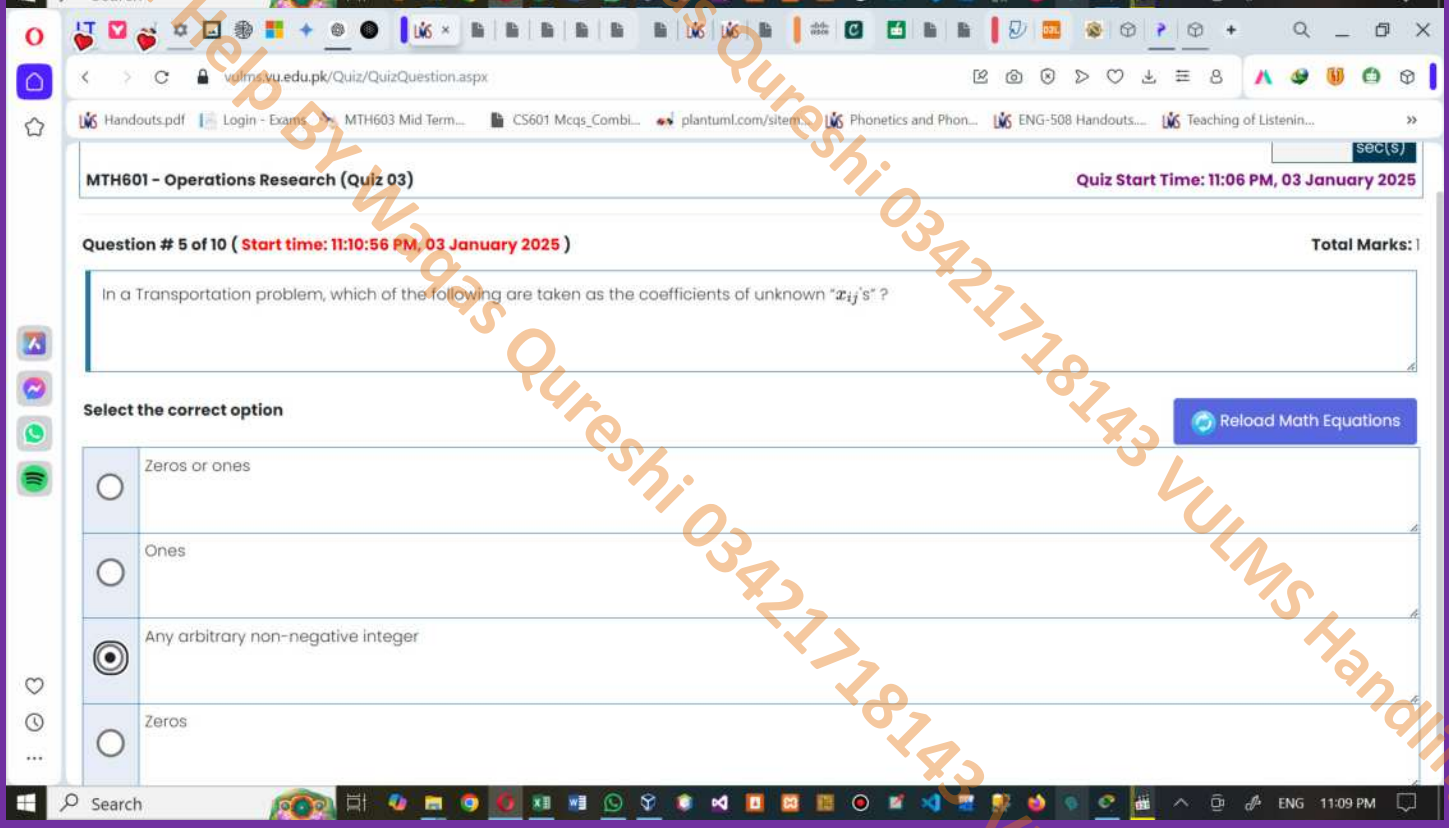
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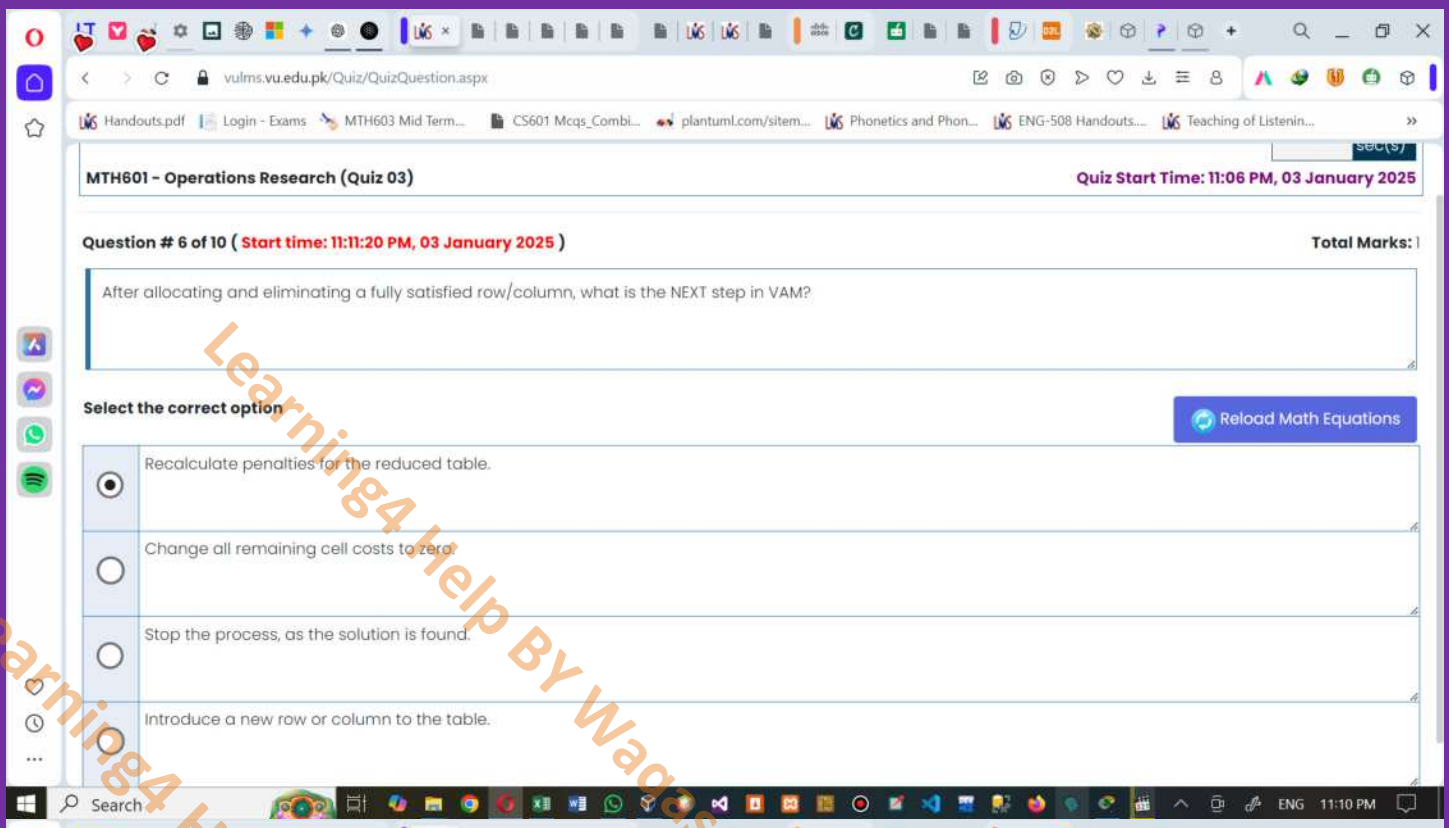


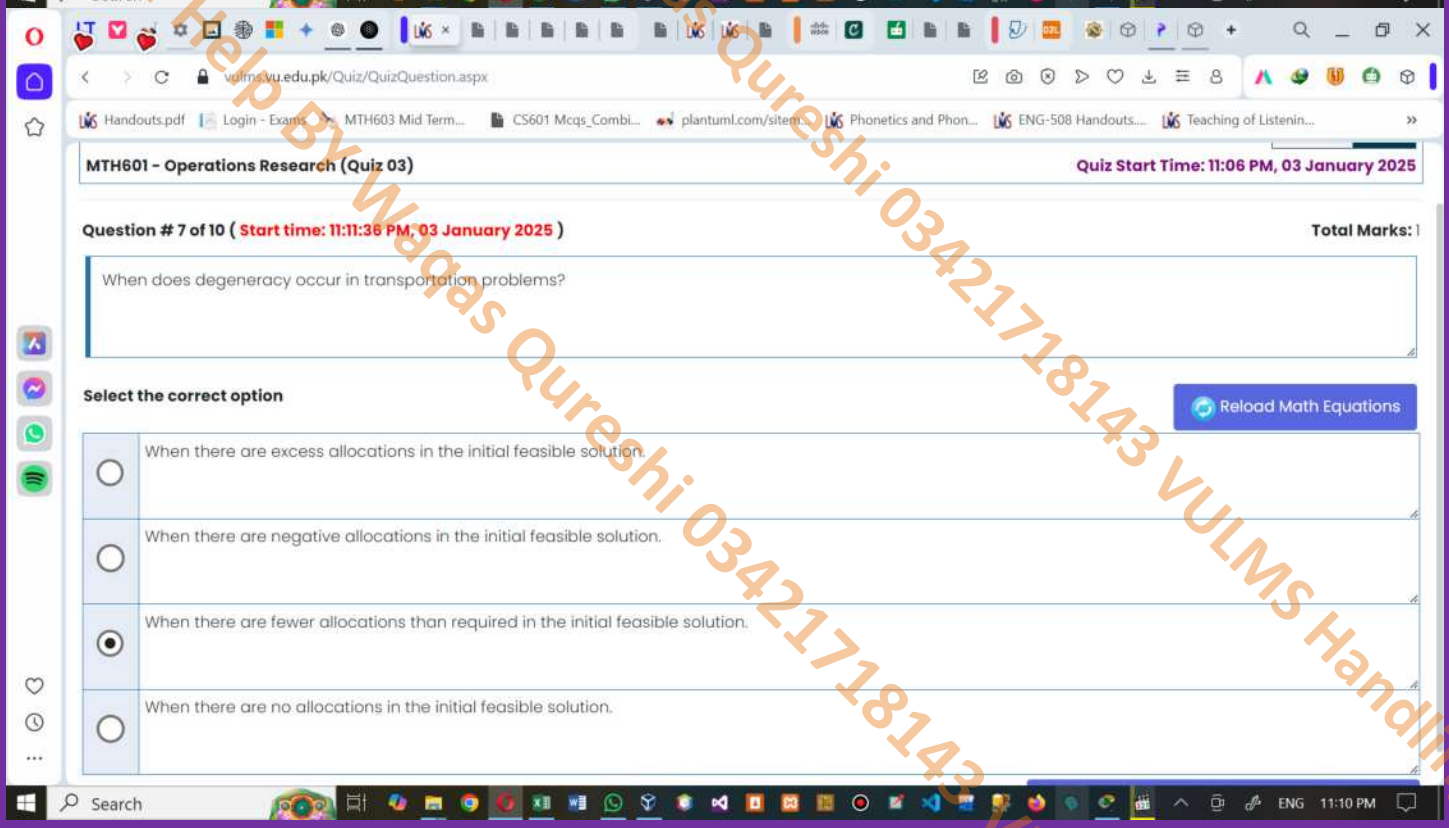
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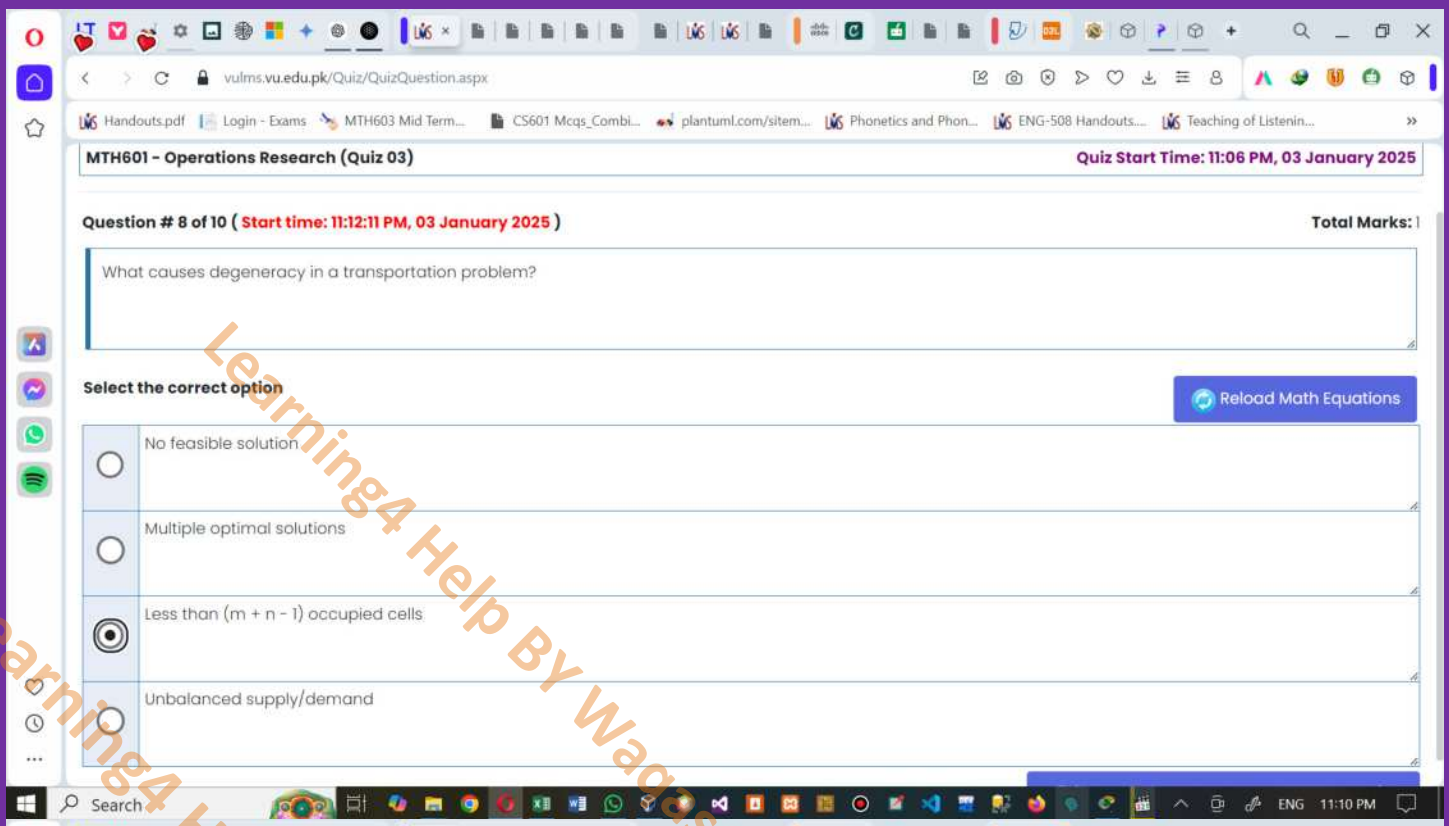


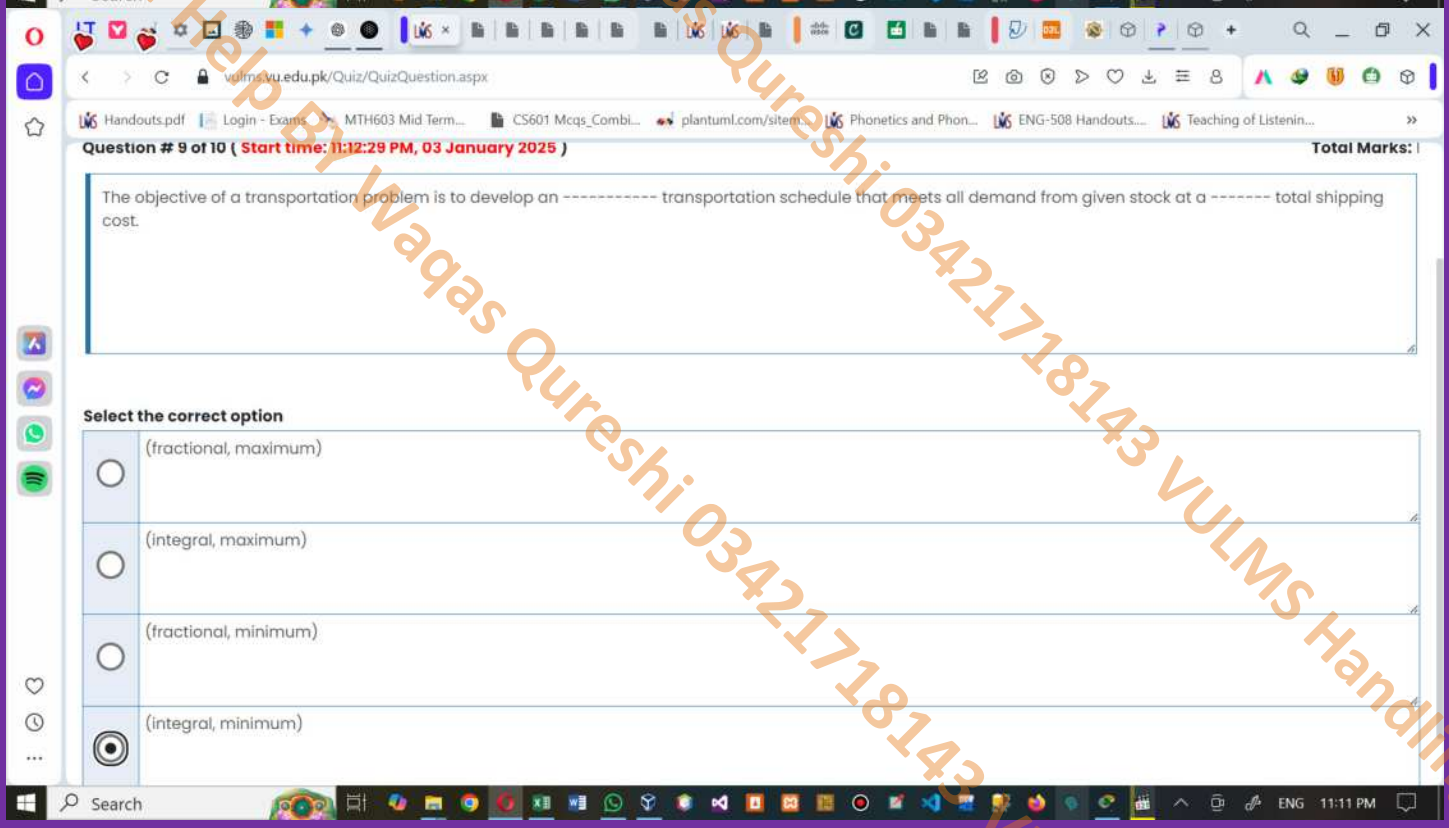
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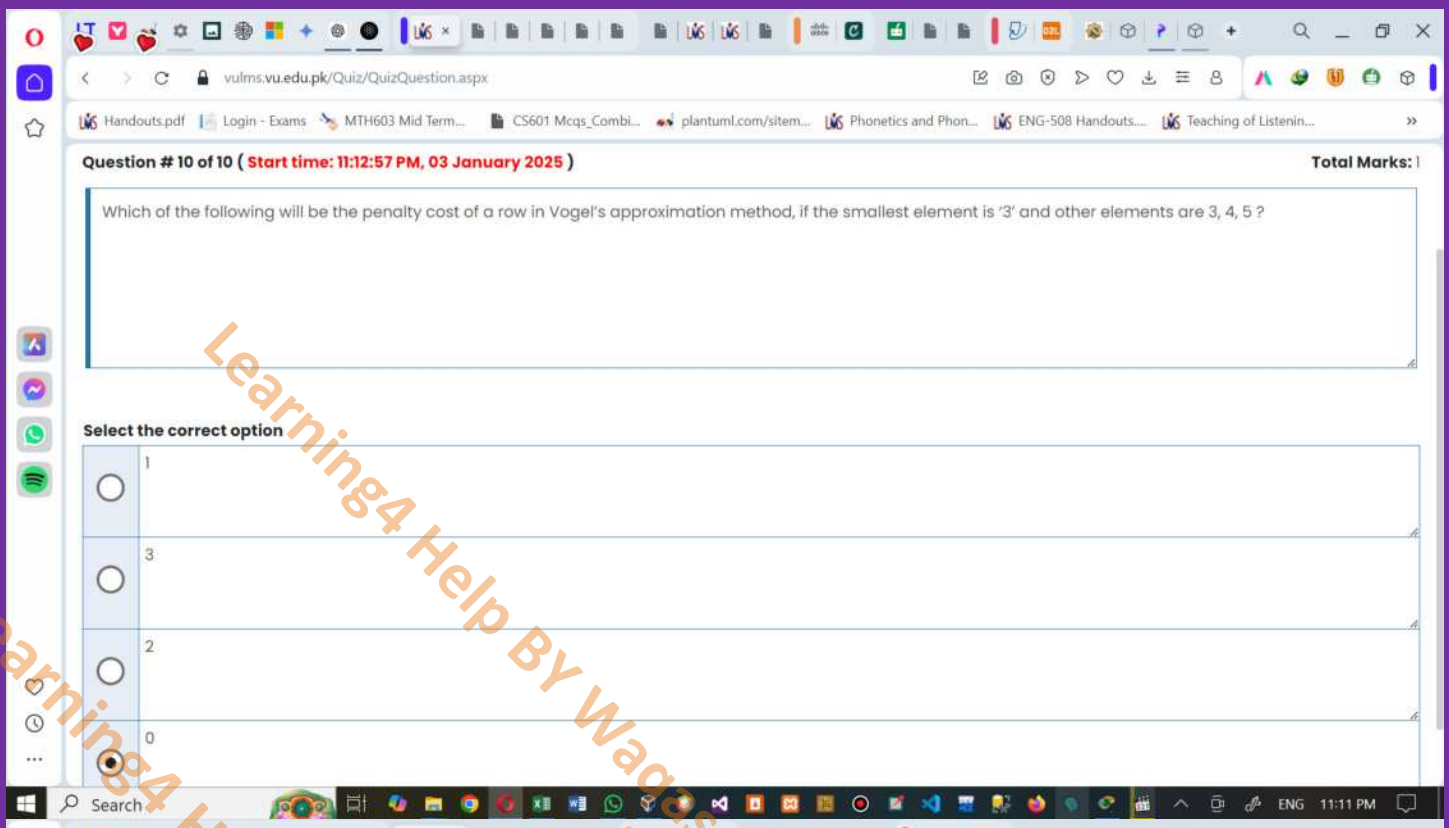


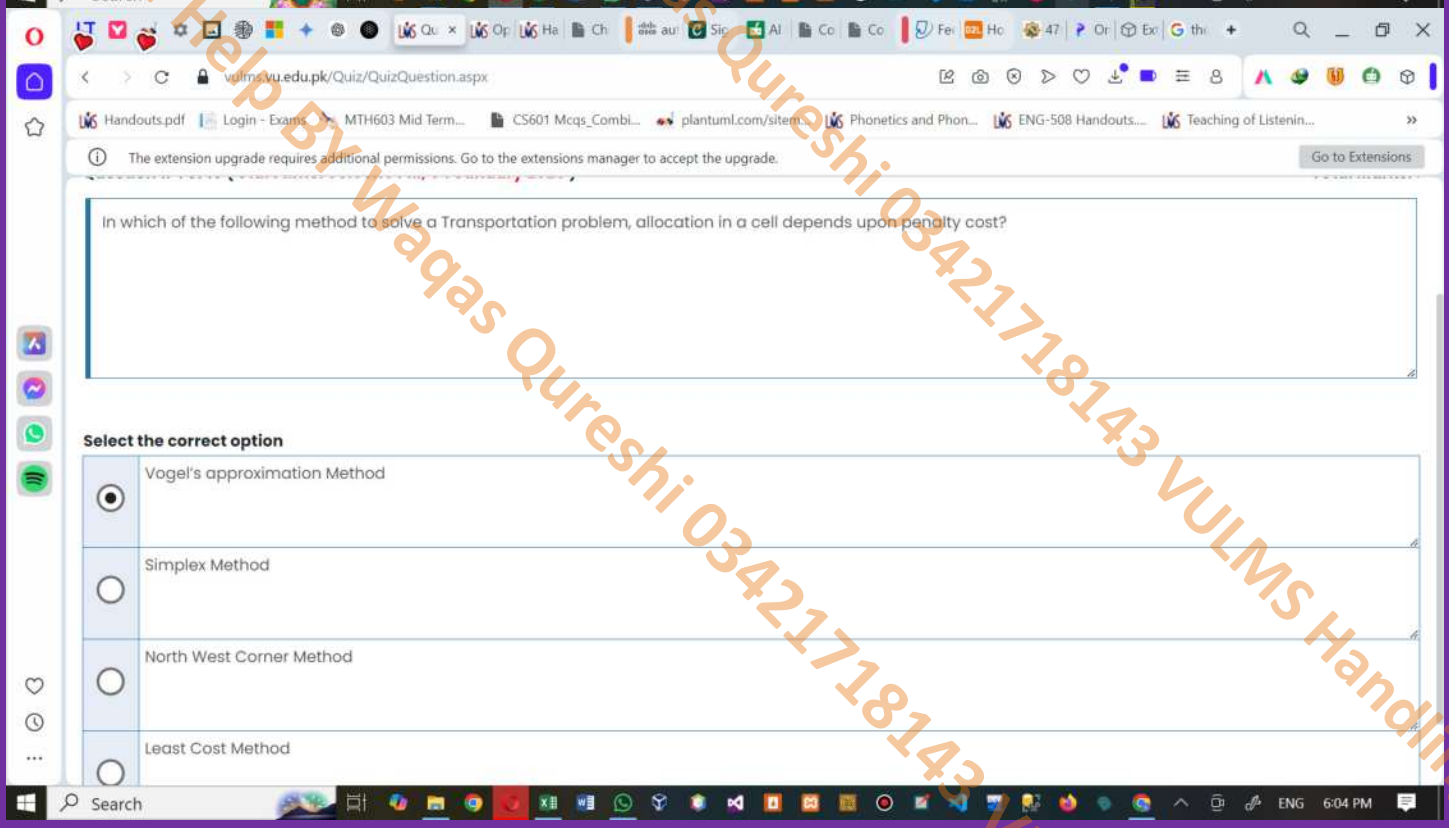
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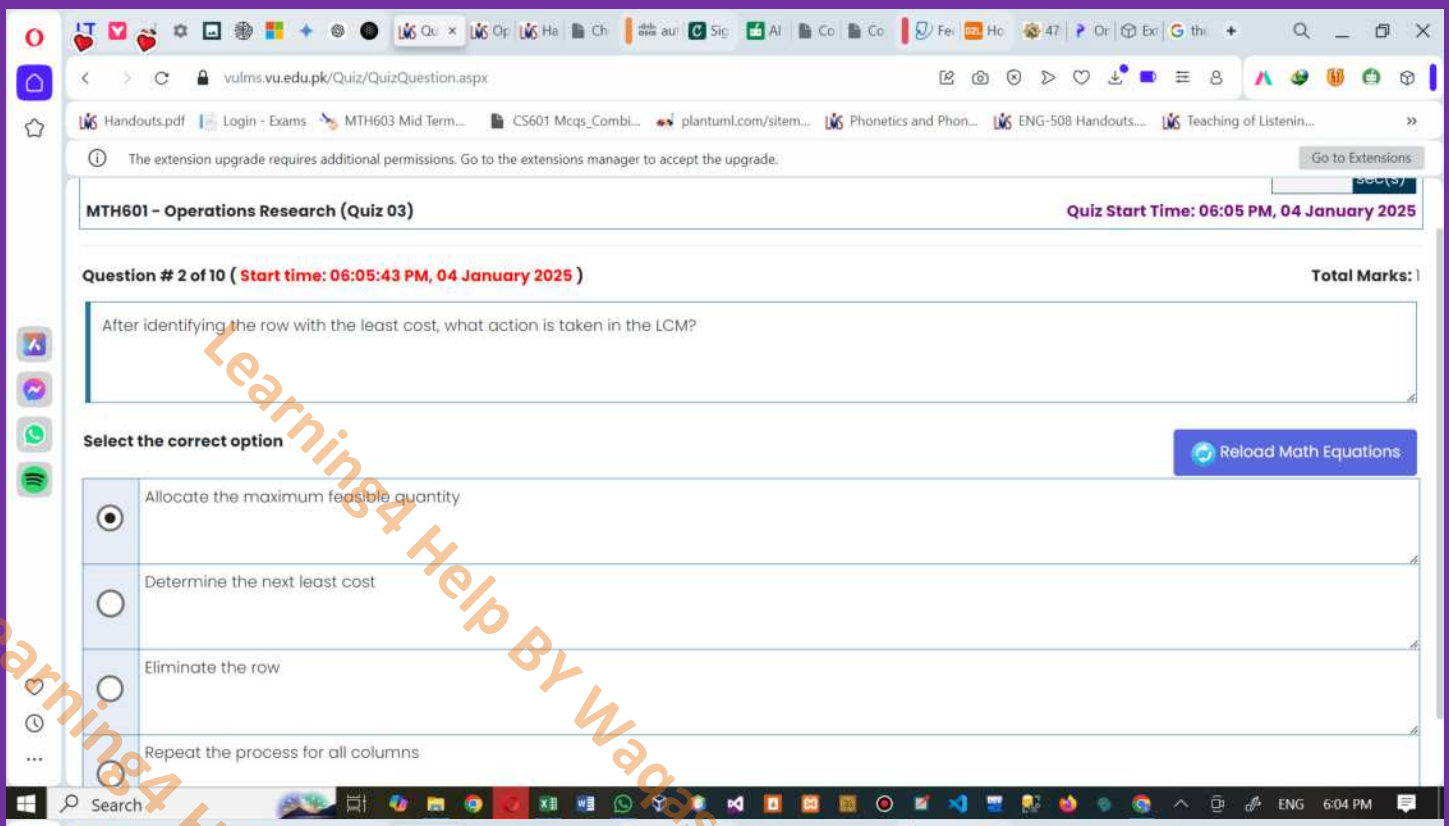


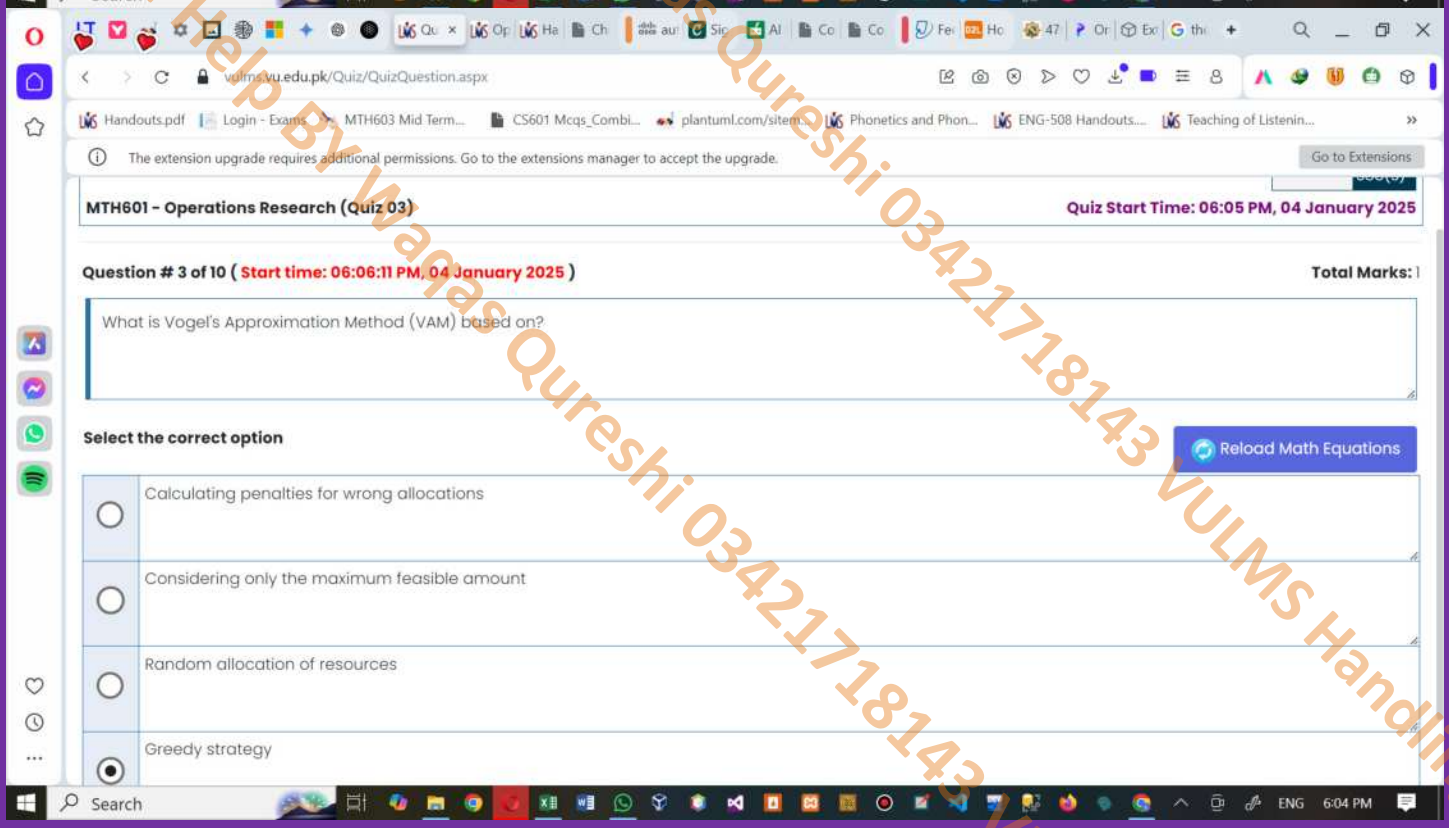
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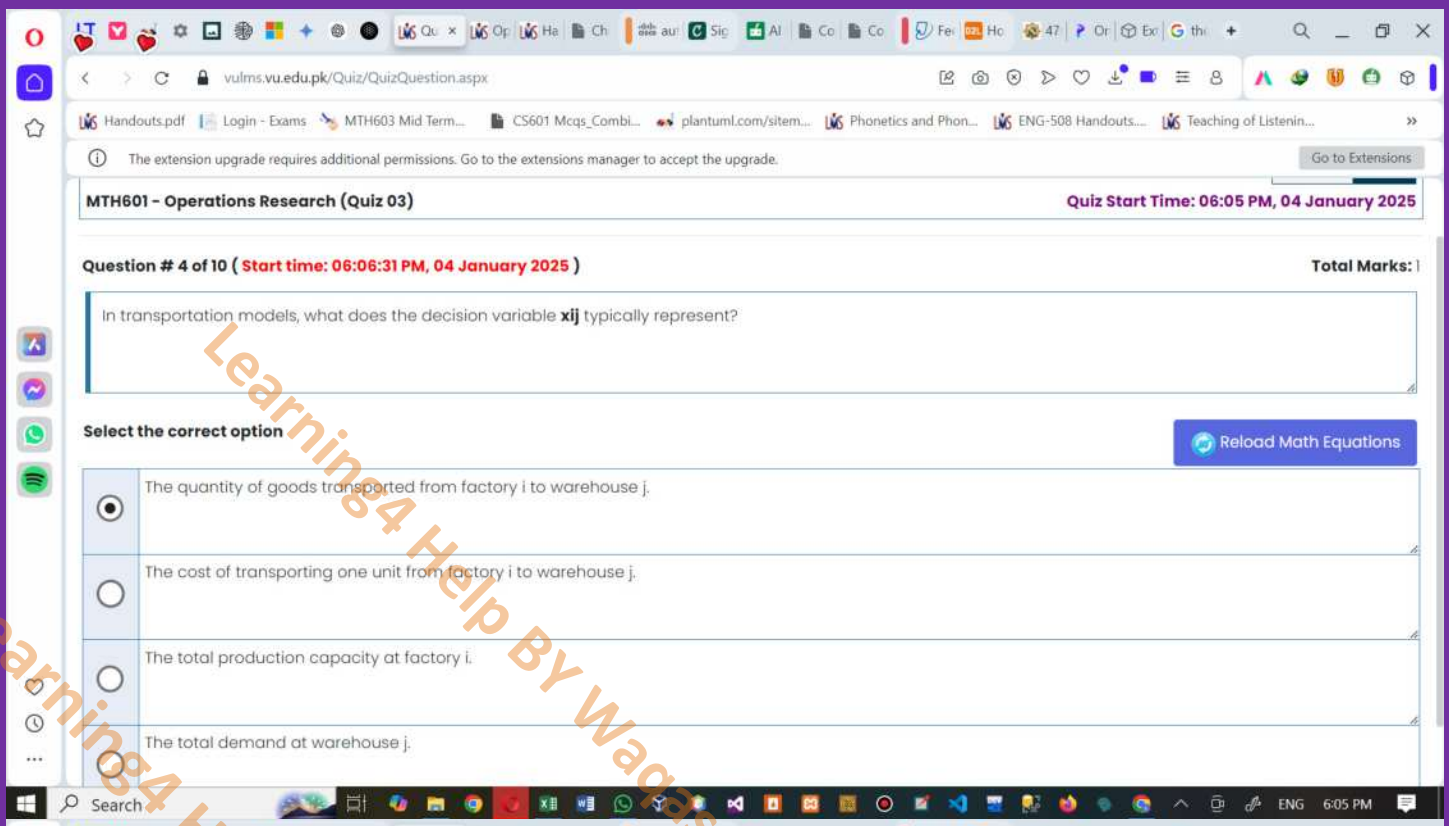


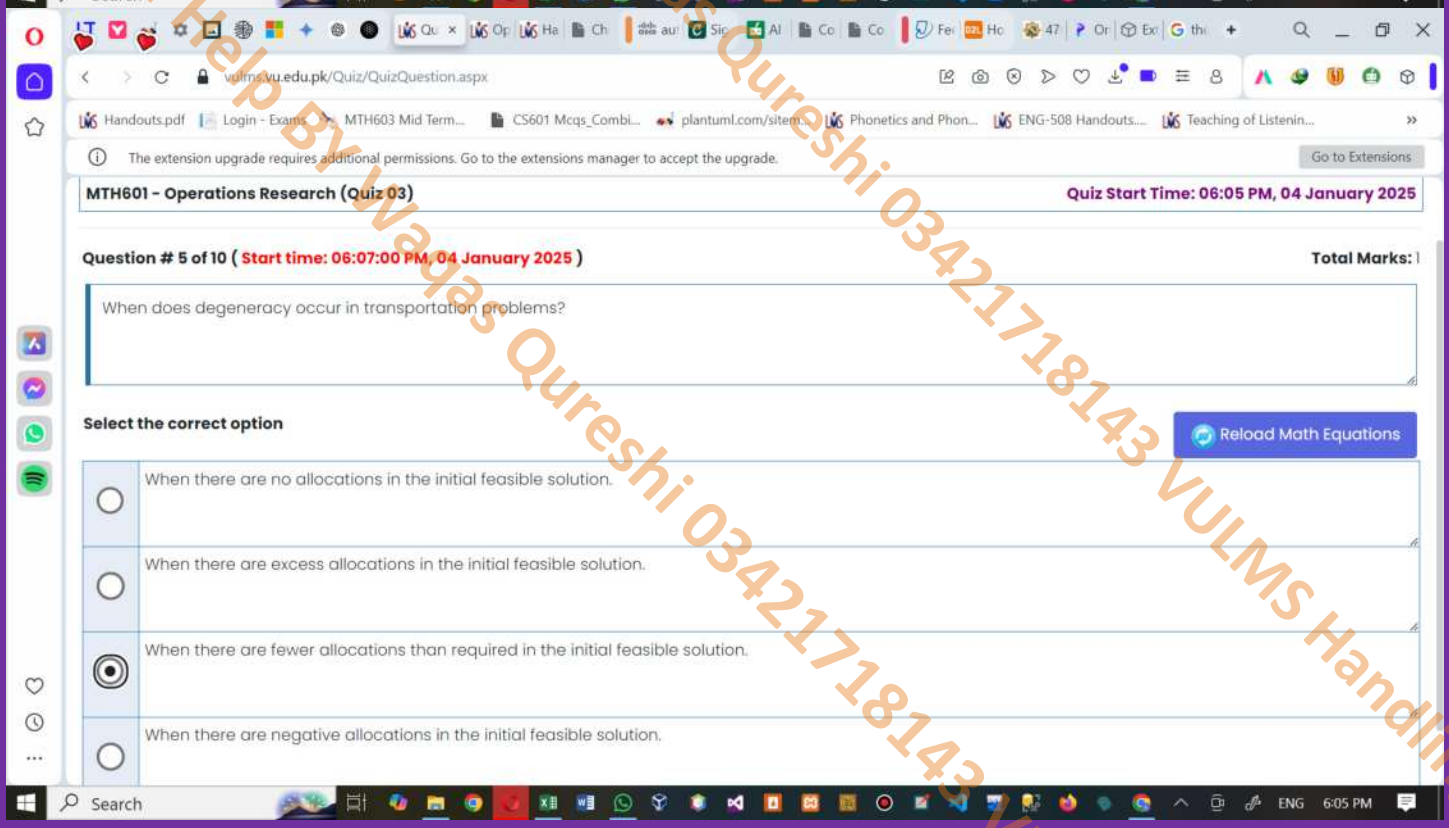
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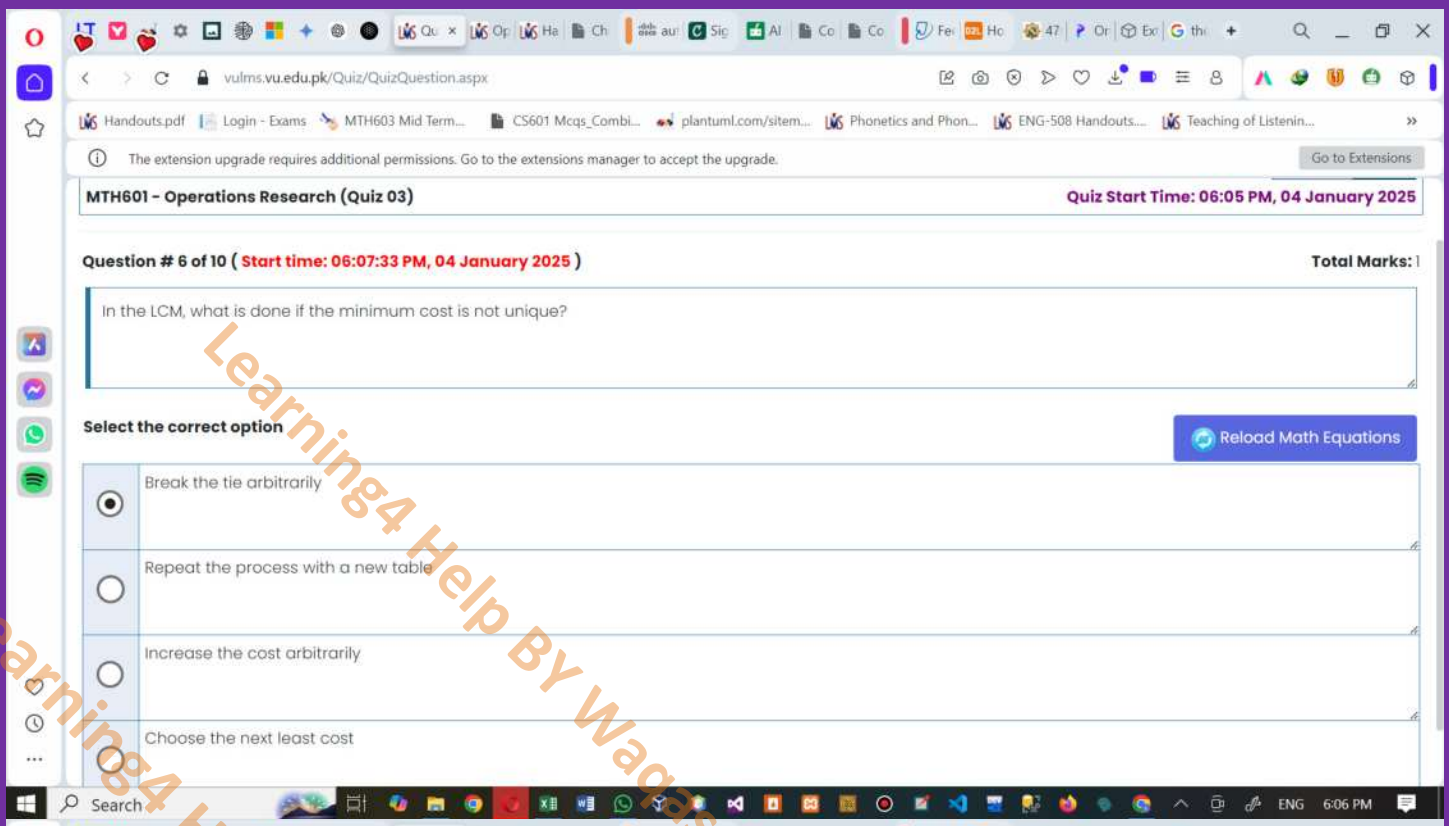


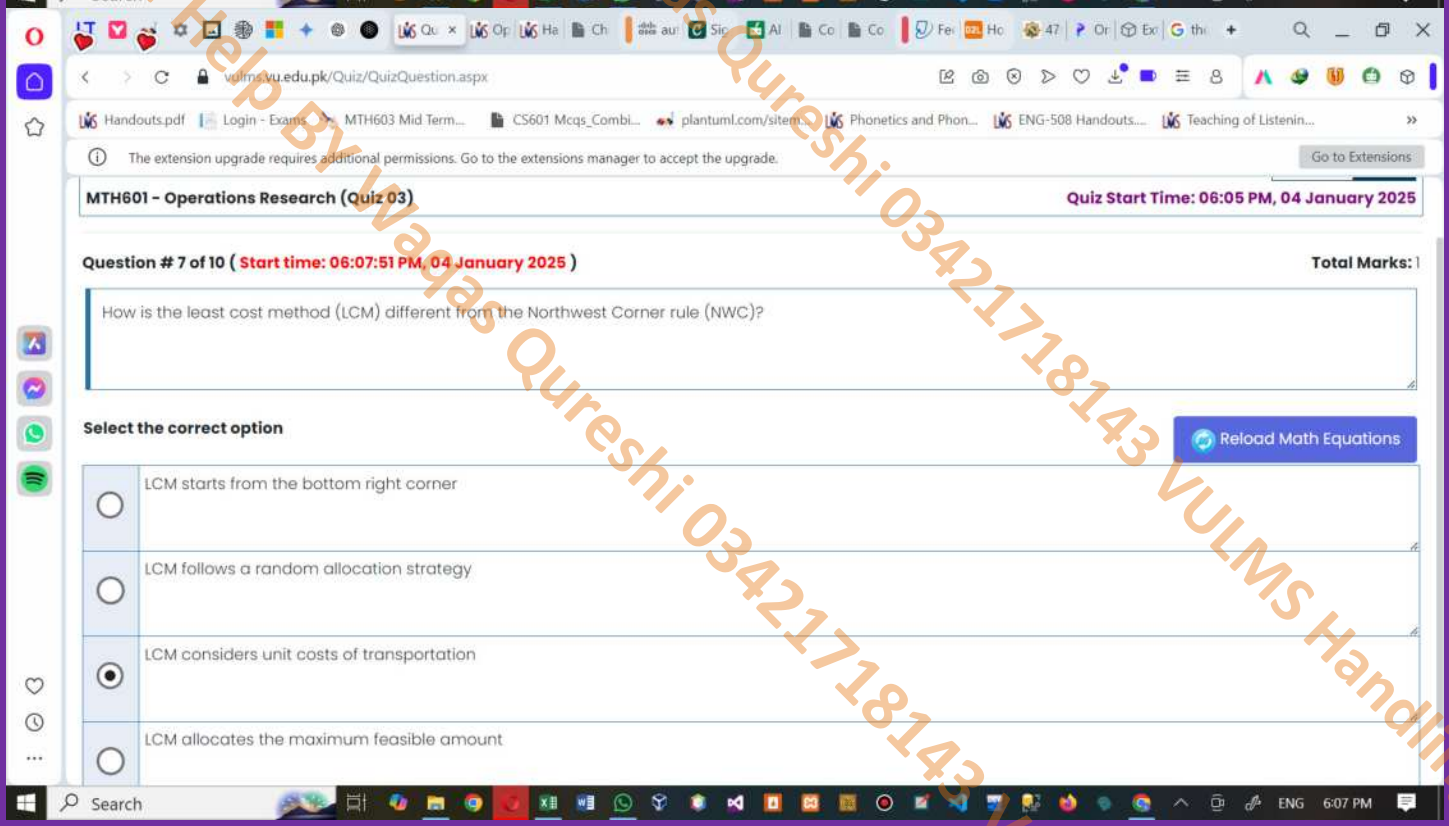
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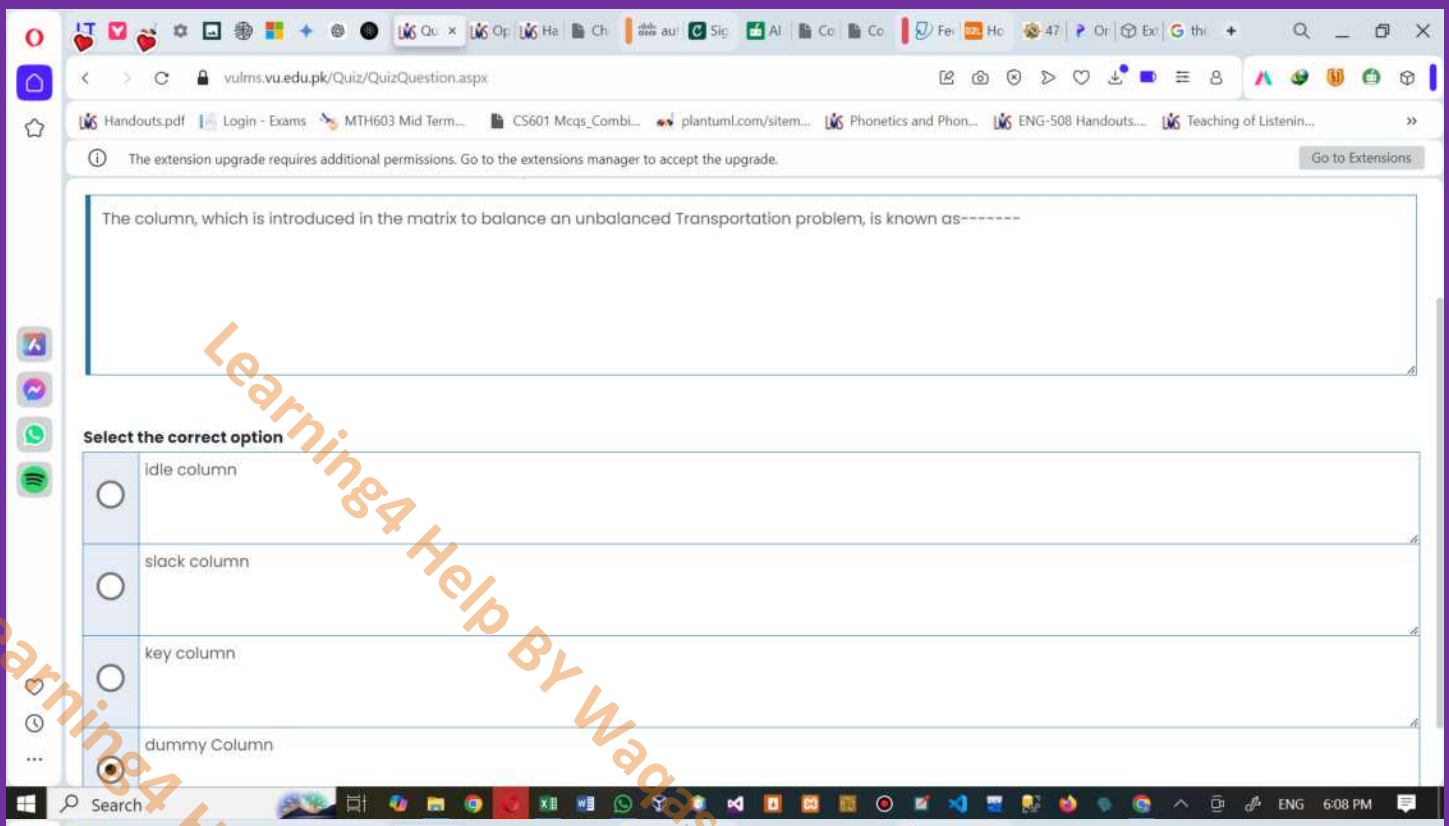


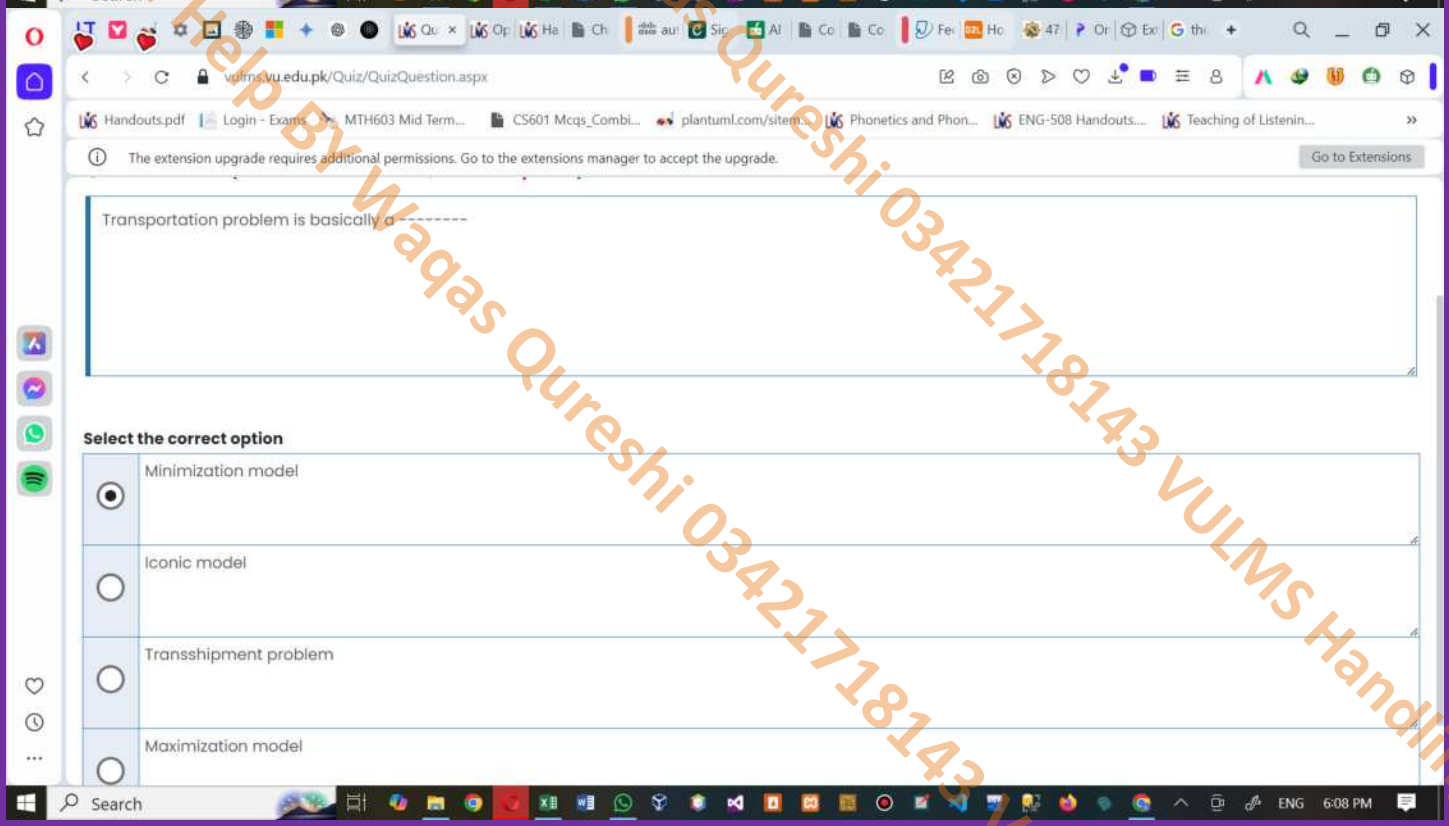
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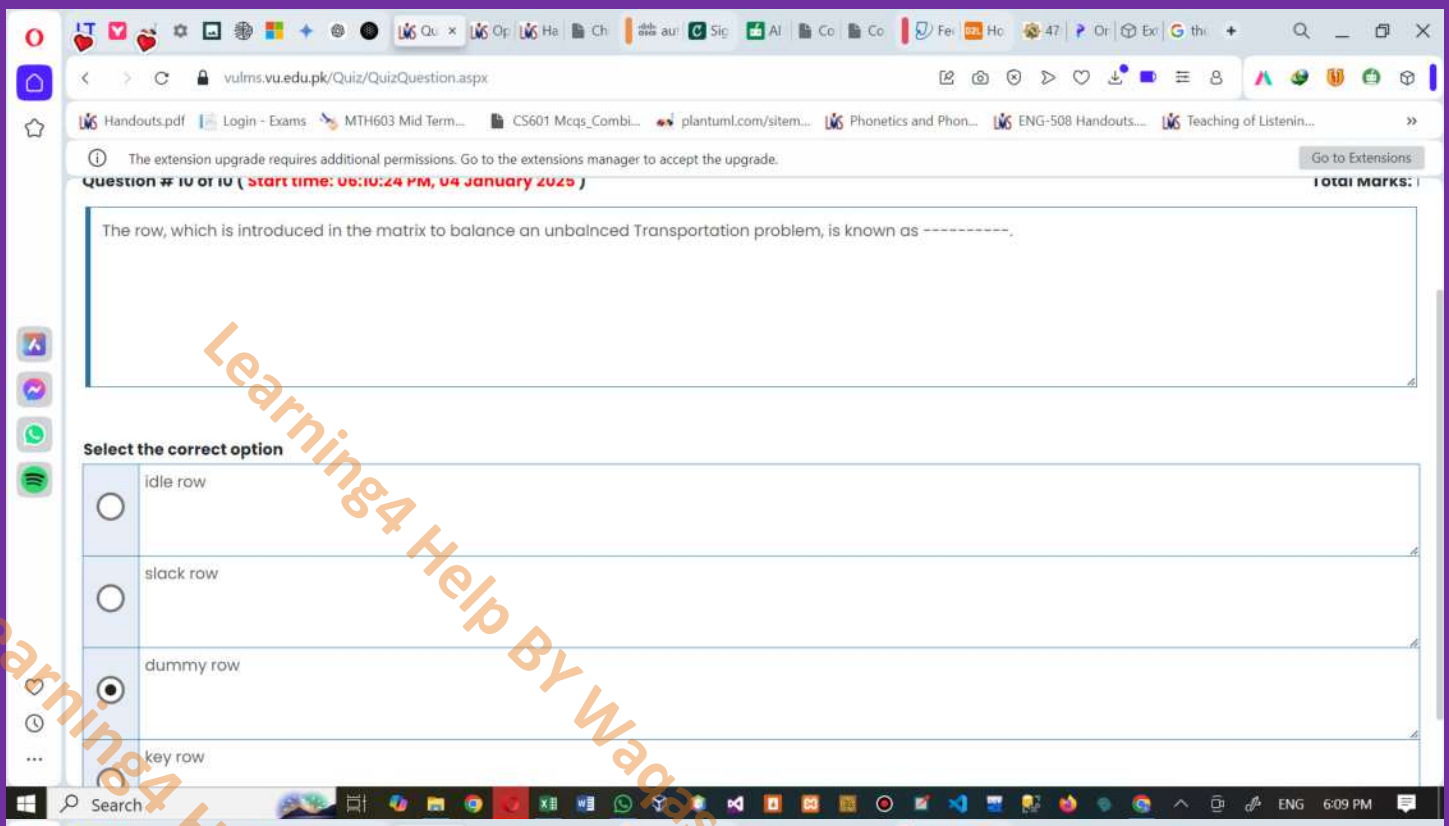


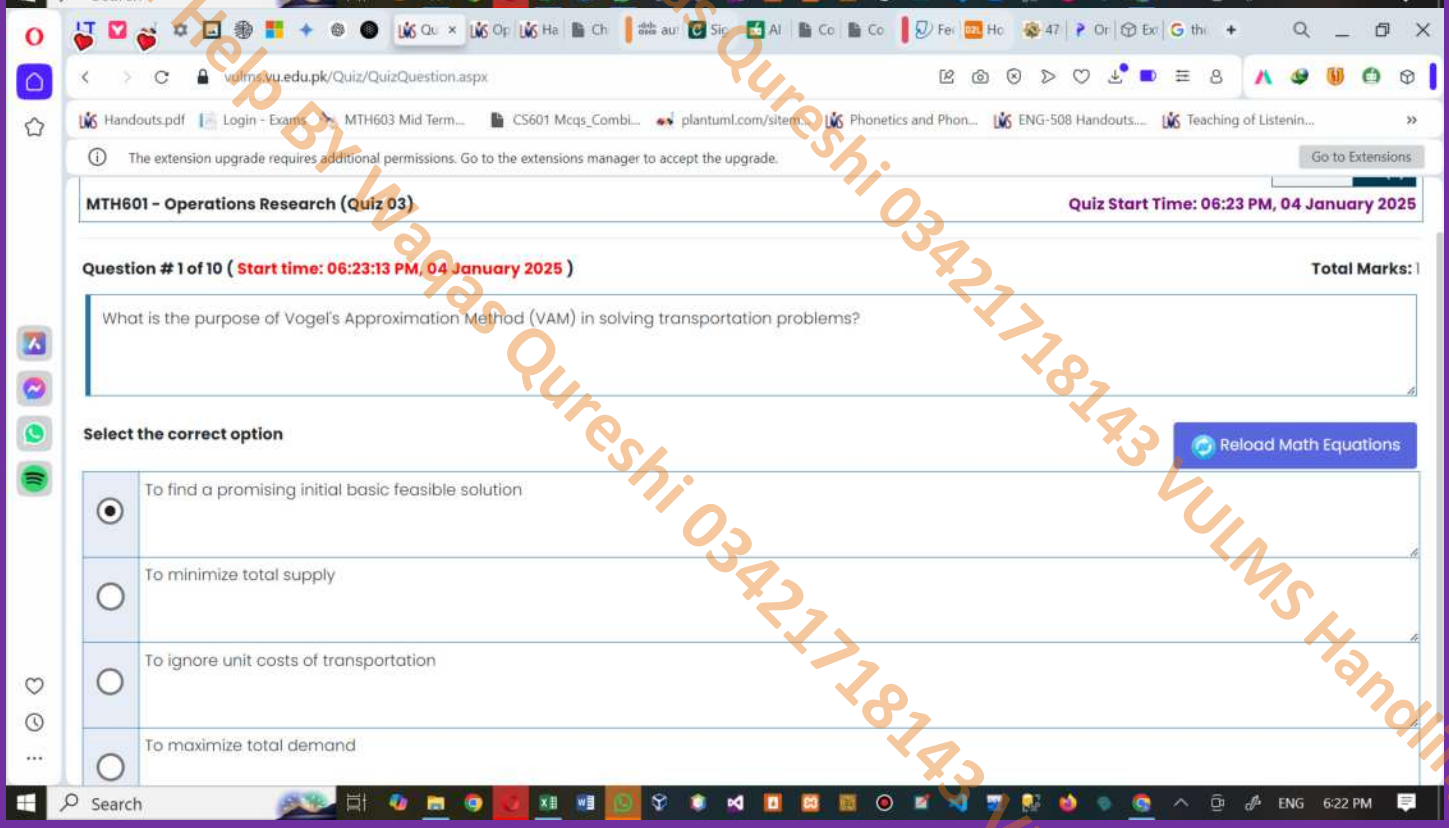
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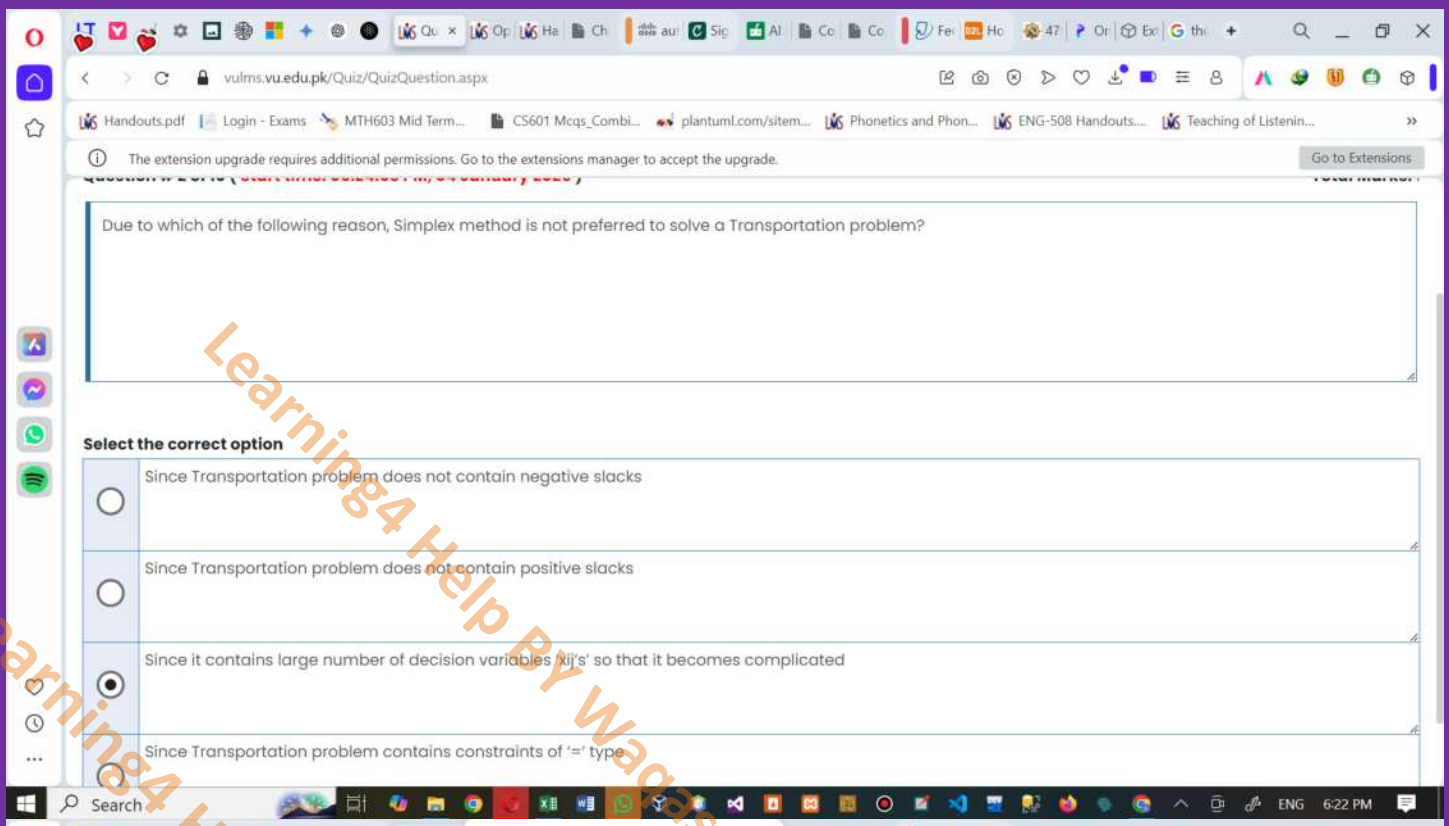


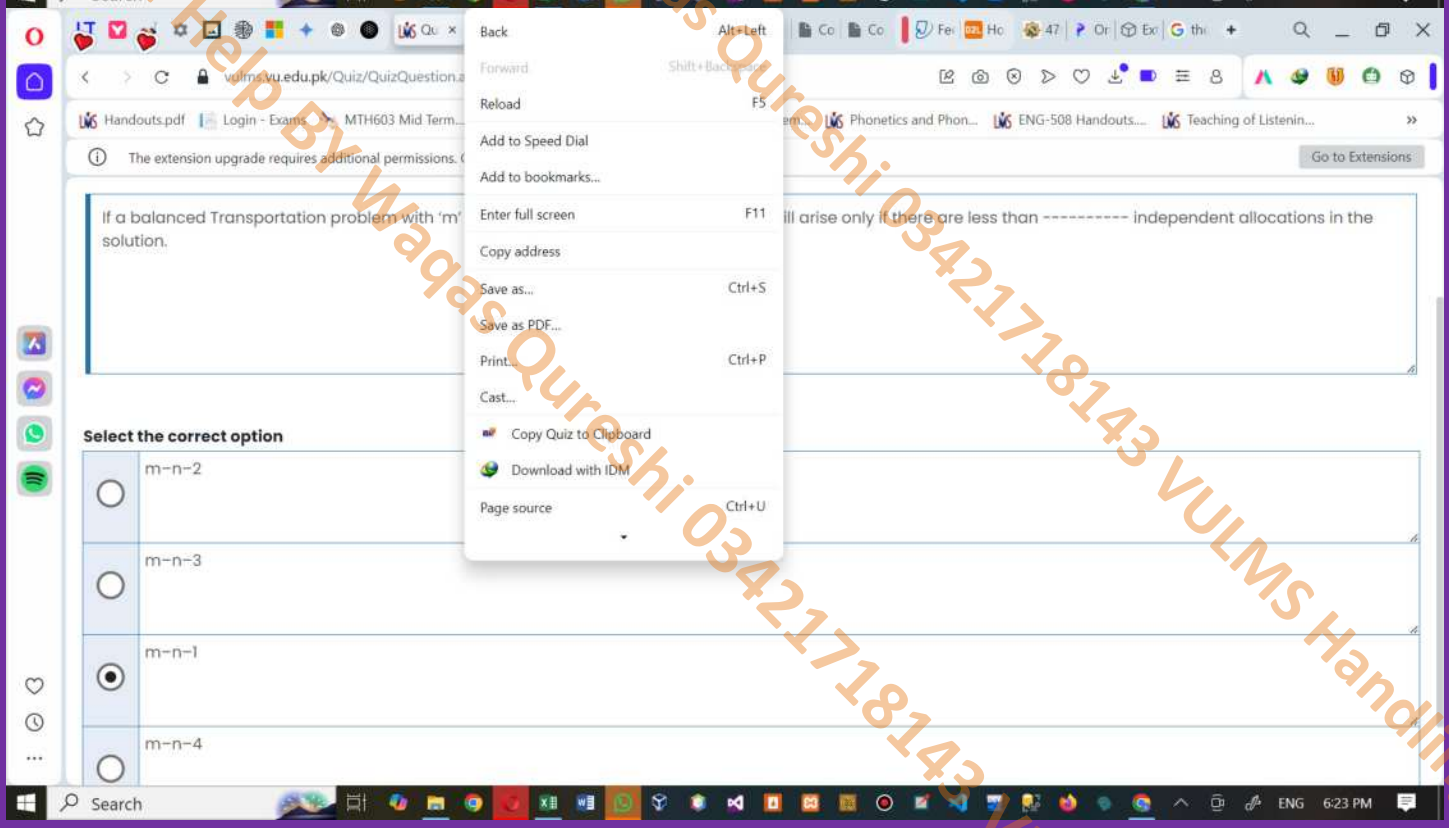
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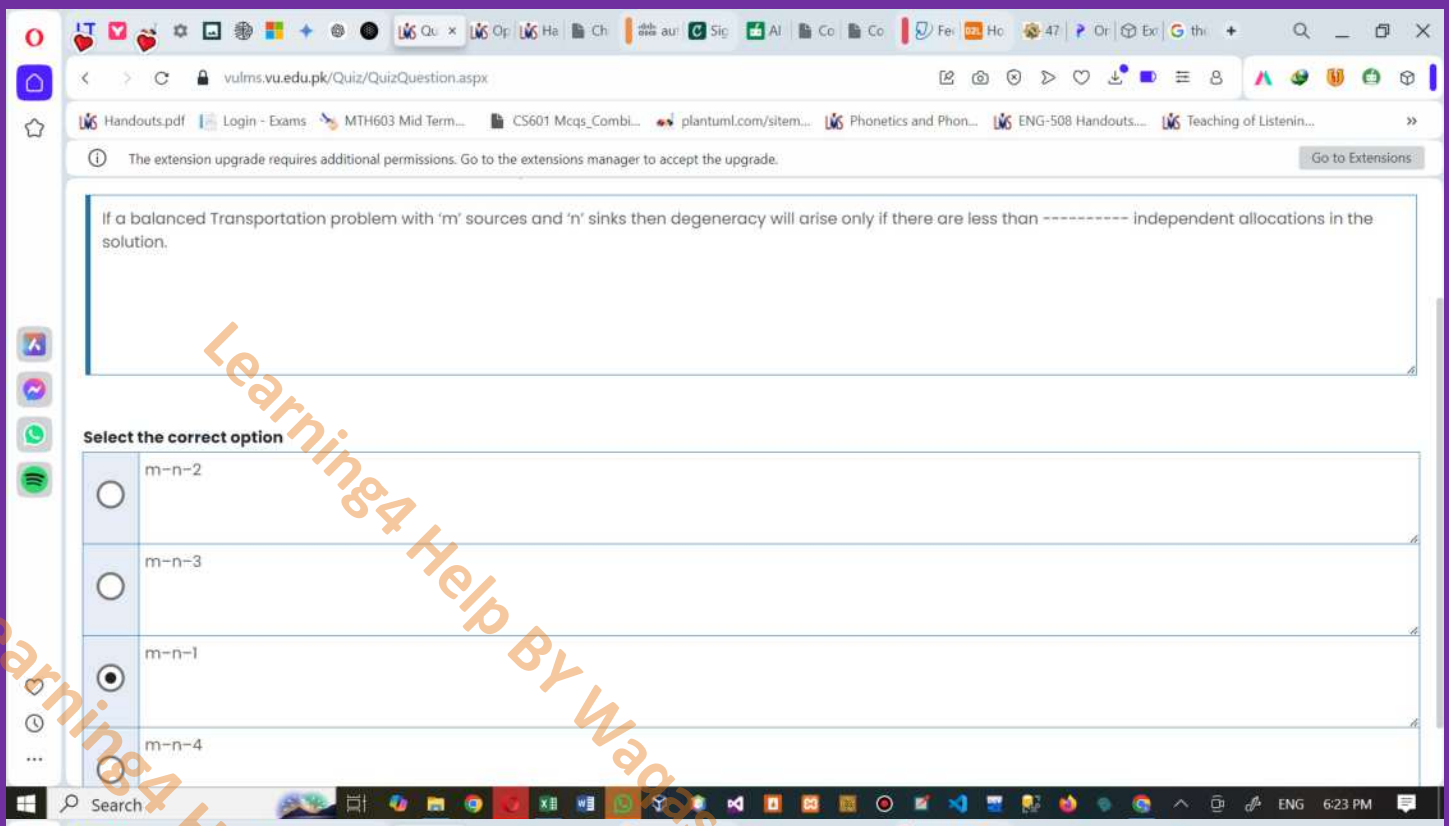


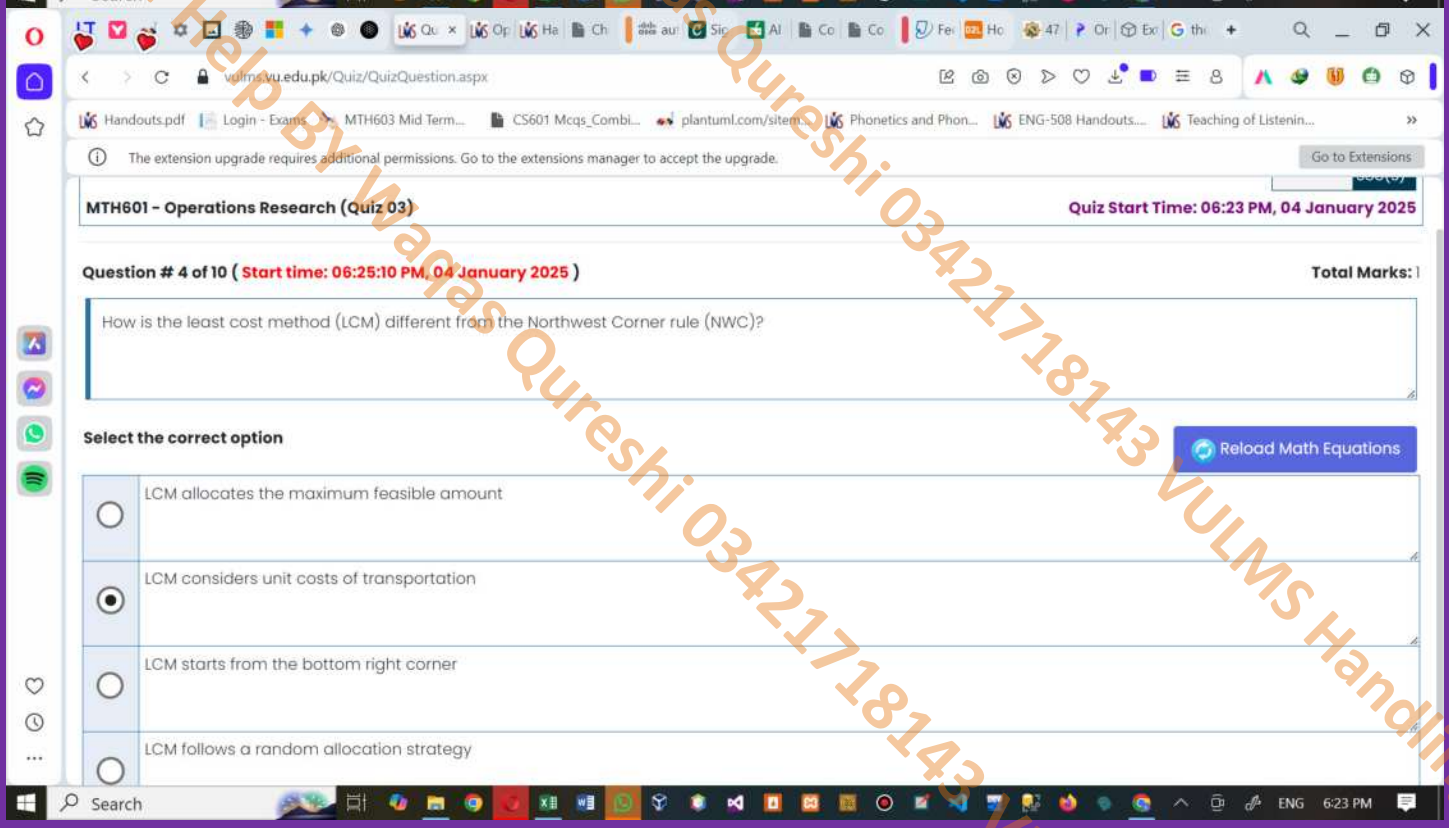
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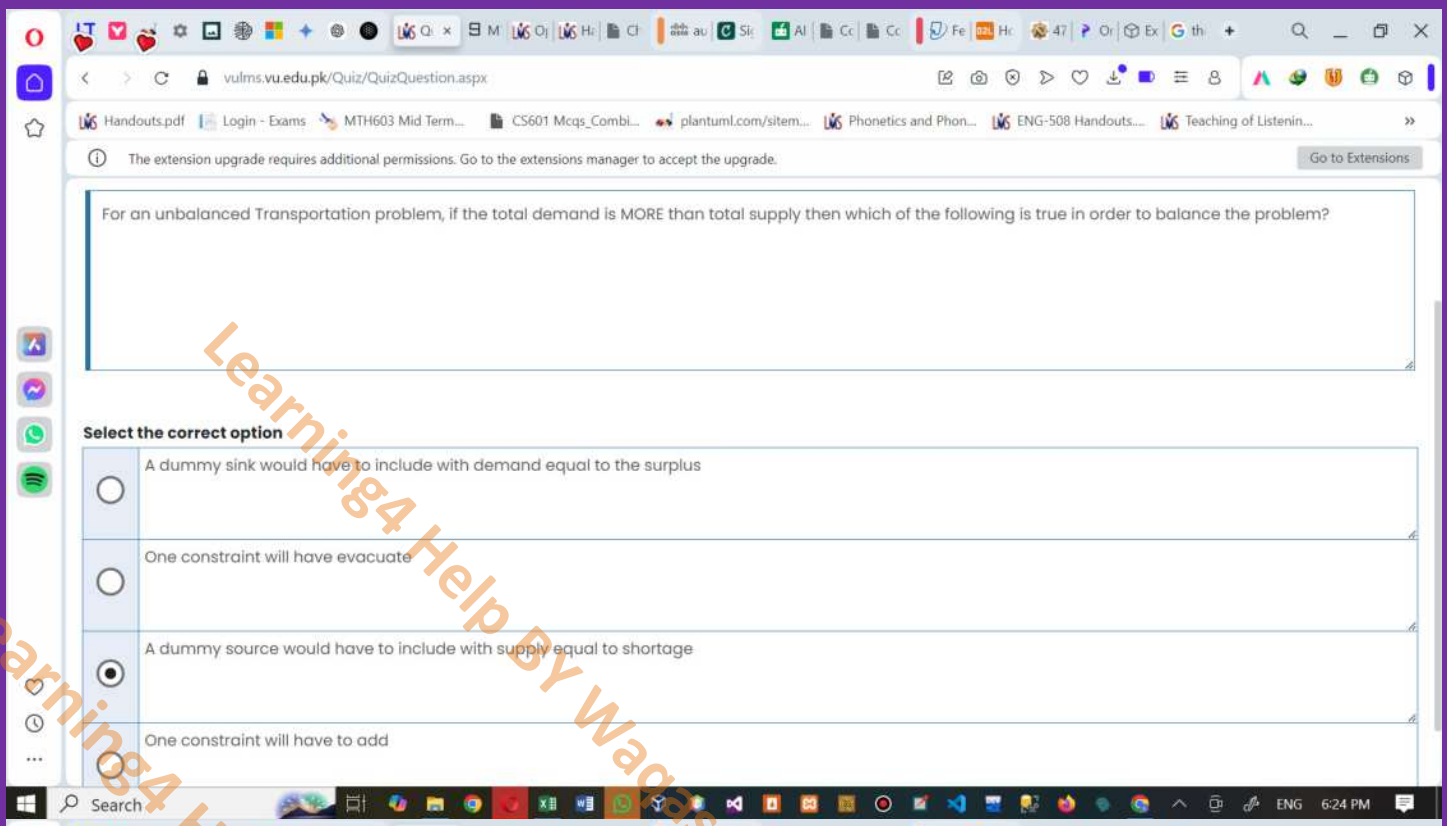


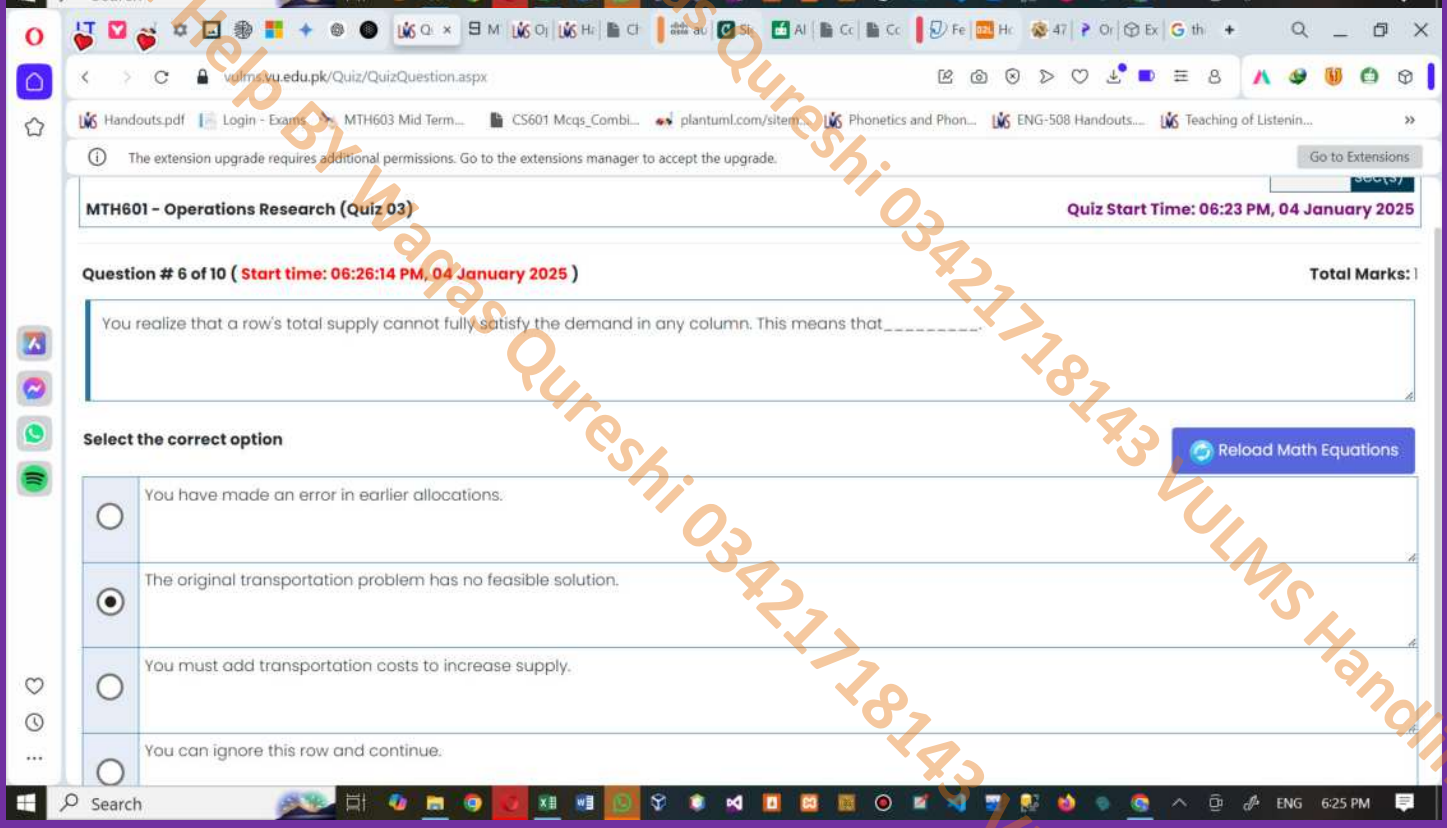
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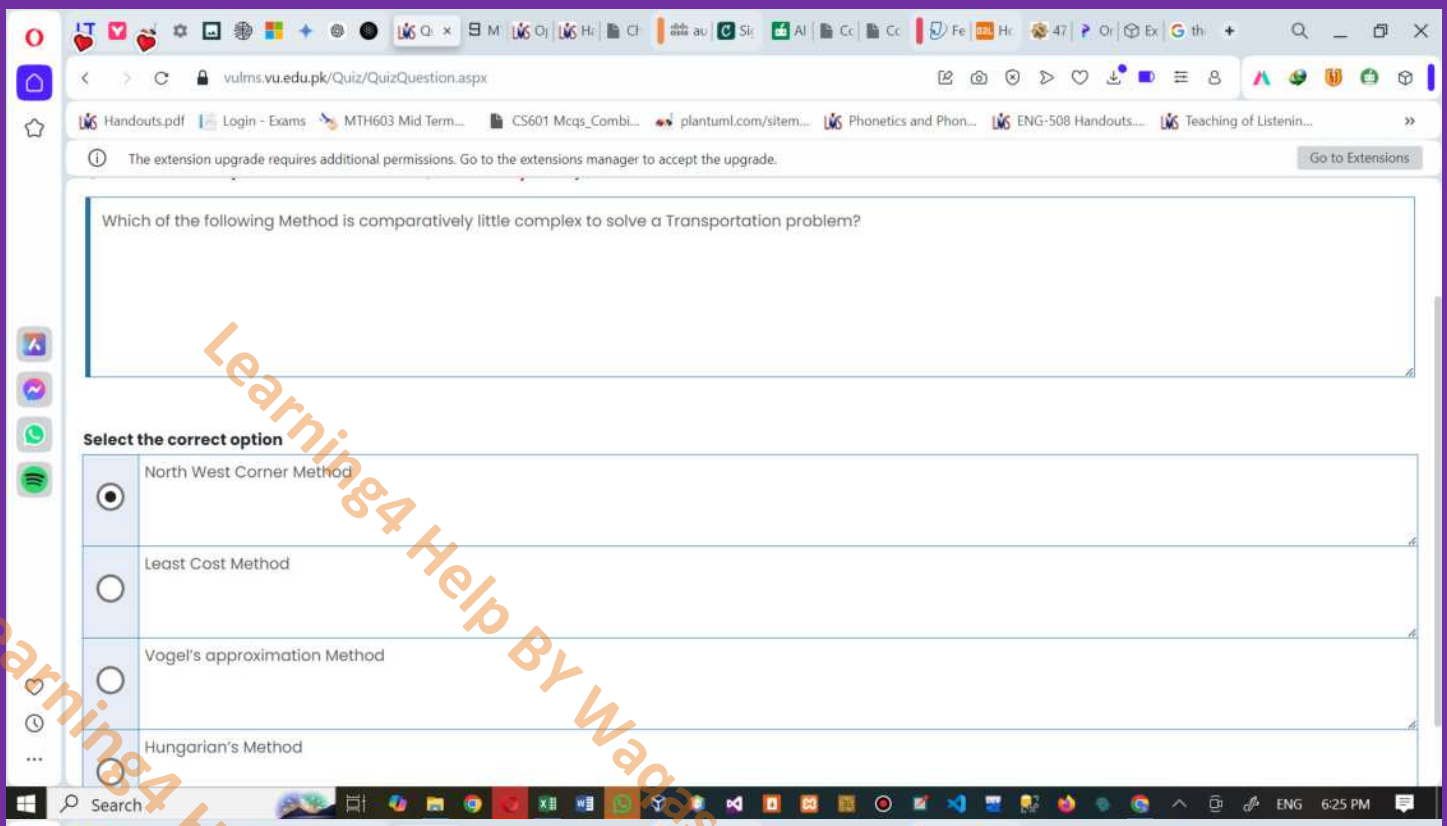


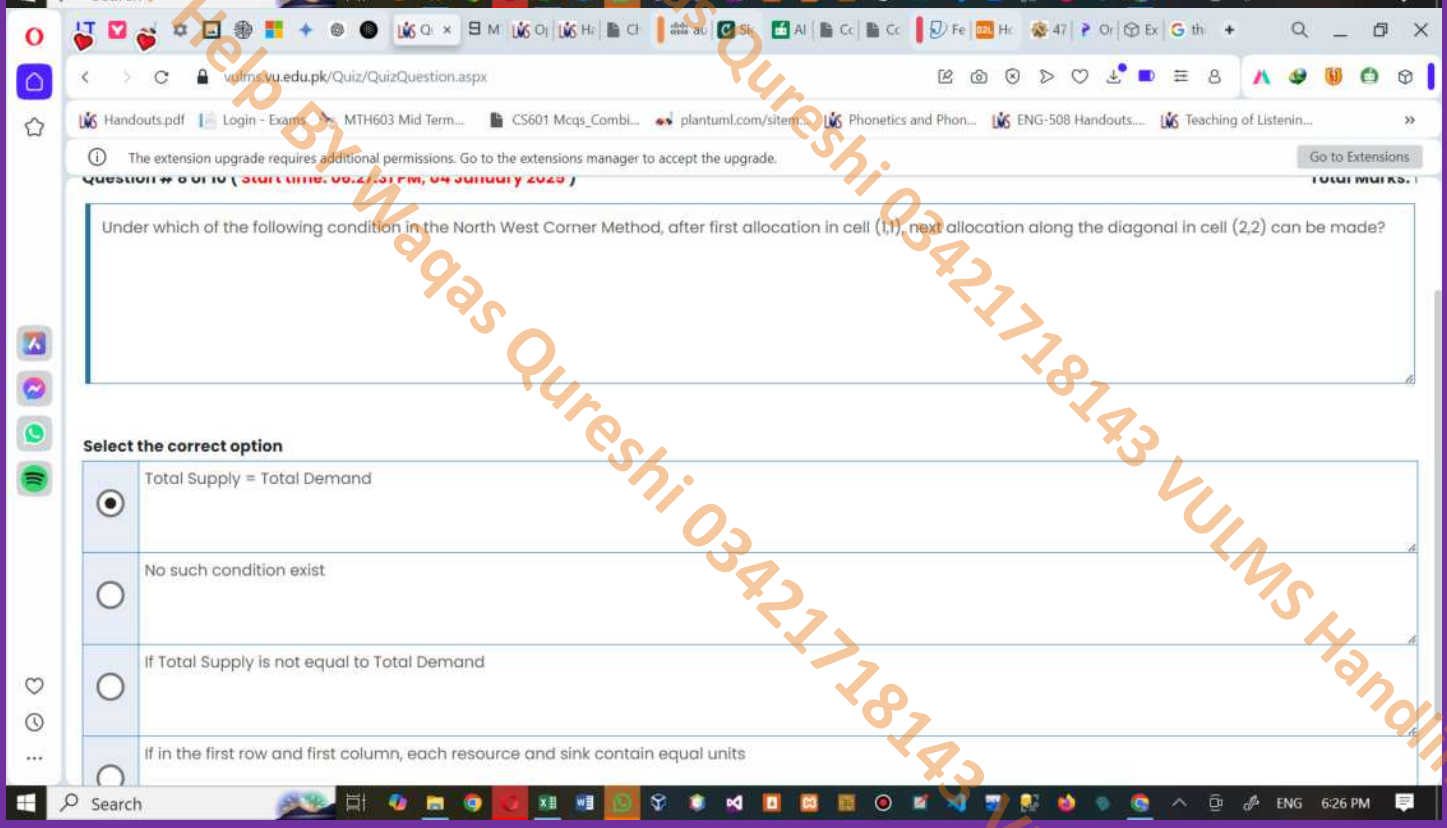
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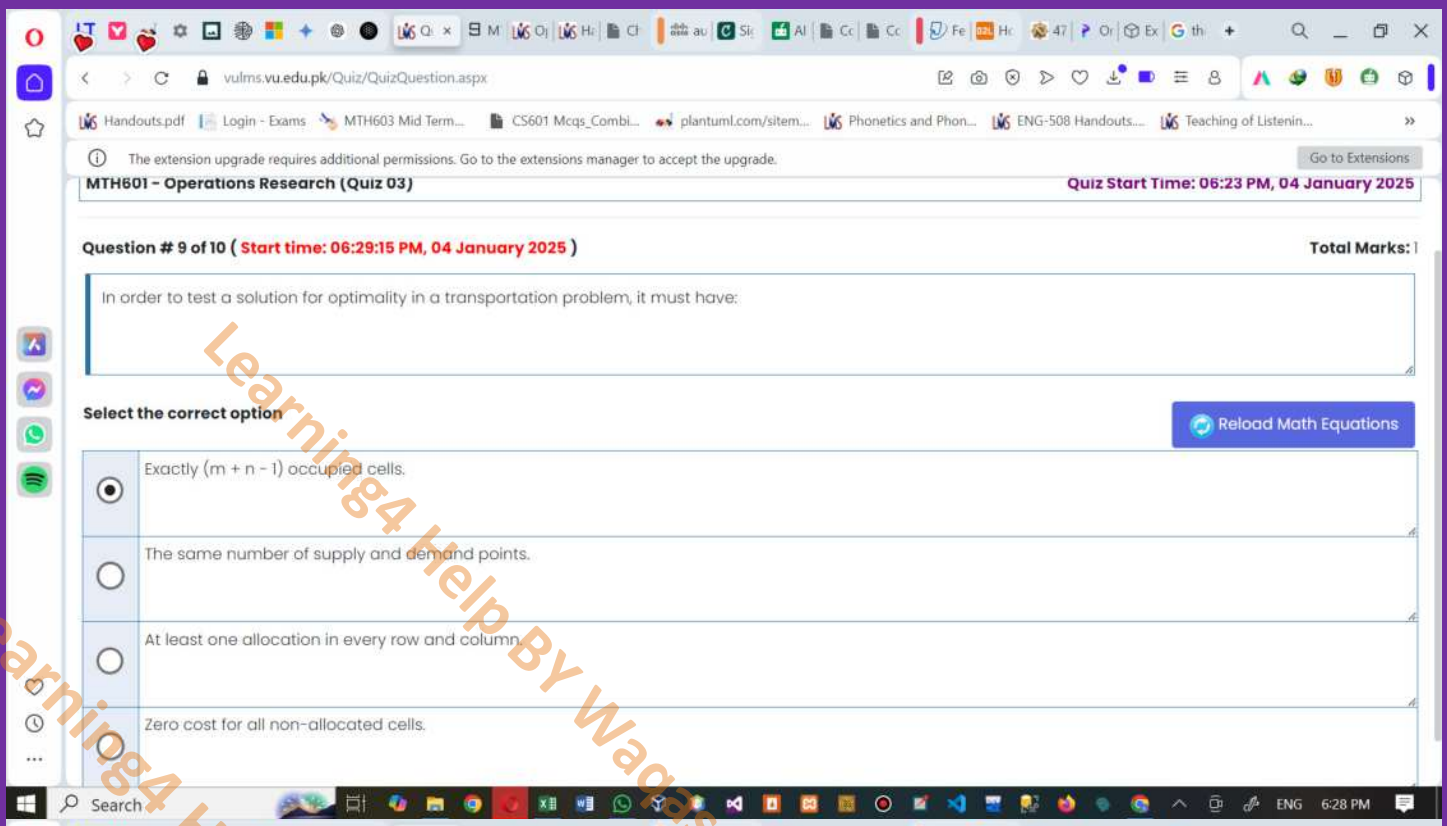


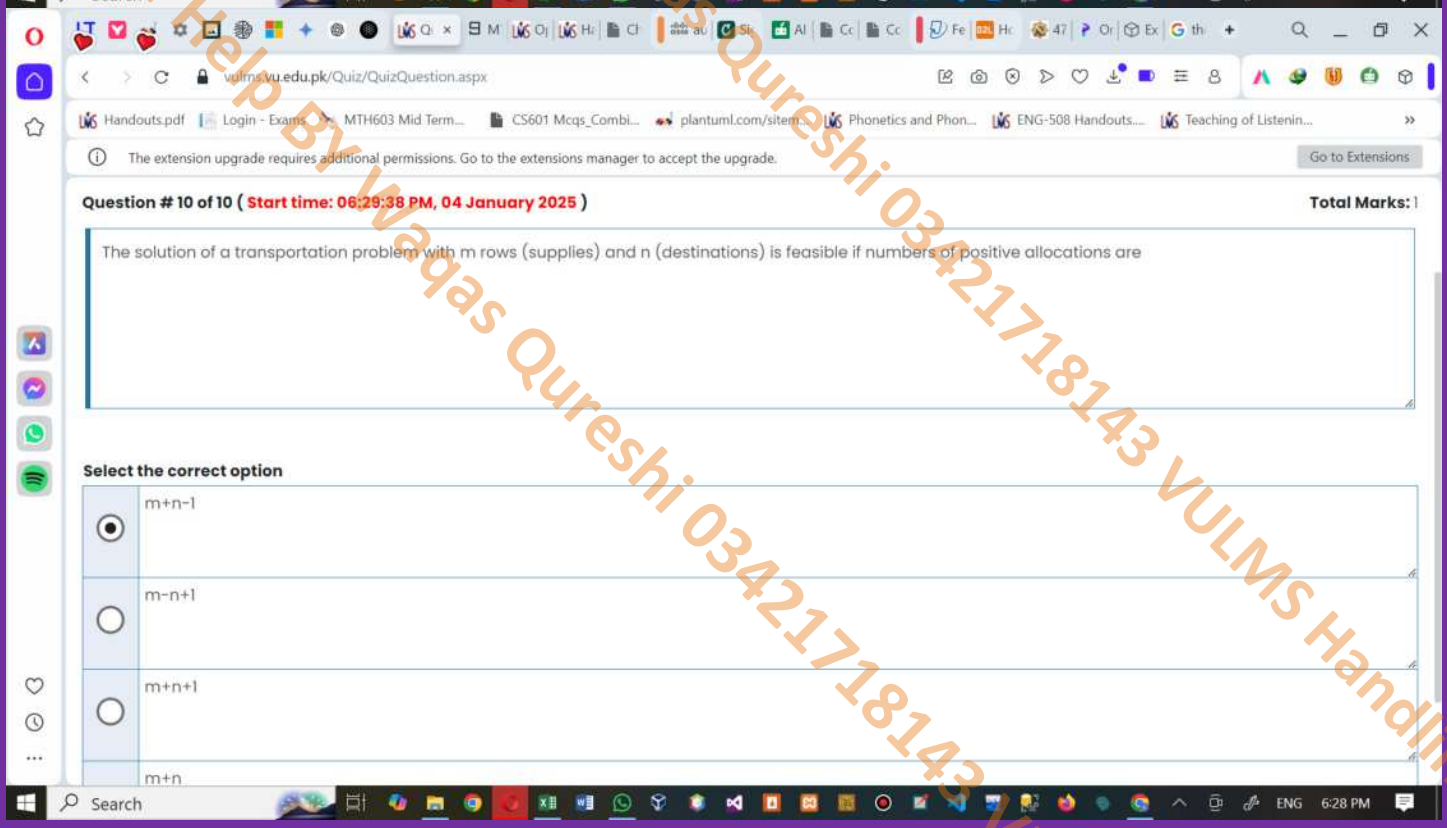
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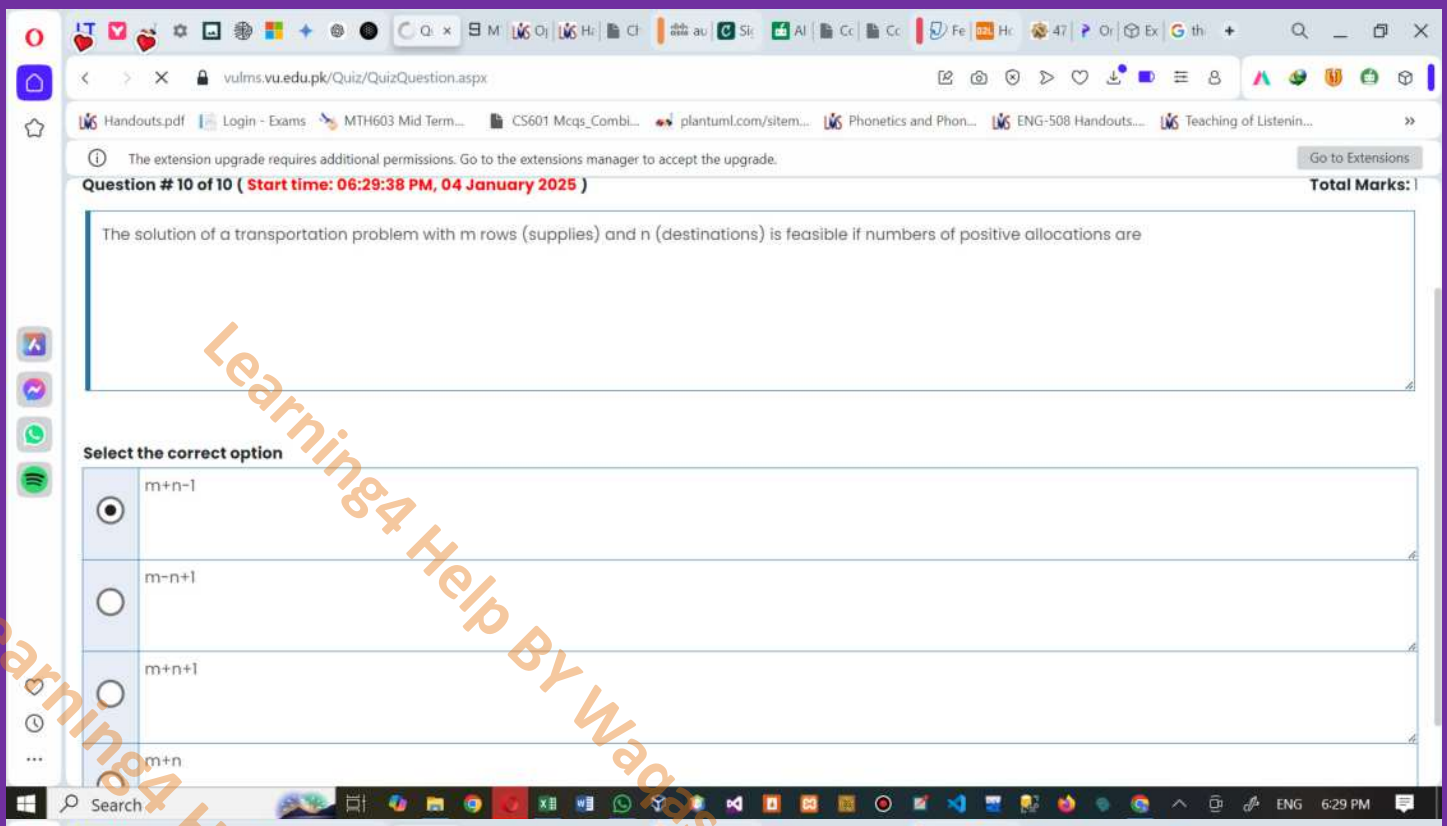


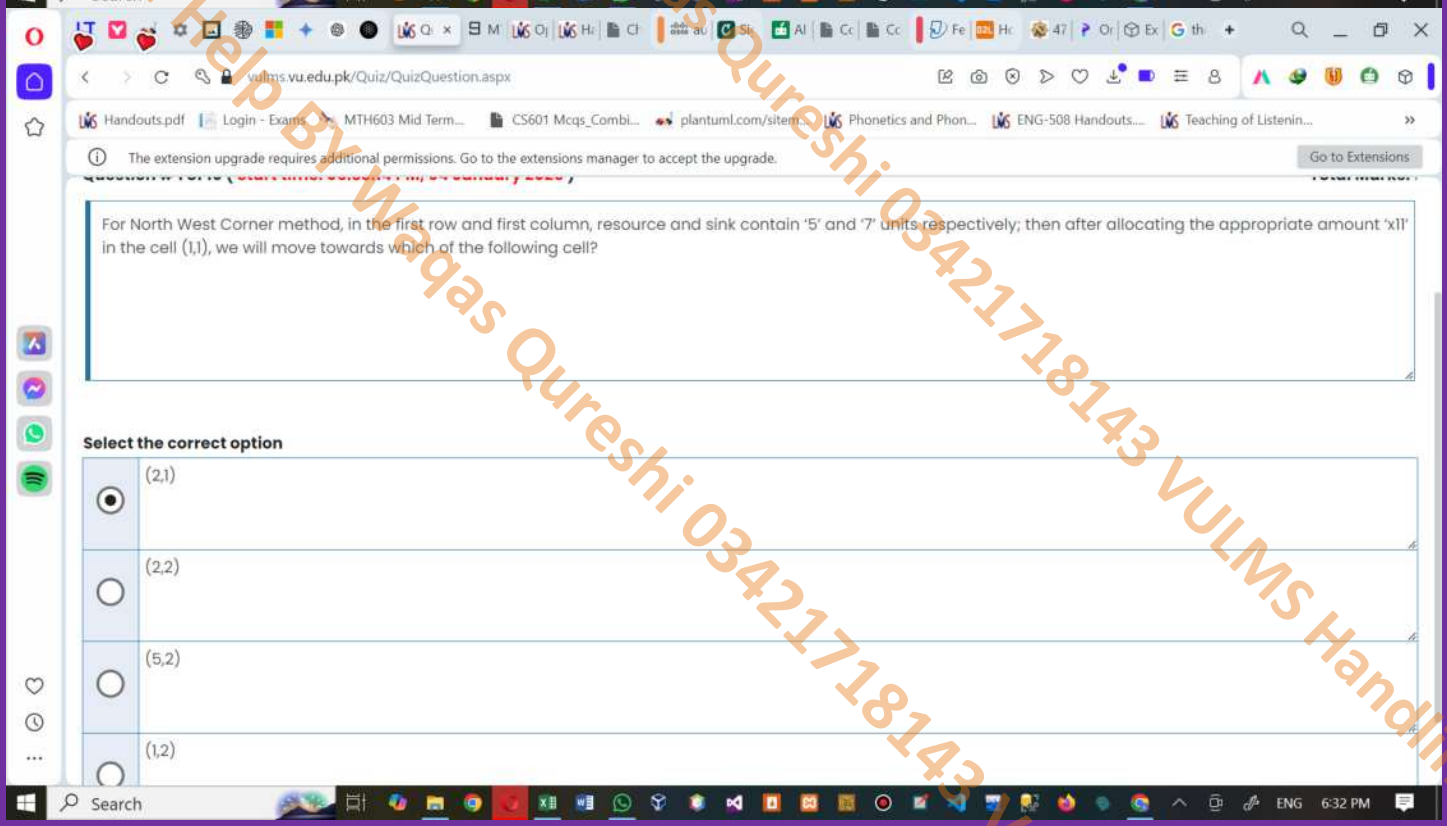
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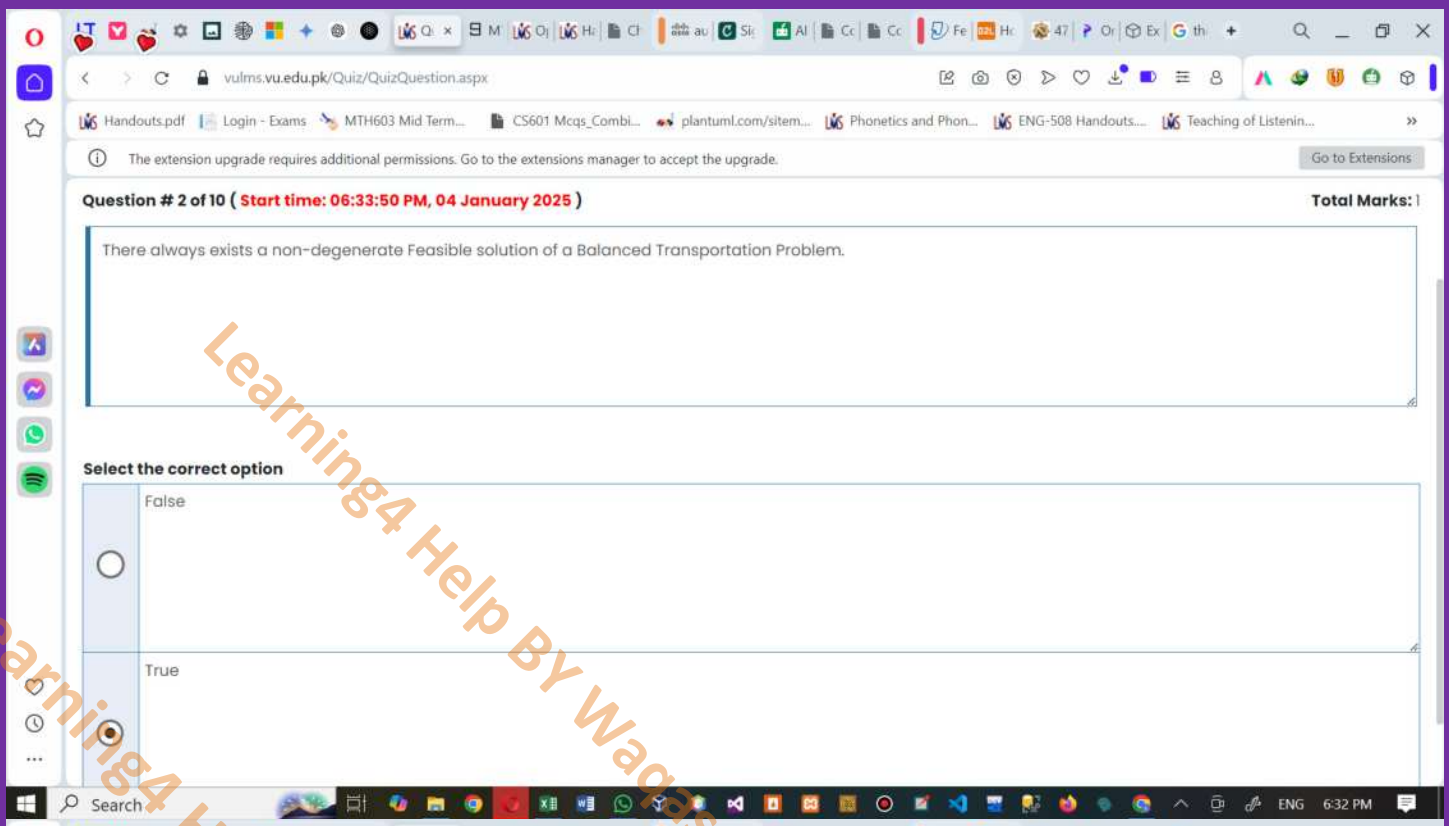


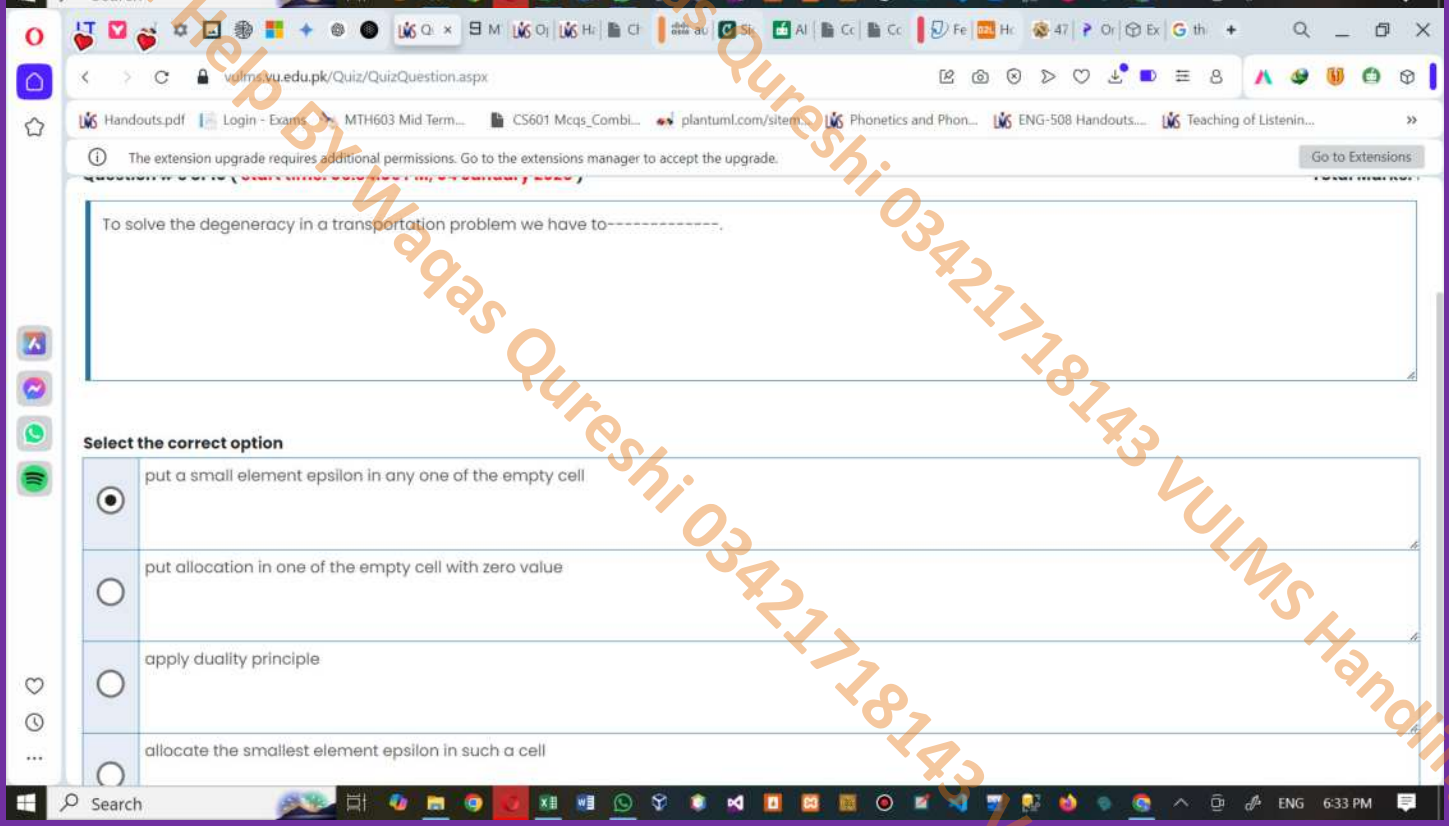
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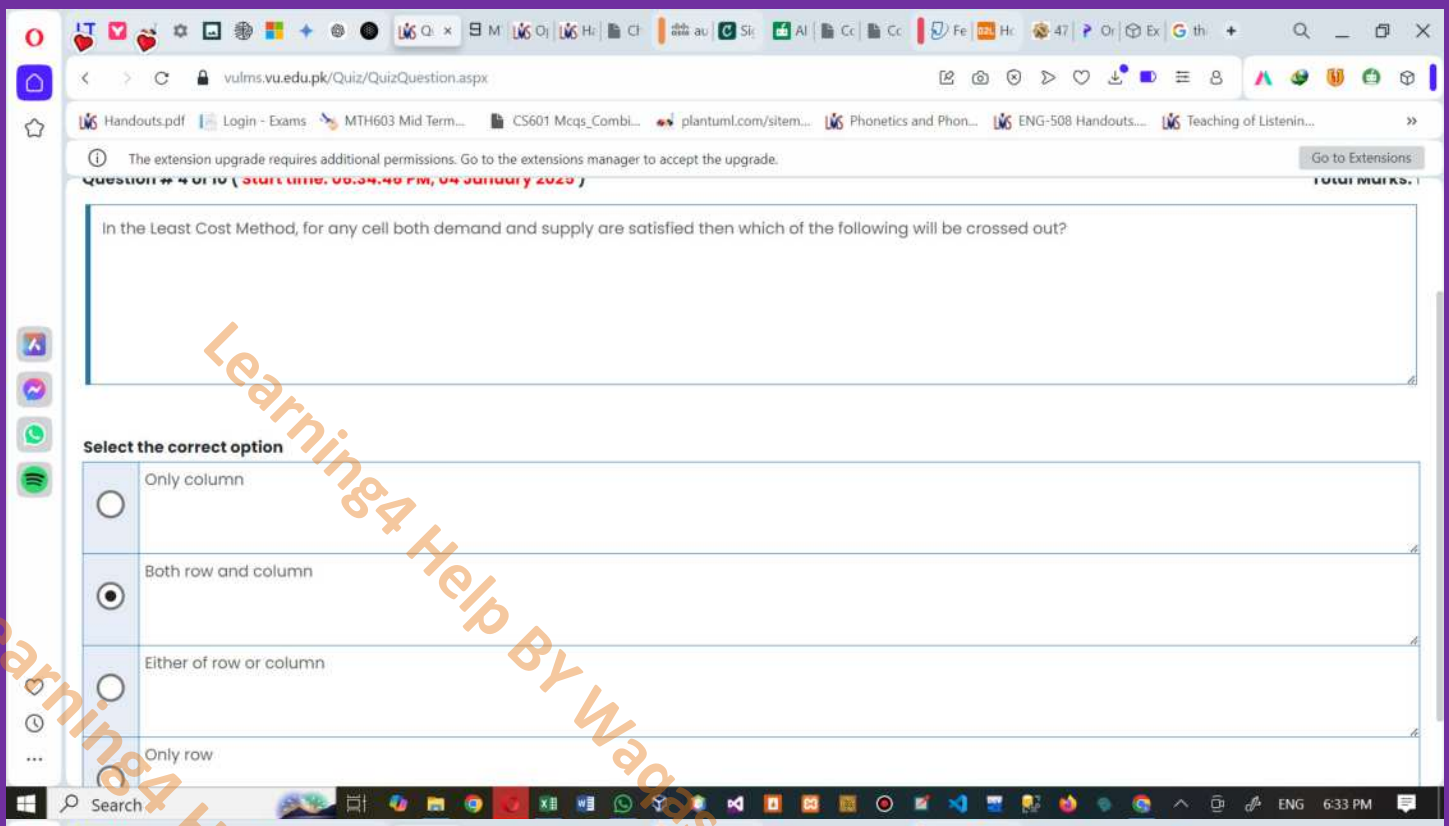


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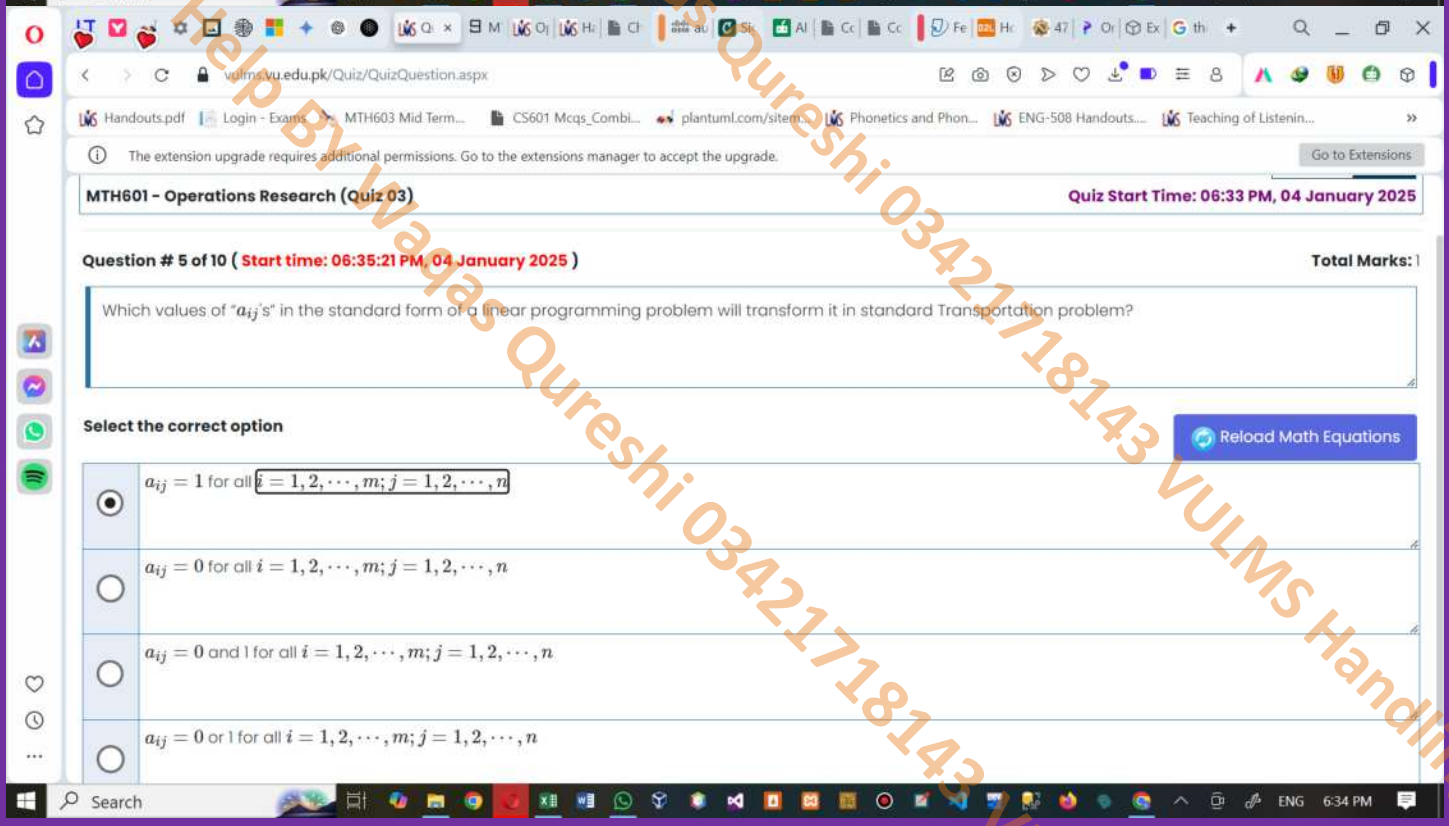
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In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

Select the correct option

- Only column
- Both row and column
- Either of row or column
- Only row



MTH601 - Operations Research (Quiz 03) — **Quiz Start Time: 06:33 PM, 04 January 2025**

Question # 5 of 10 (Start time: 06:35:21 PM, 04 January 2025) — **Total Marks:** 1

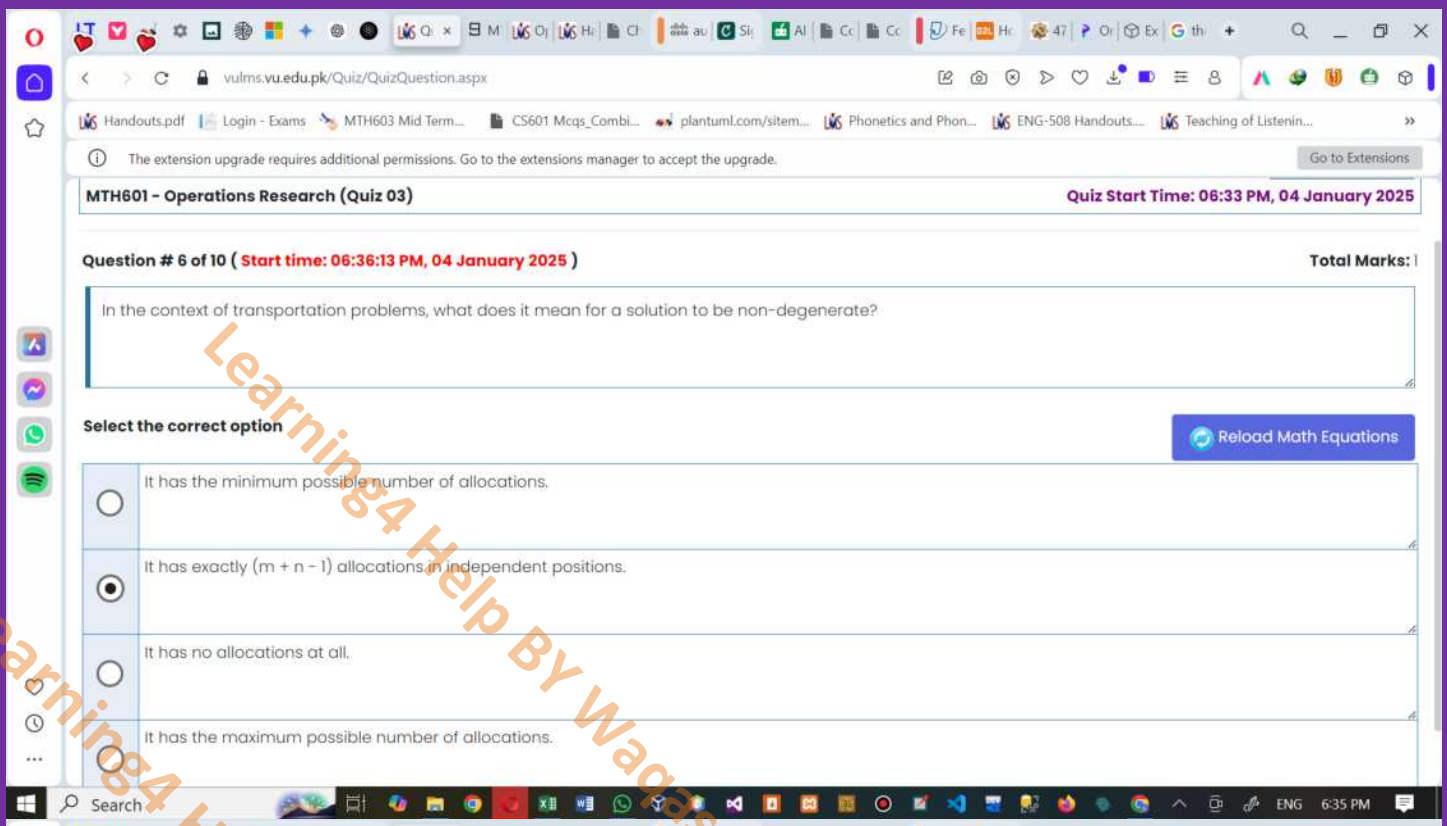
Which values of "a_{ij}'s" in the standard form of a linear programming problem will transform it in standard Transportation problem?

Select the correct option

Reload Math Equations

- $a_{ij} = 1$ for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- $a_{ij} = 0$ for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- $a_{ij} = 0$ and 1 for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$
- $a_{ij} = 0$ or 1 for all $i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$

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MTH601 - Operations Research (Quiz 03)

Quiz Start Time: 06:33 PM, 04 January 2025

Question # 6 of 10 (Start time: 06:36:13 PM, 04 January 2025)

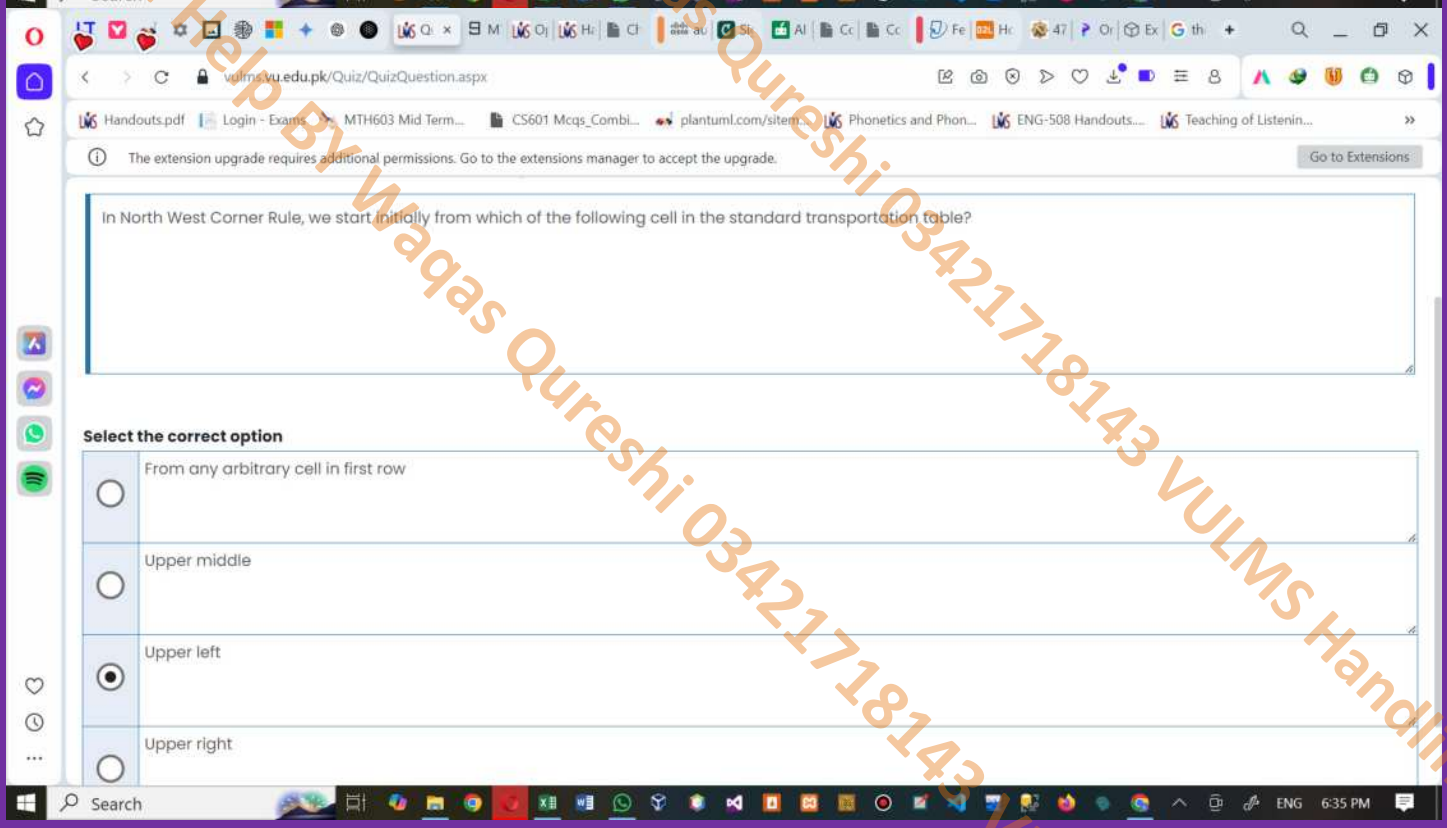
Total Marks:

In the context of transportation problems, what does it mean for a solution to be non-degenerate?

Select the correct option

Reload Math Equations

- It has the minimum possible number of allocations.
- It has exactly (m + n - 1) allocations in independent positions.
- It has no allocations at all.
- It has the maximum possible number of allocations.

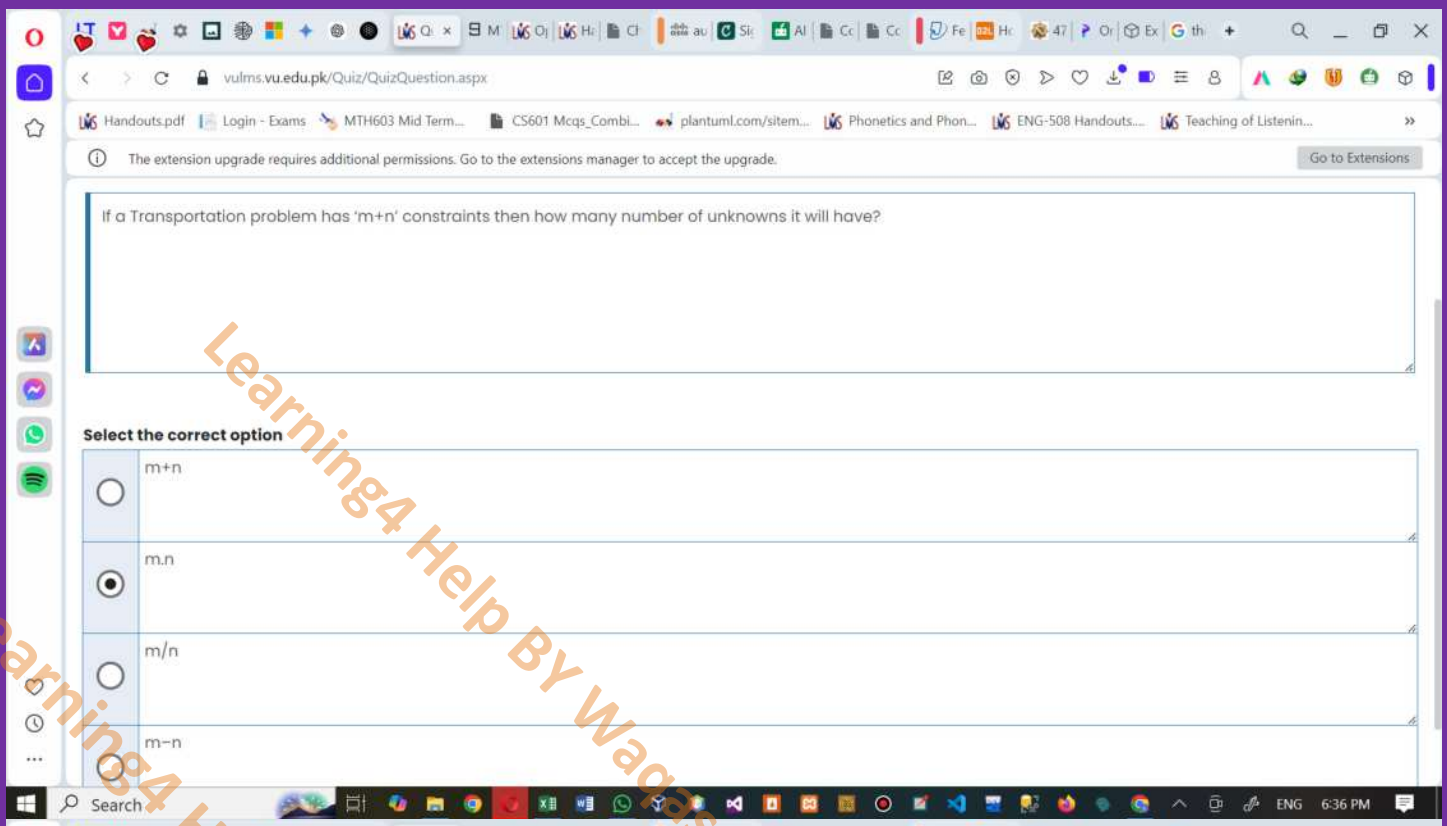


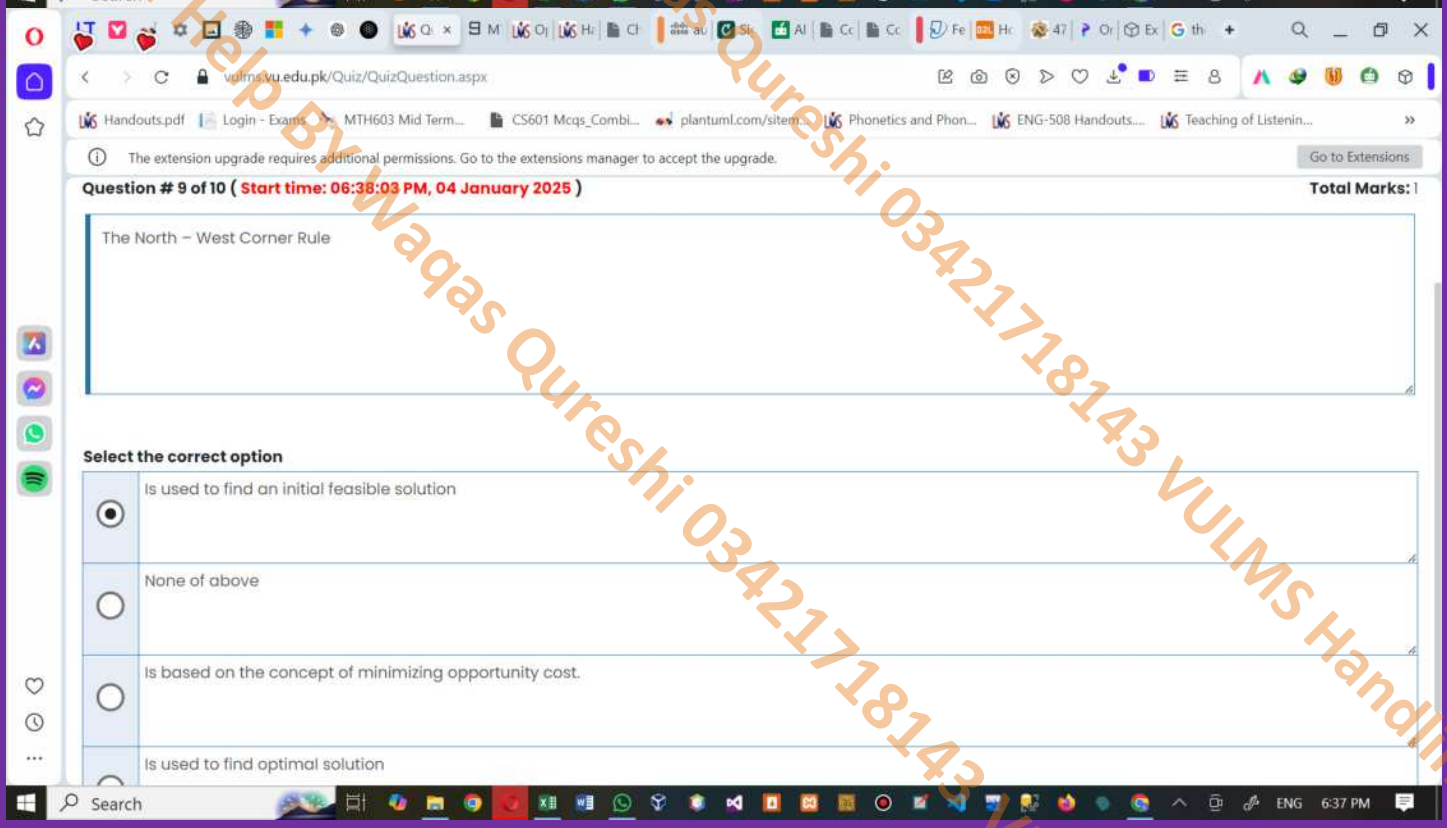
In North West Corner Rule, we start initially from which of the following cell in the standard transportation table?

Select the correct option

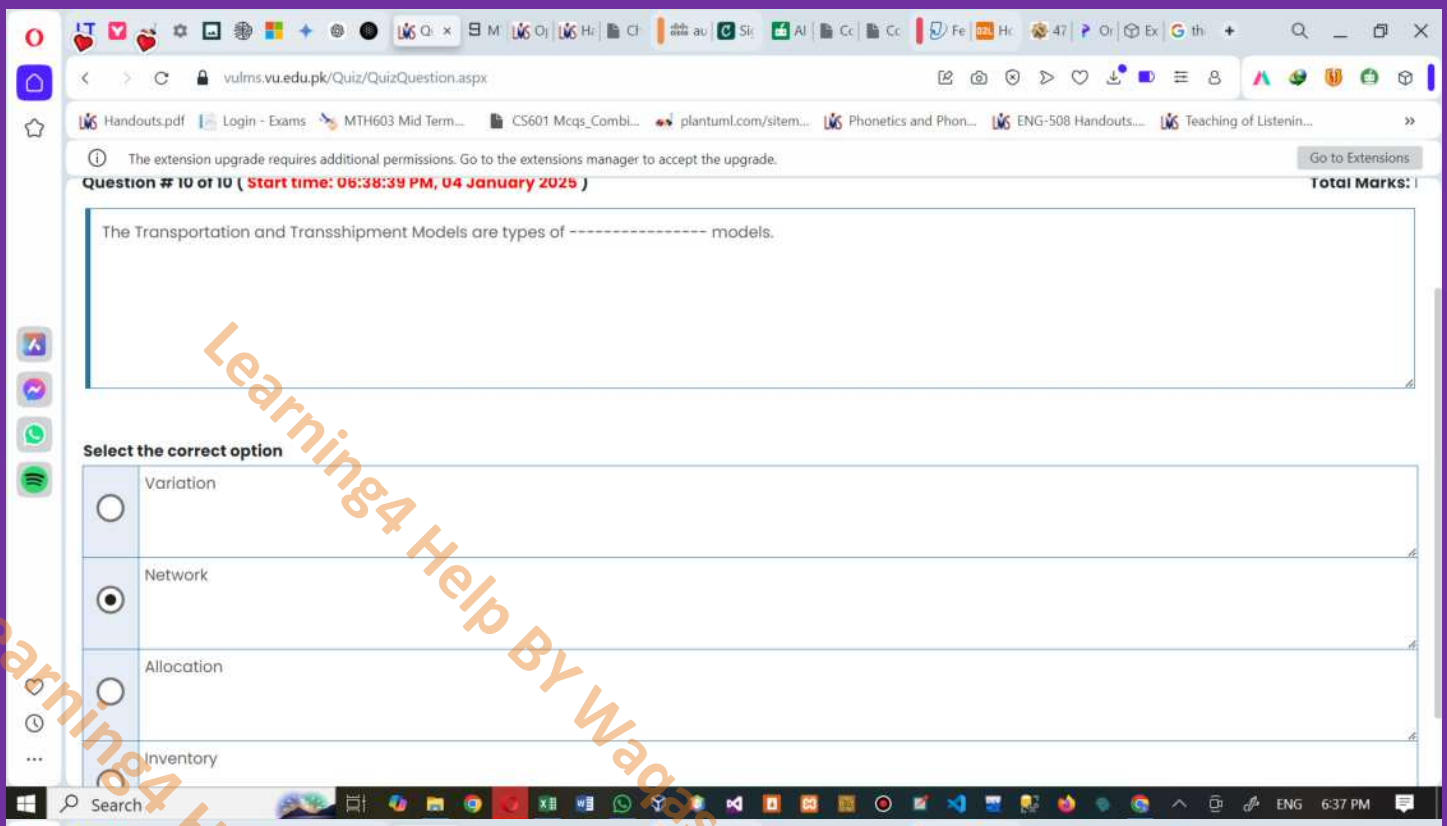
- From any arbitrary cell in first row
- Upper middle
- Upper left
- Upper right

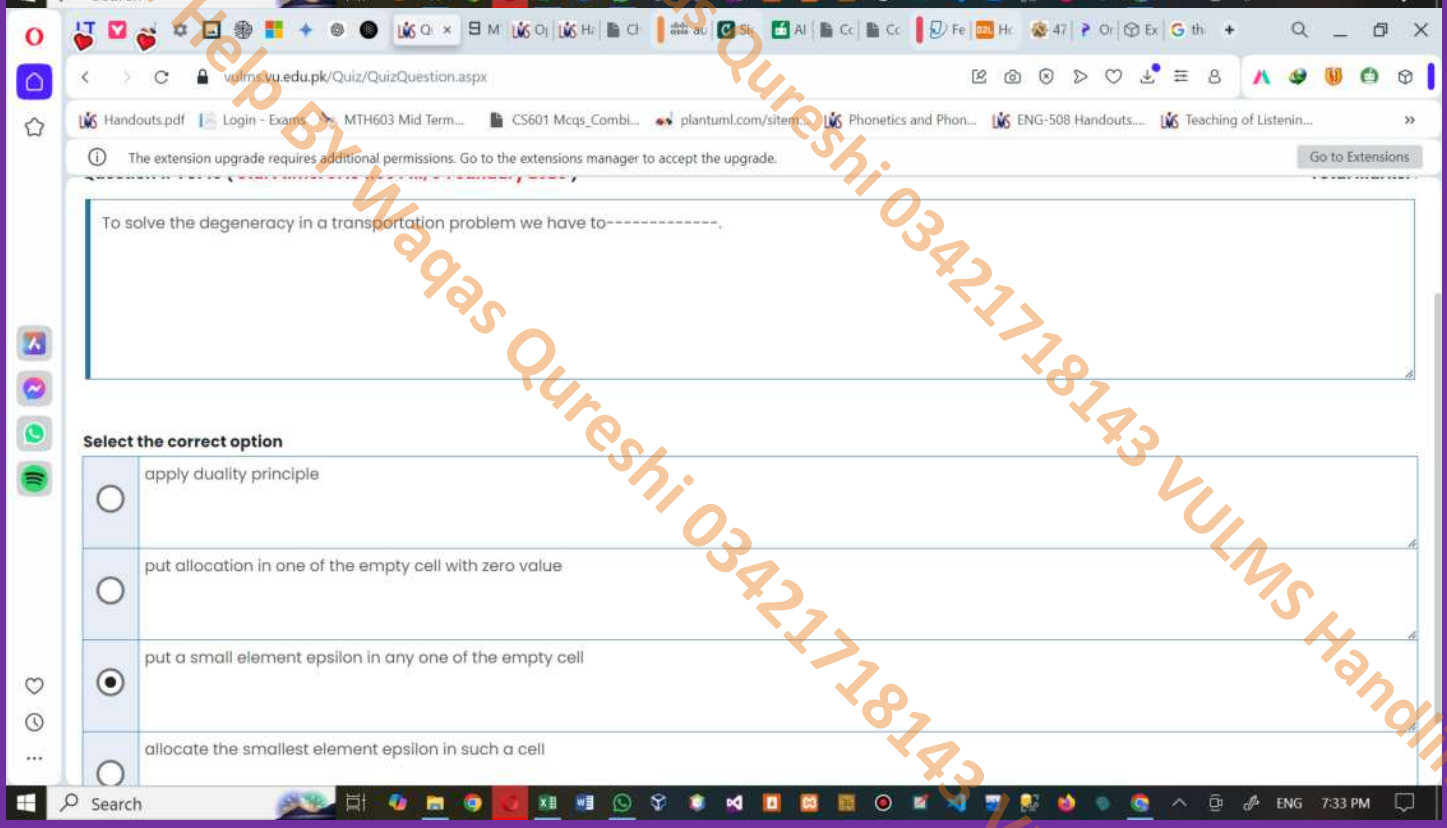
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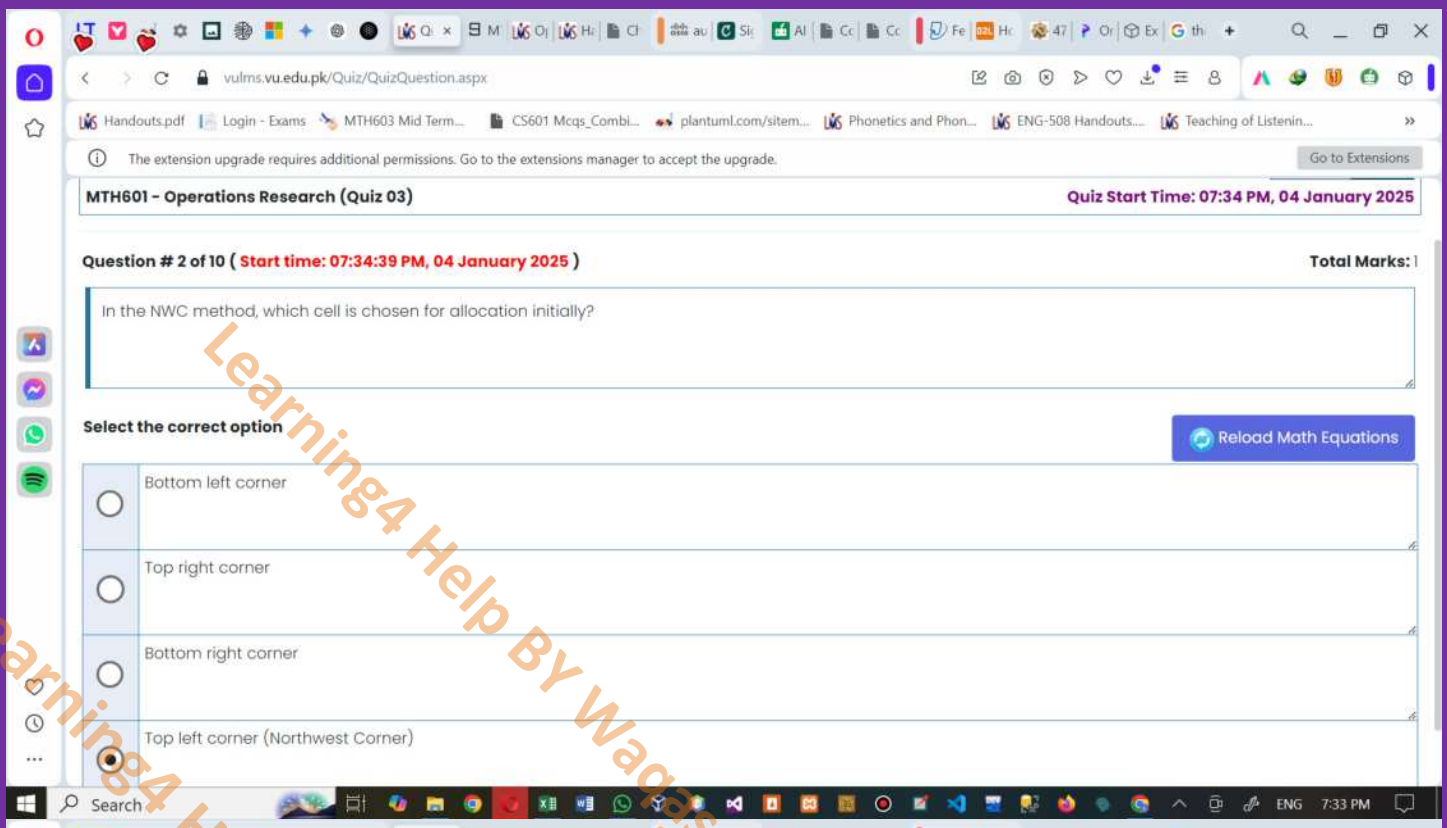


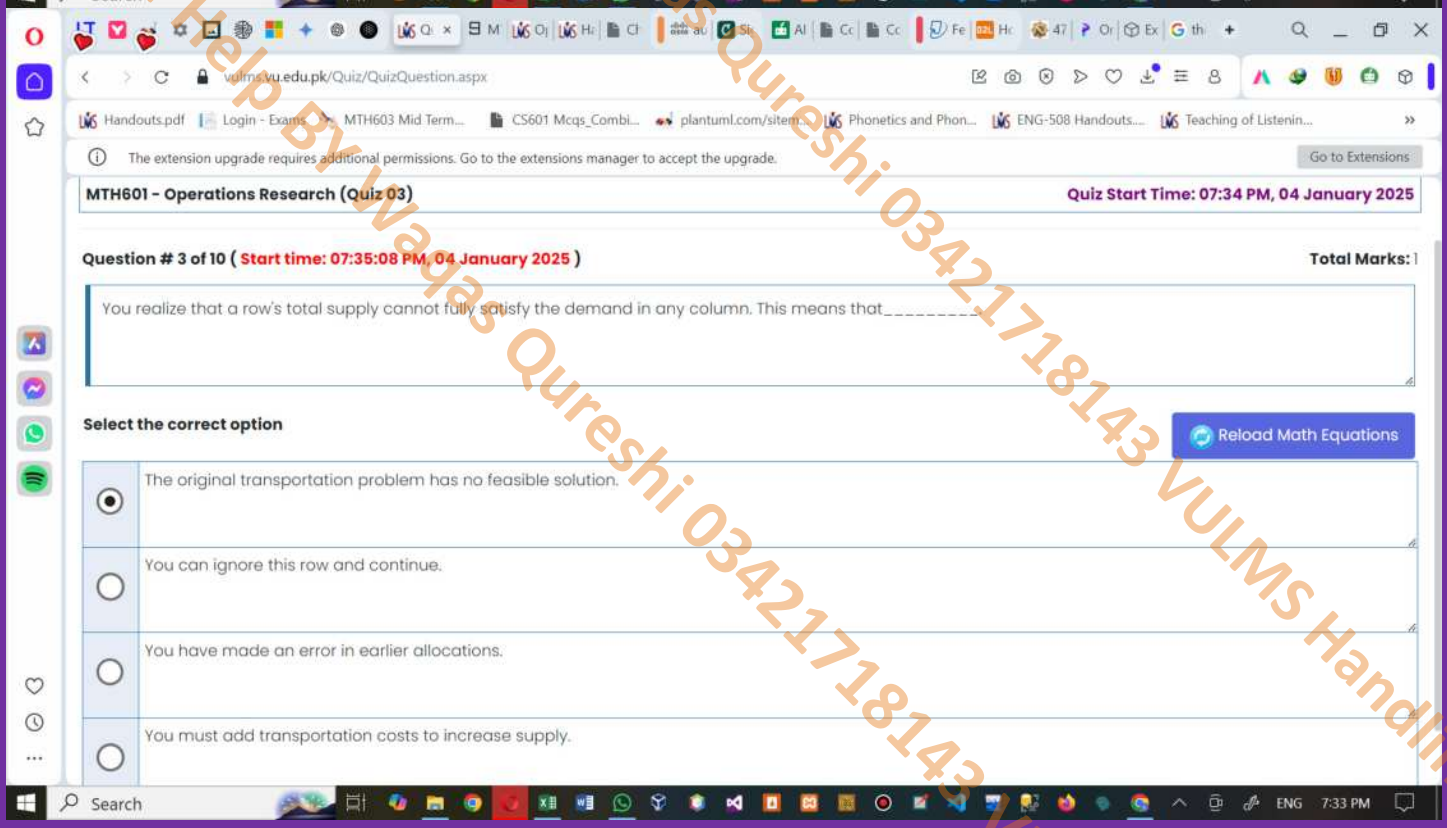
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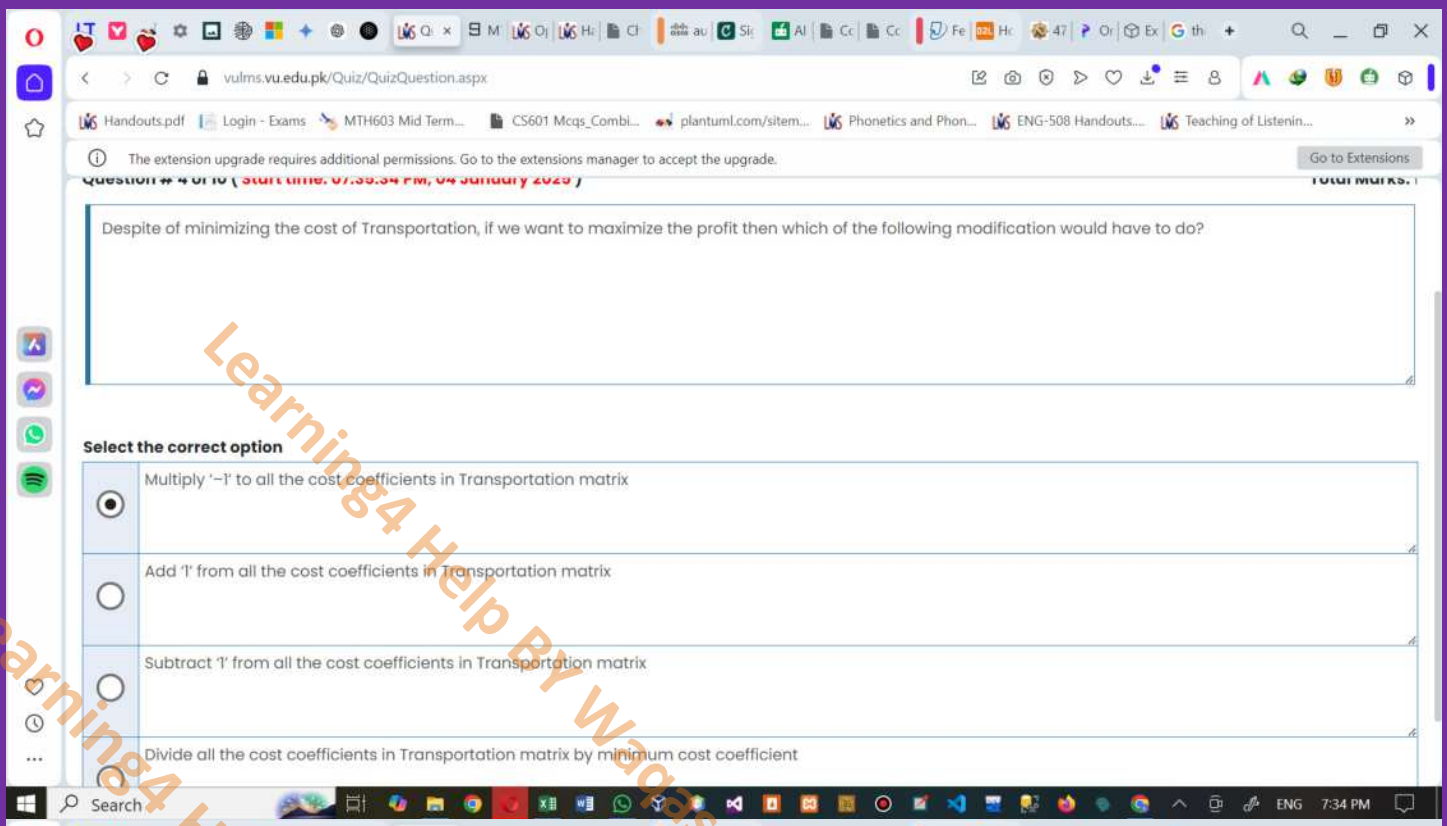


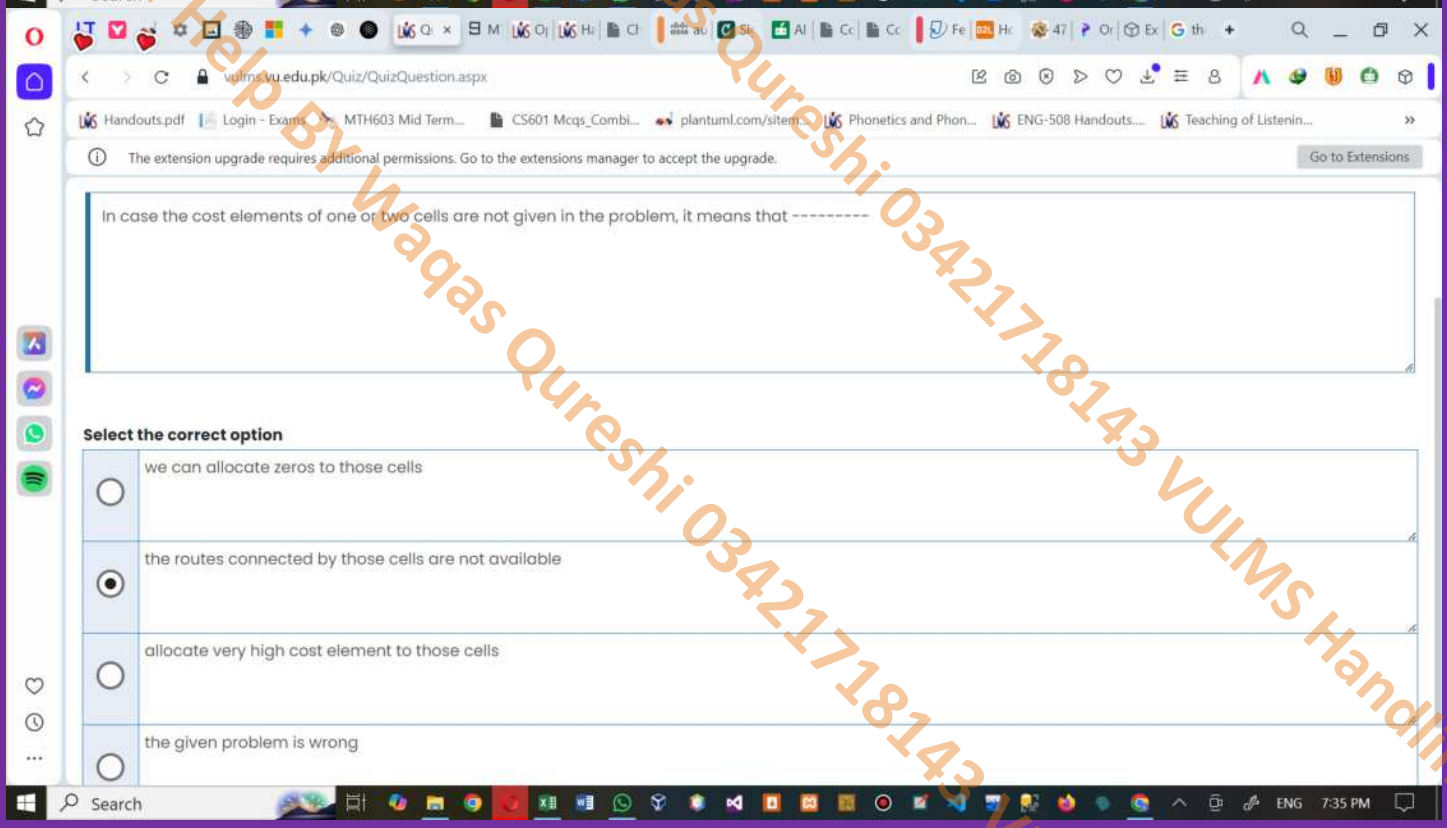
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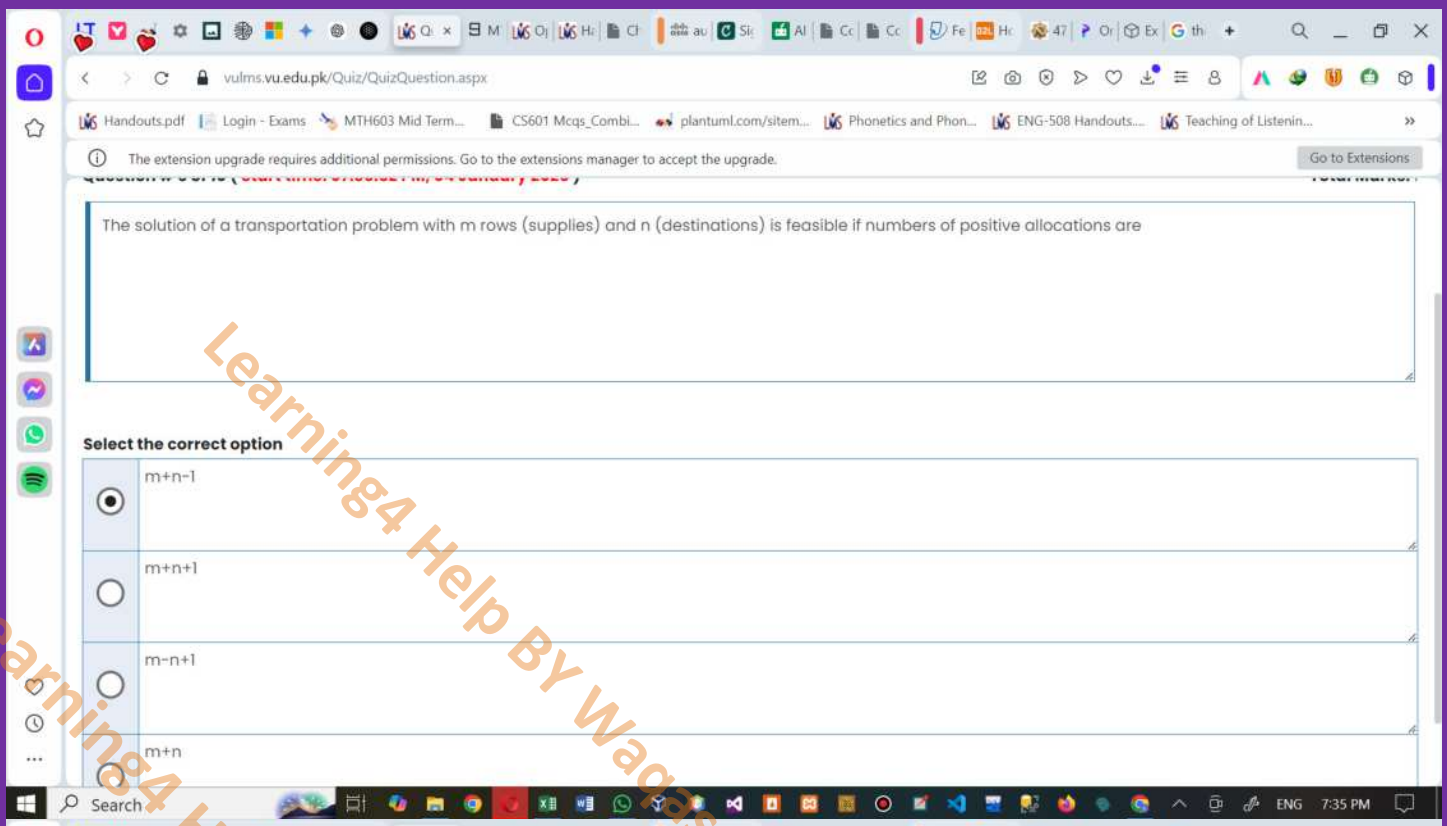


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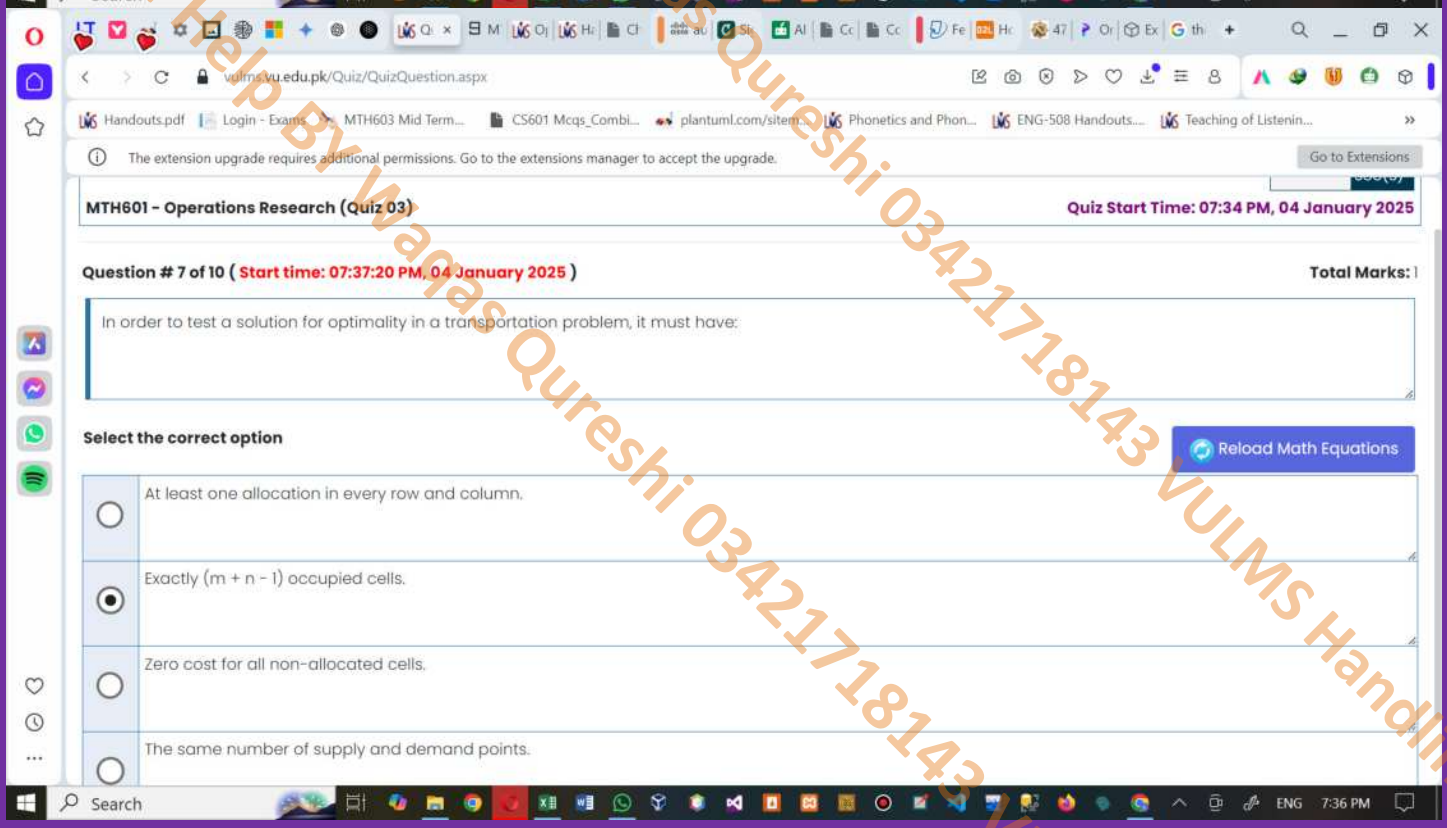
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The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

Select the correct option

- m+n-1
- m+n+1
- m-n+1
- m+n



MTH601 - Operations Research (Quiz 03) **Quiz Start Time: 07:34 PM, 04 January 2025**

Question # 7 of 10 (Start time: 07:37:20 PM, 04 January 2025) **Total Marks:**

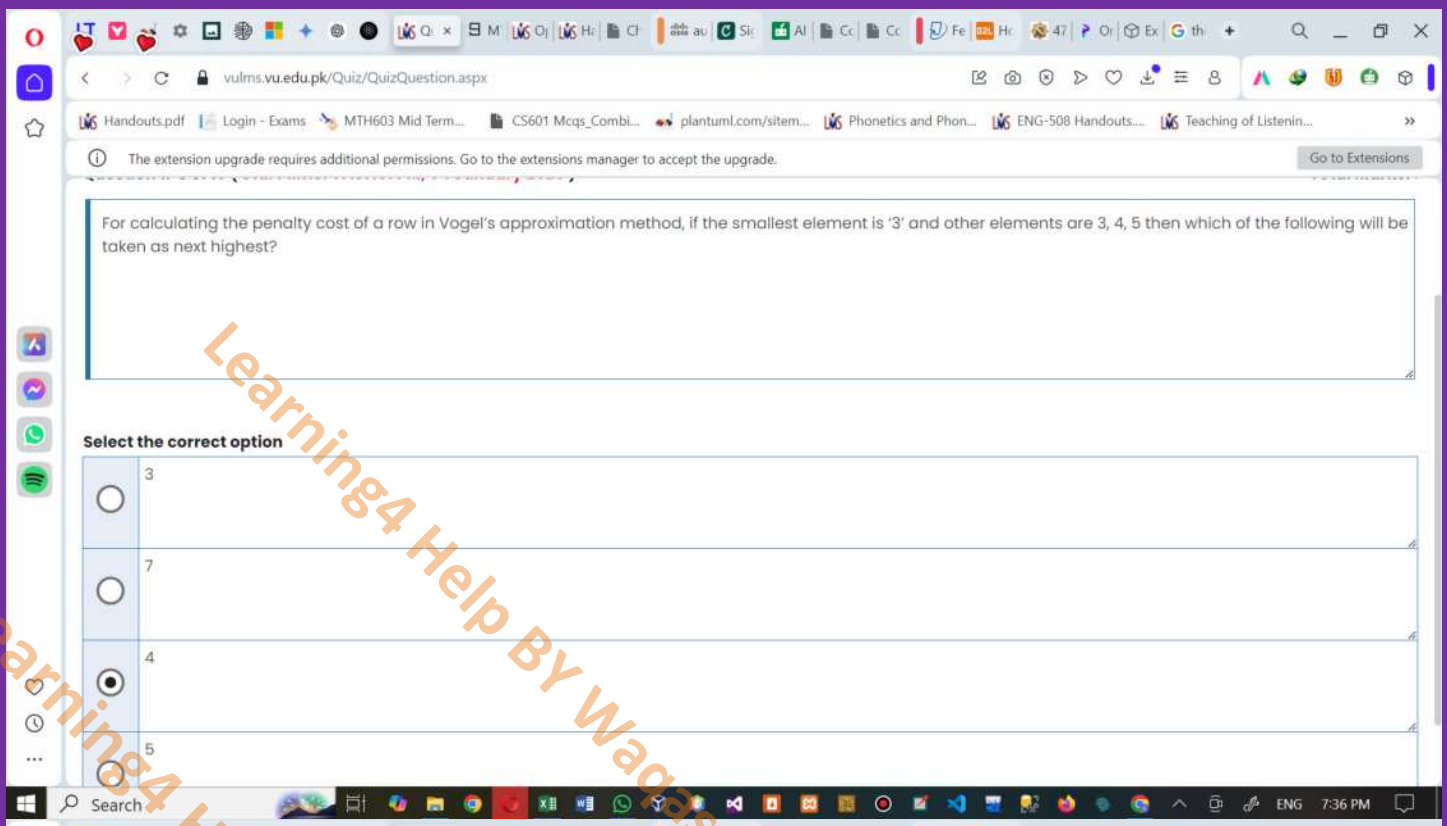
In order to test a solution for optimality in a transportation problem, it must have:

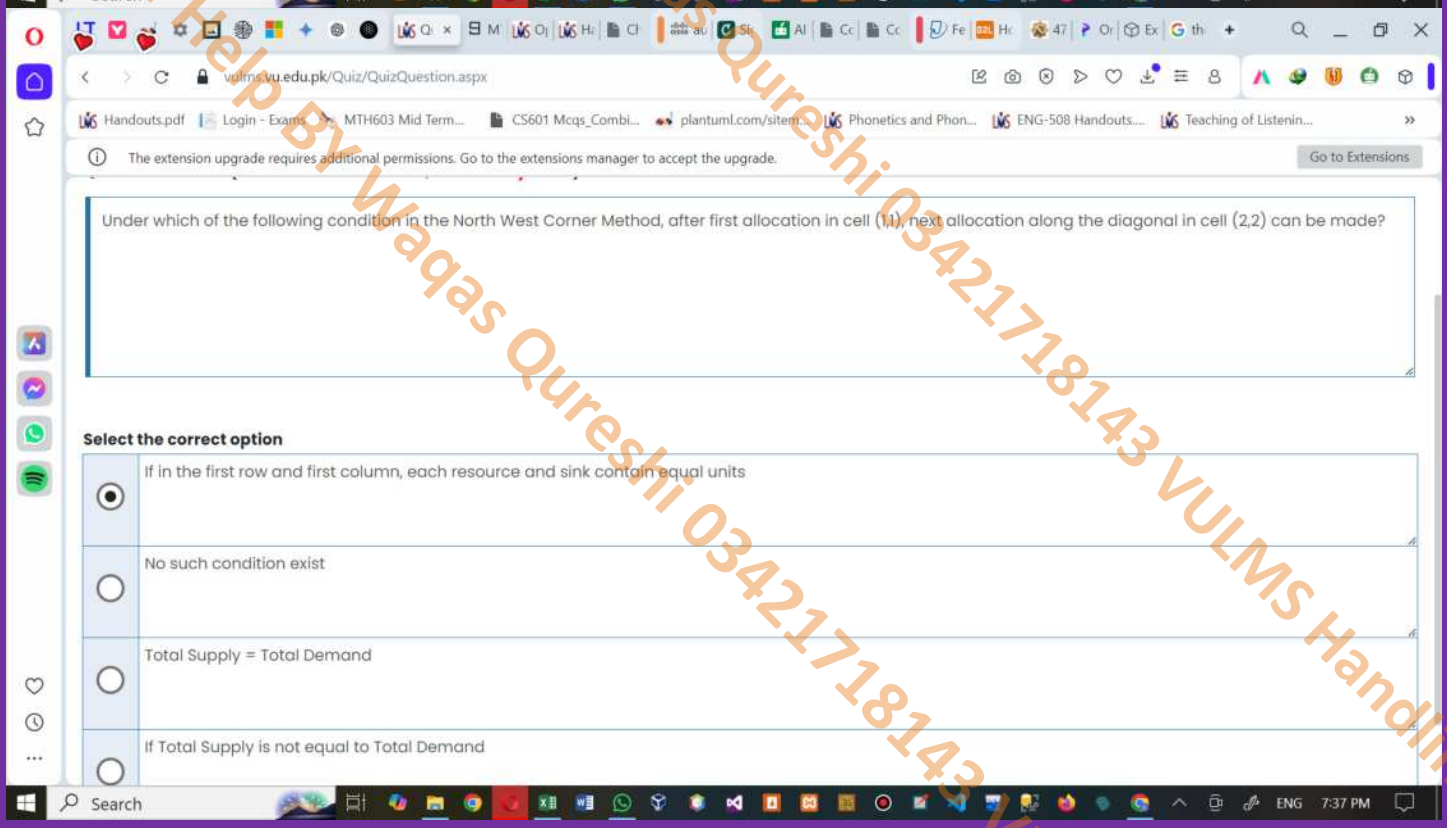
Select the correct option

Reload Math Equations

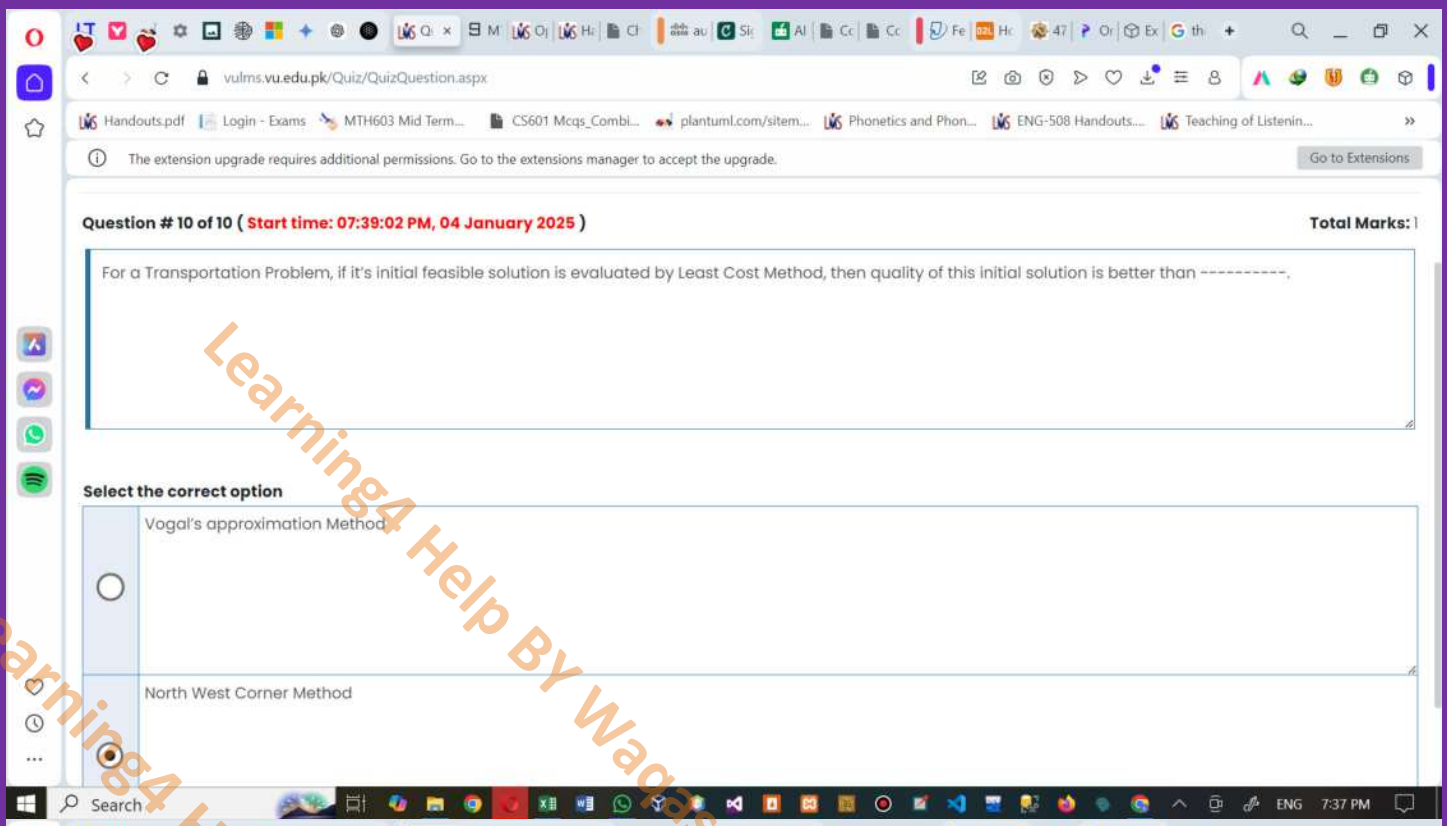
- At least one allocation in every row and column.
- Exactly (m + n - 1) occupied cells.
- Zero cost for all non-allocated cells.
- The same number of supply and demand points.

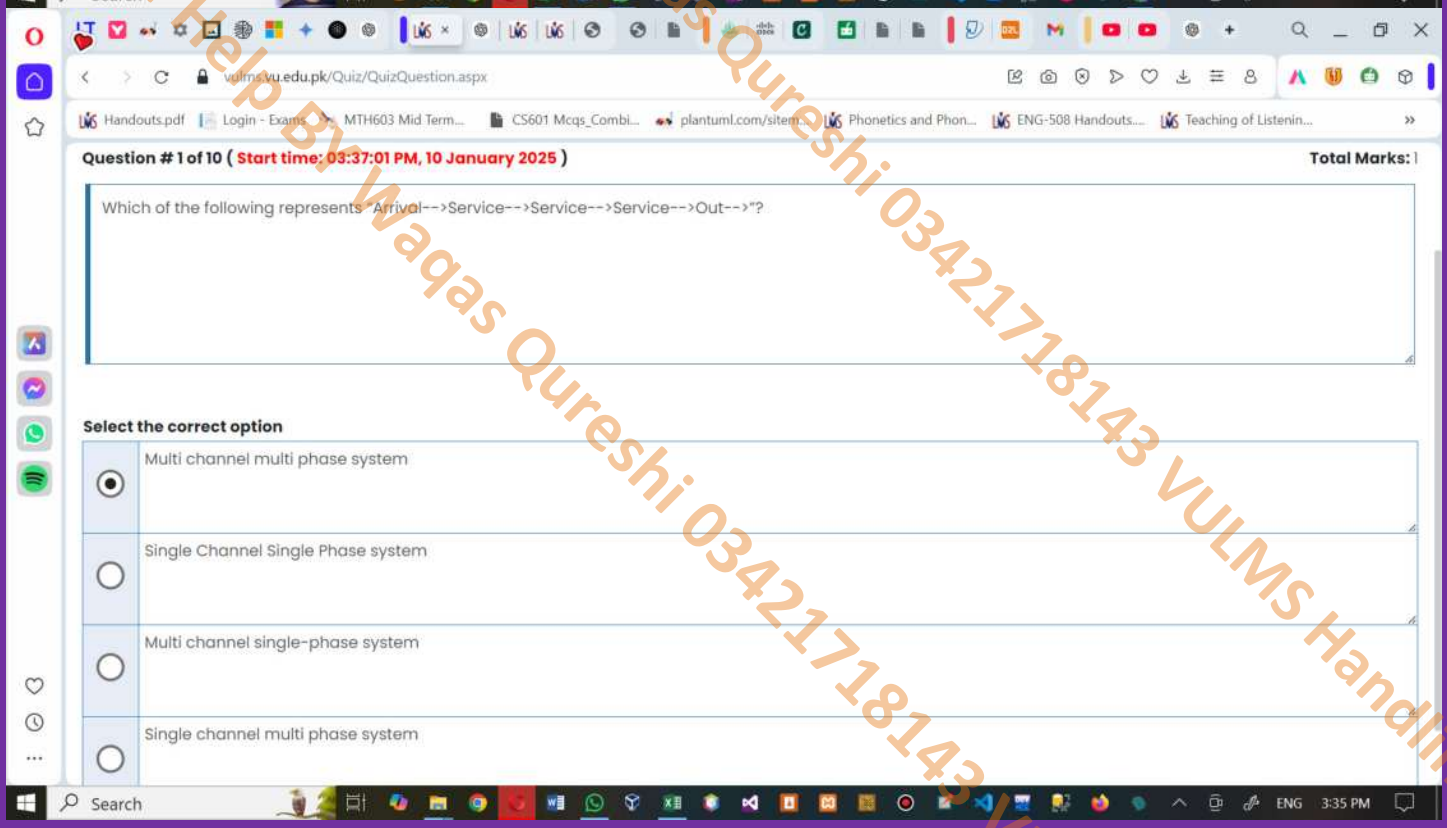
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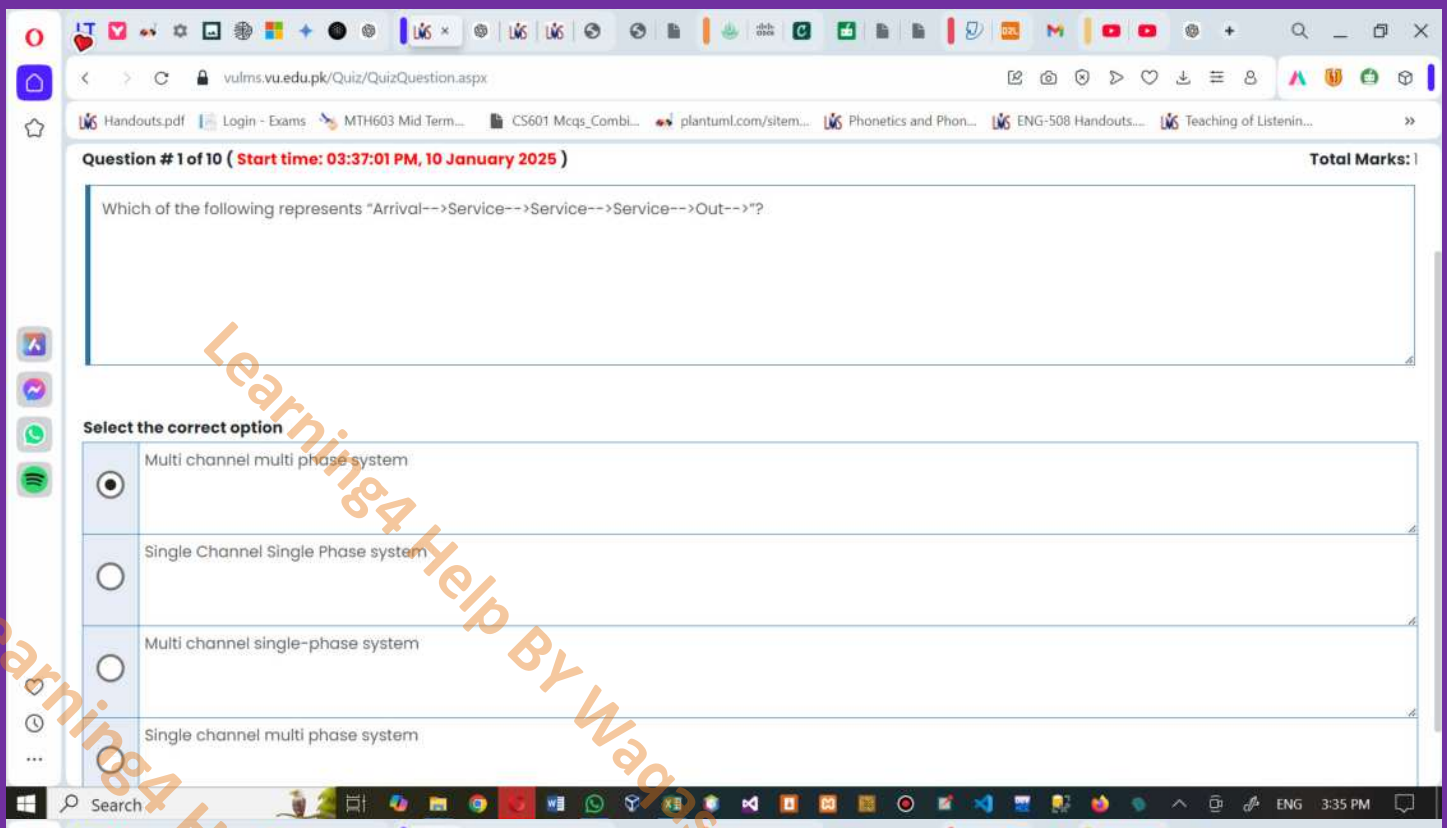


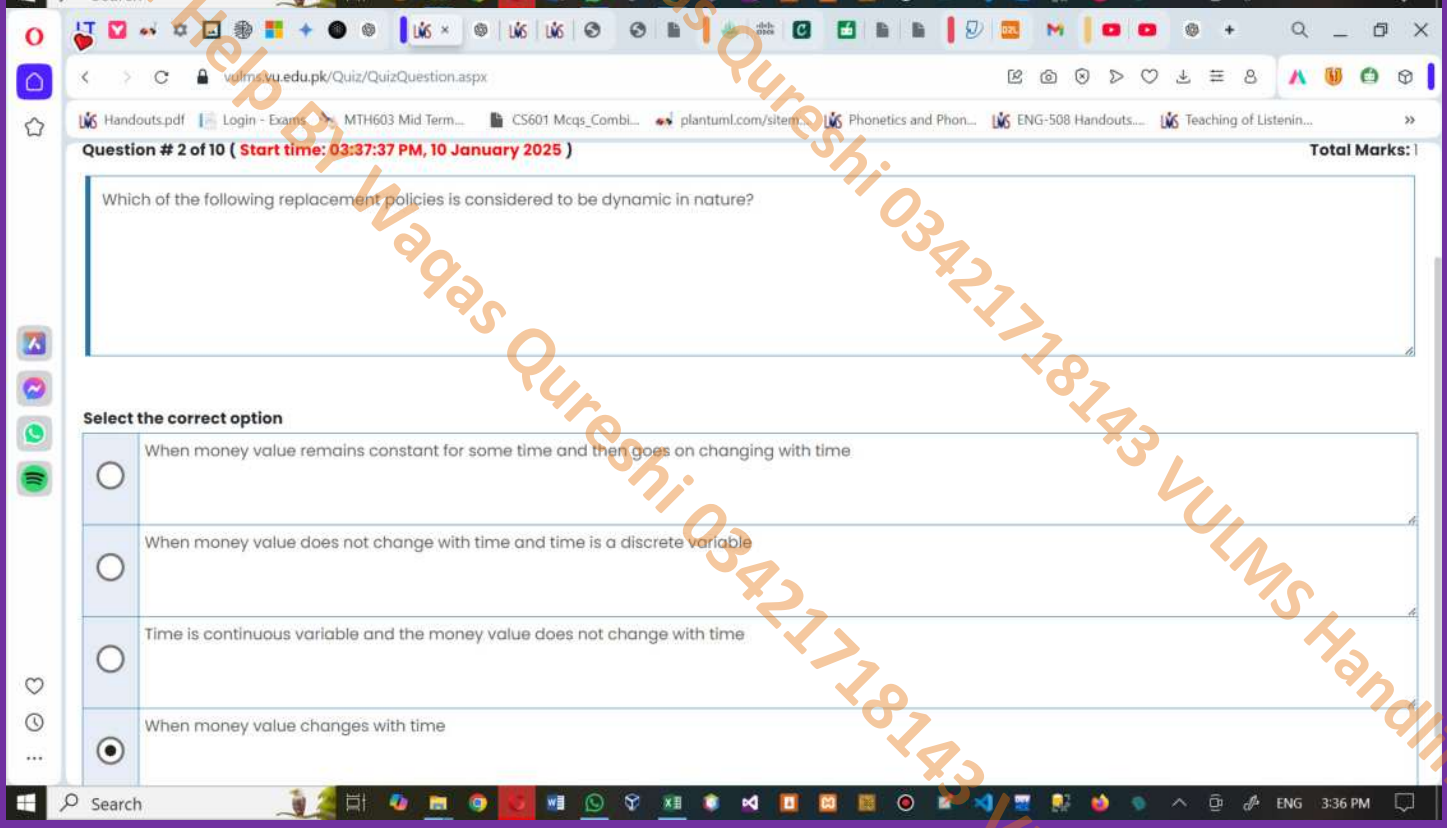
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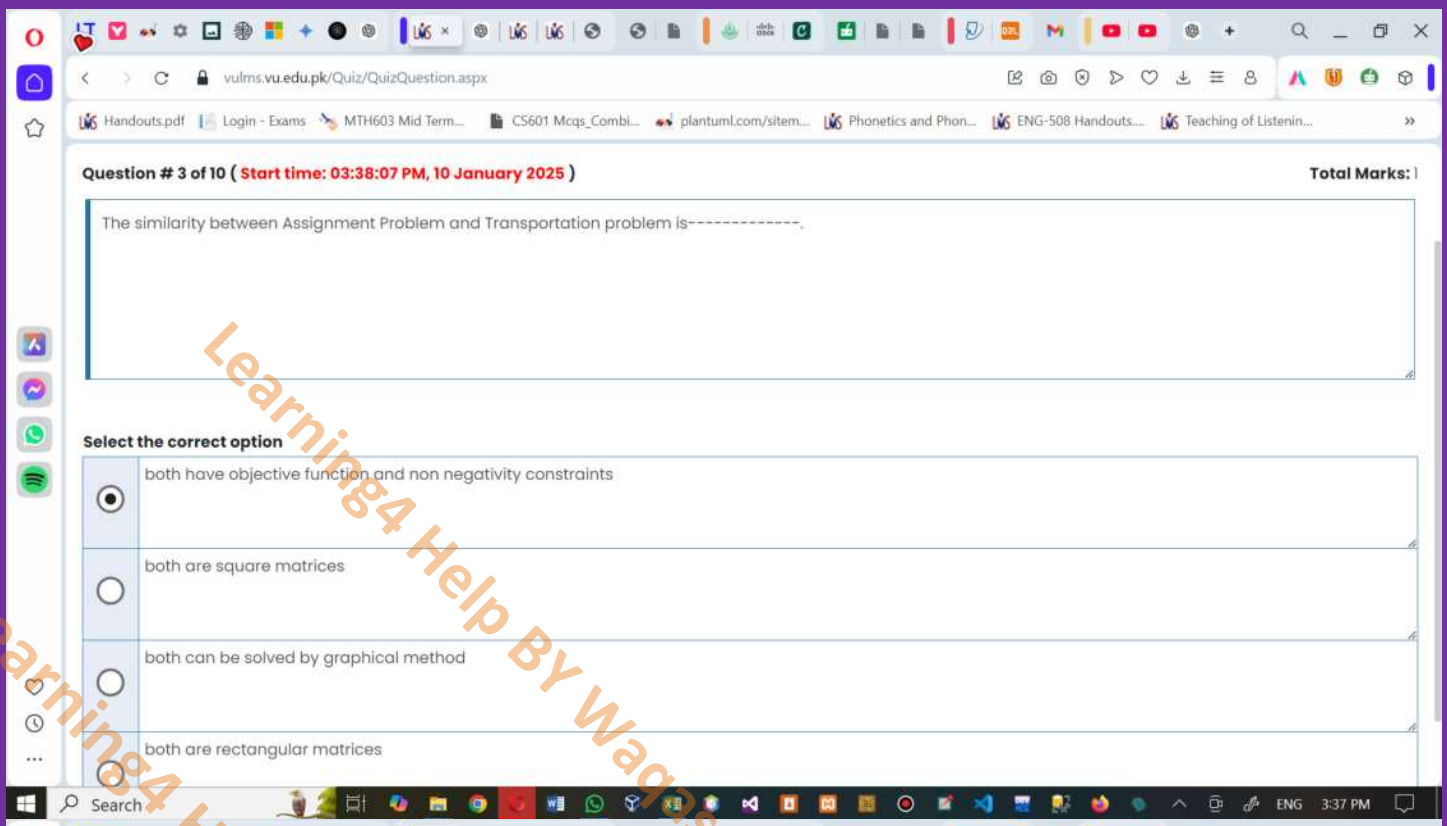


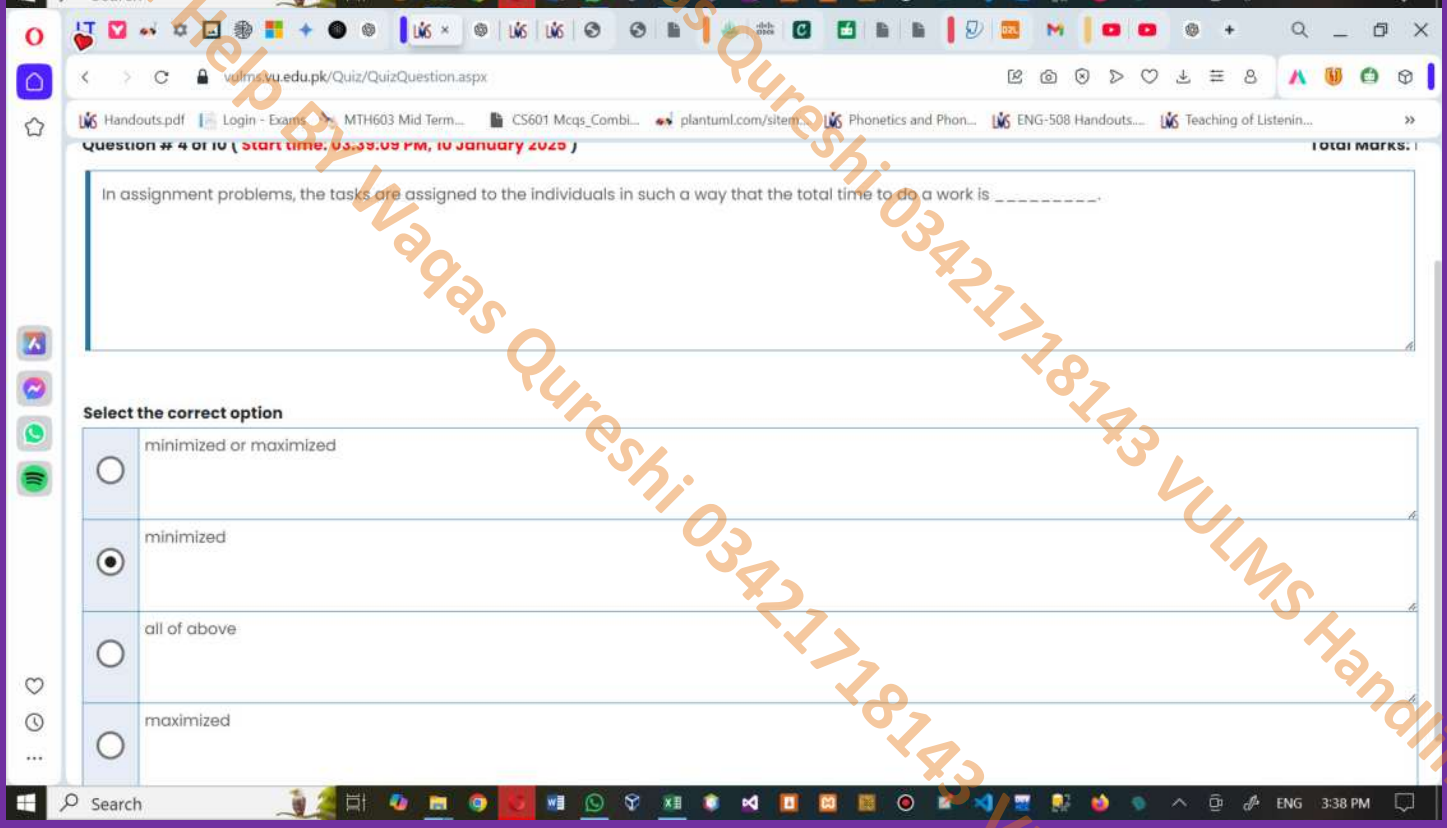
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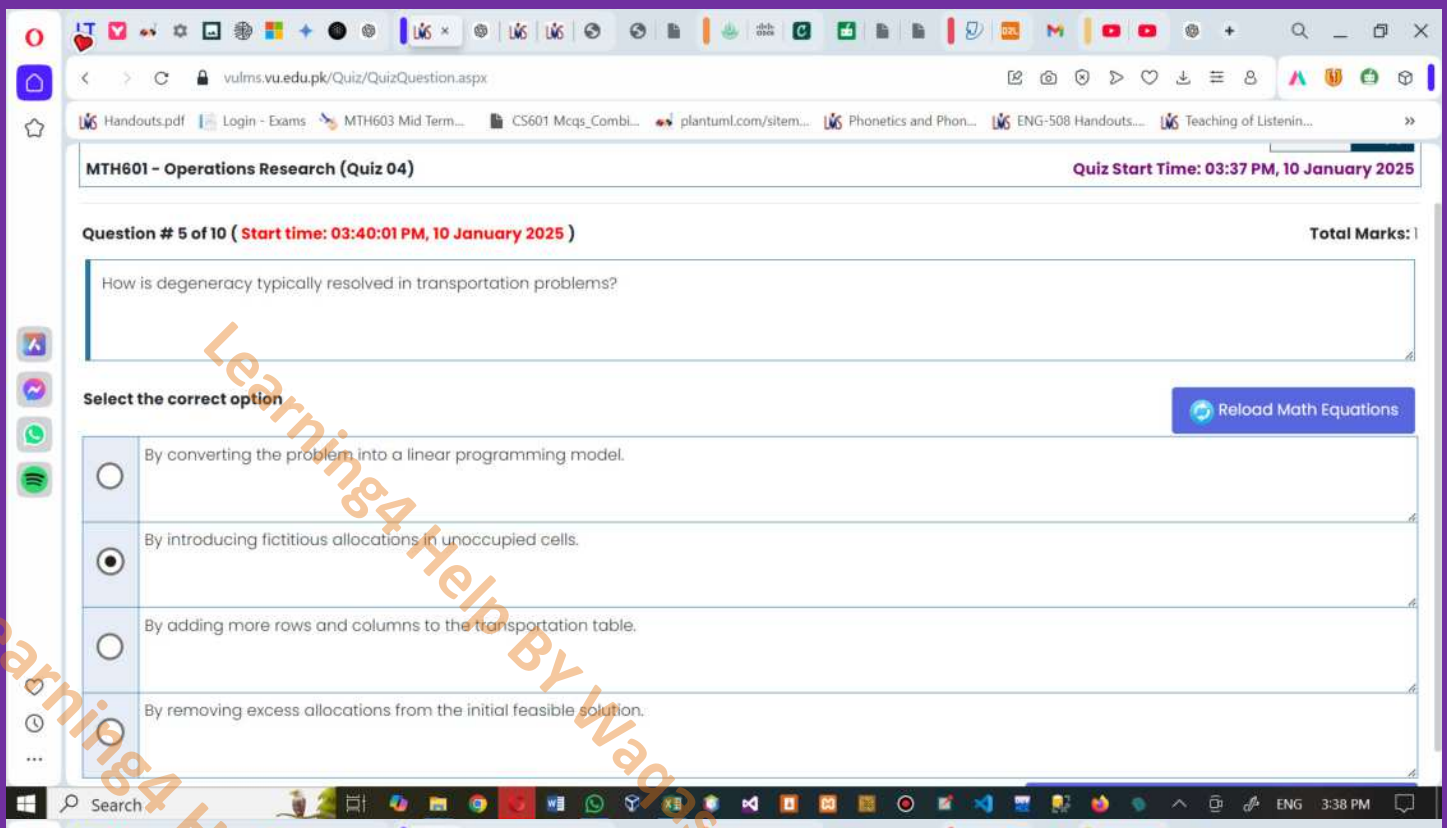


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MTH601 - Operations Research (Quiz 04)

Quiz Start Time: 03:37 PM, 10 January 2025

Question # 5 of 10 (Start time: 03:40:01 PM, 10 January 2025)

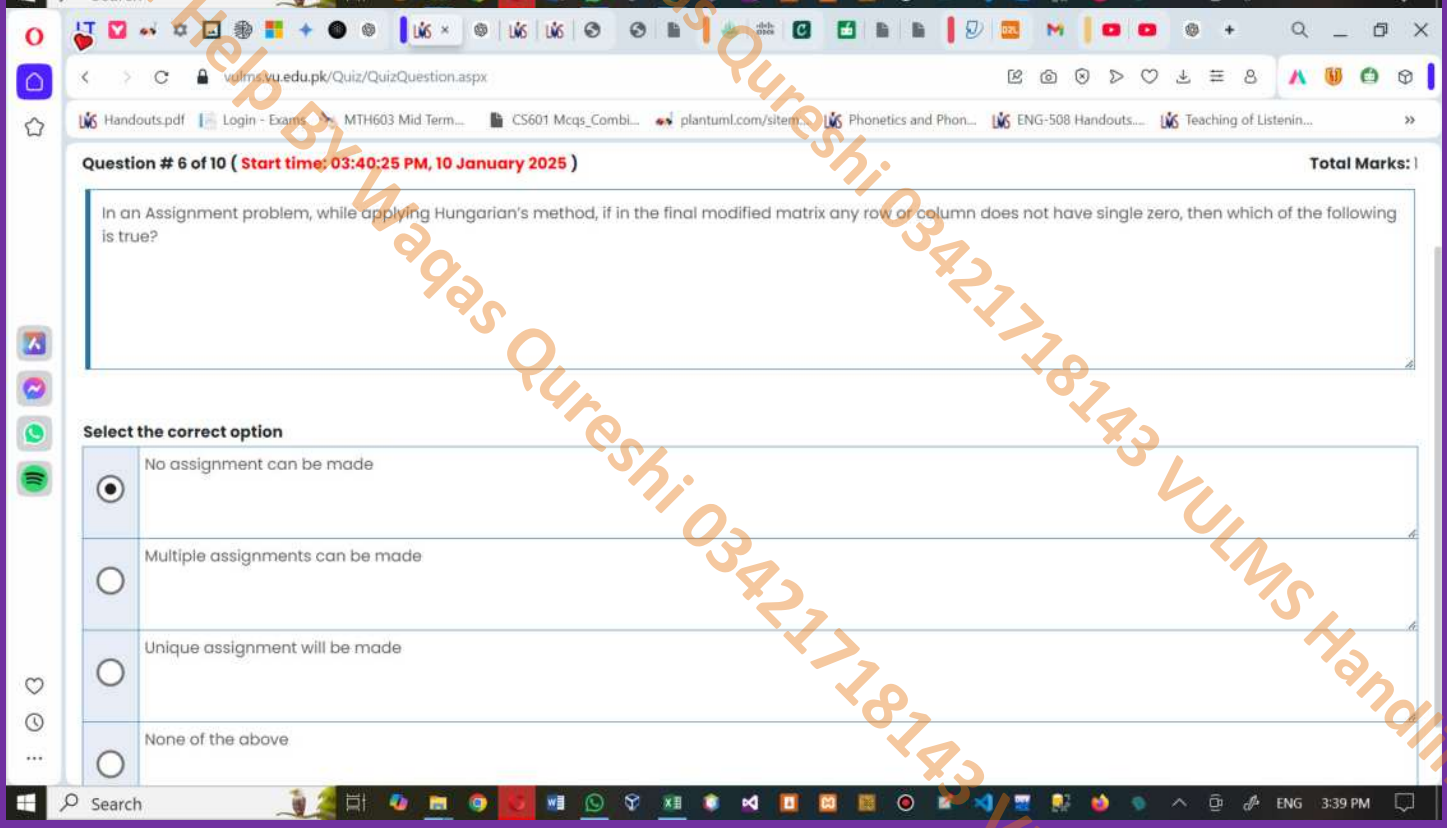
Total Marks: 1

How is degeneracy typically resolved in transportation problems?

Select the correct option

Reload Math Equations

- By converting the problem into a linear programming model.
- By introducing fictitious allocations in unoccupied cells.
- By adding more rows and columns to the transportation table.
- By removing excess allocations from the initial feasible solution.



Question # 6 of 10 (Start time: 03:40:25 PM, 10 January 2025)

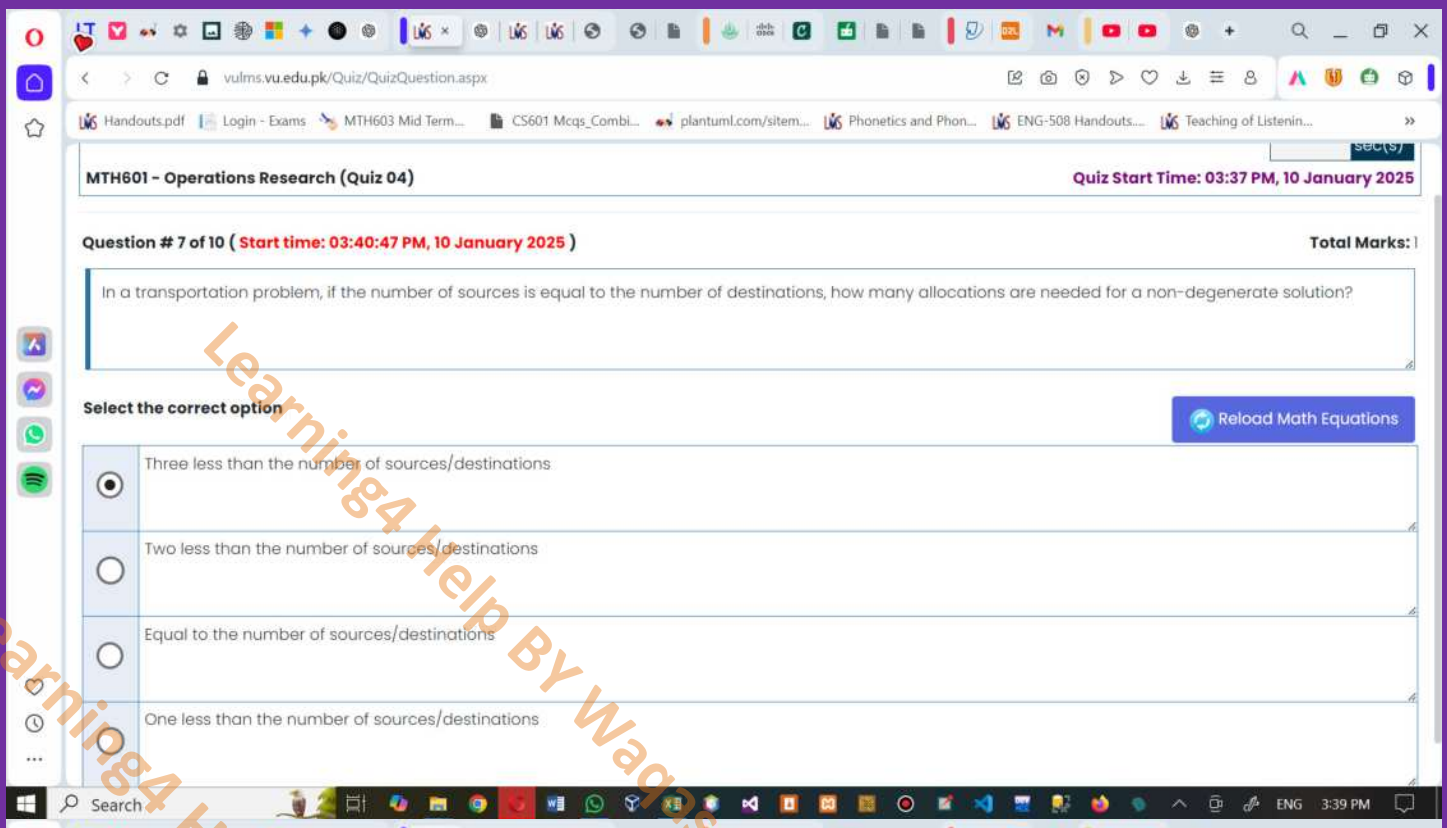
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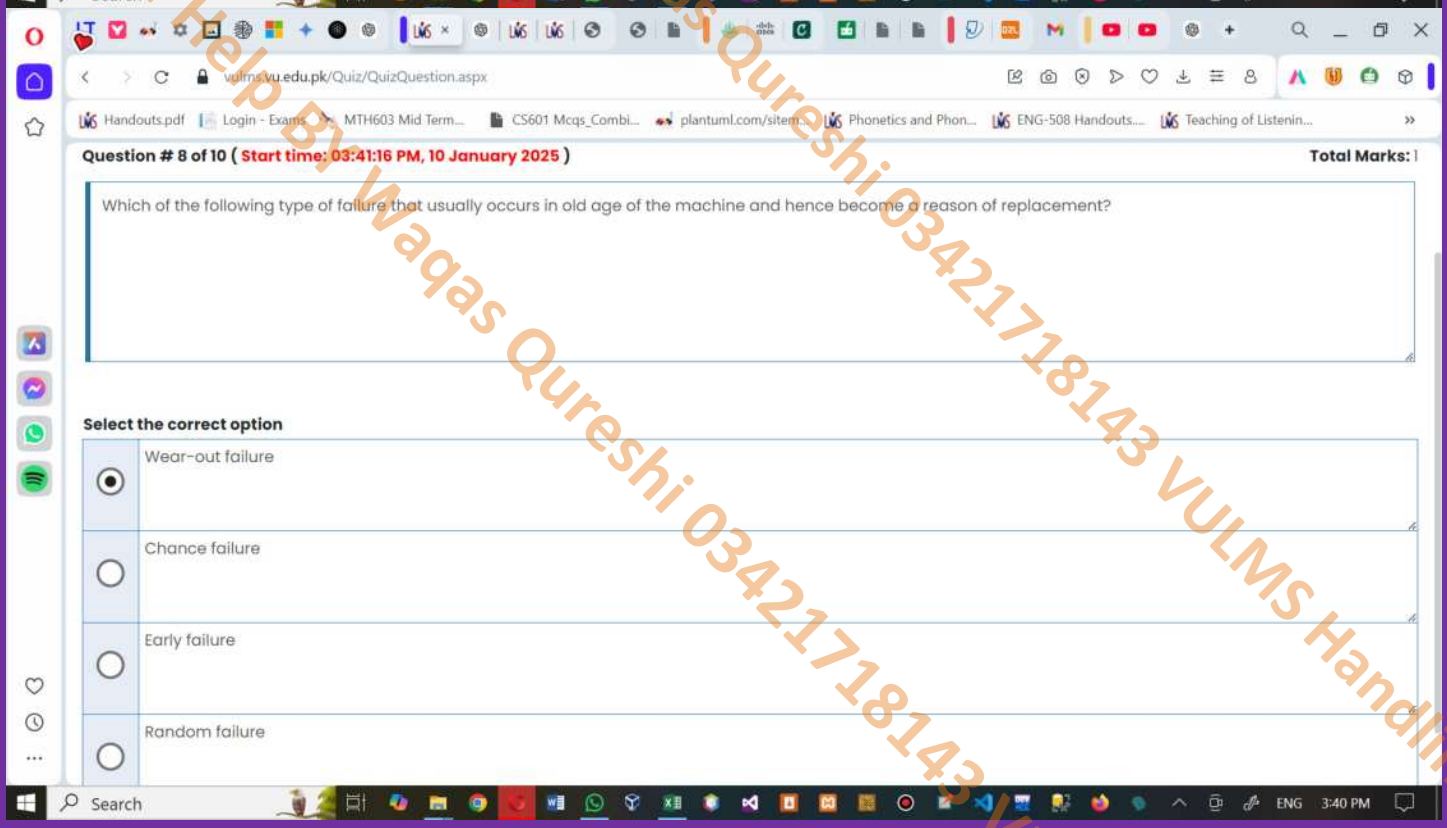
In an Assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?

Select the correct option

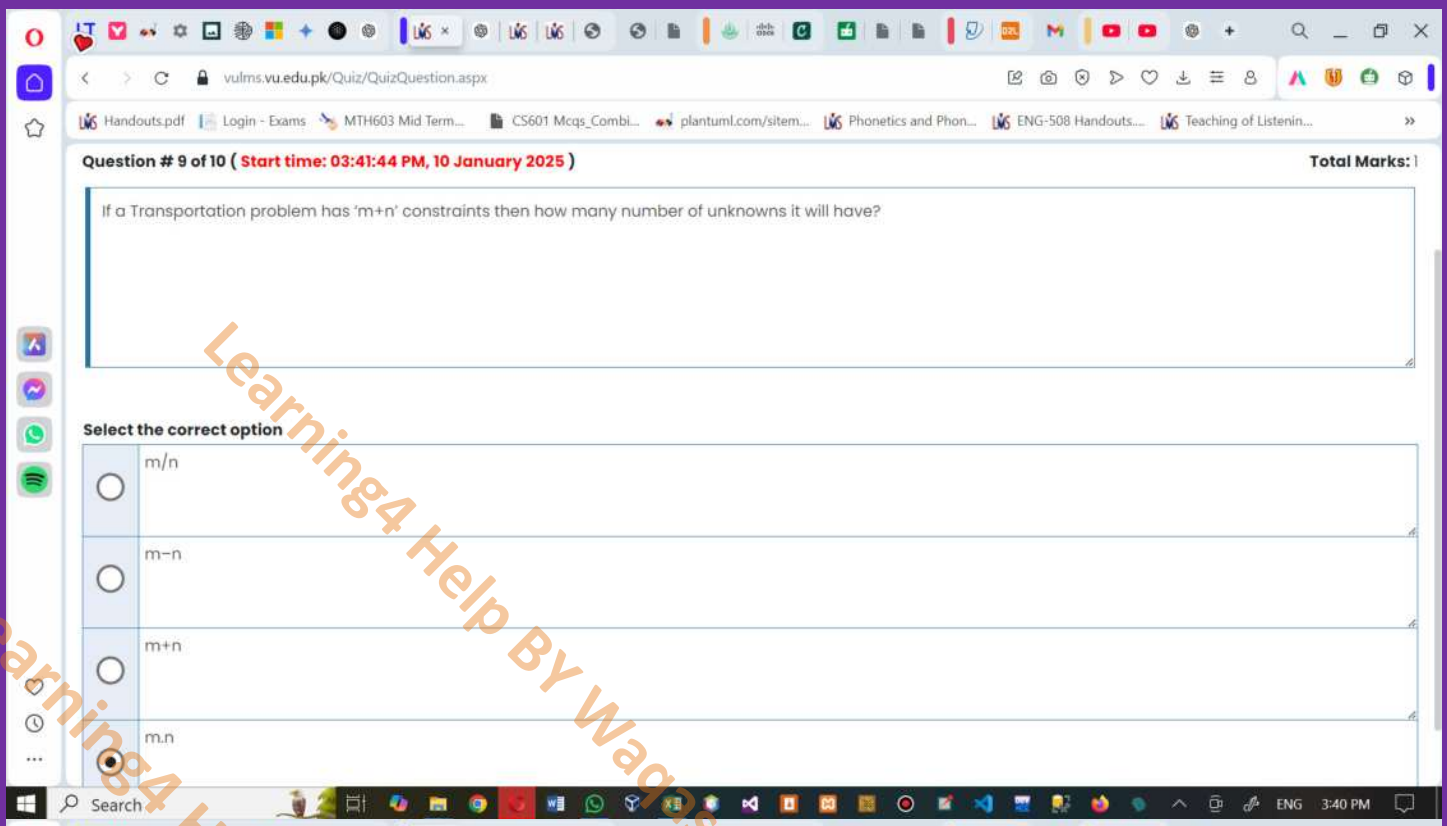
- No assignment can be made
- Multiple assignments can be made
- Unique assignment will be made
- None of the above

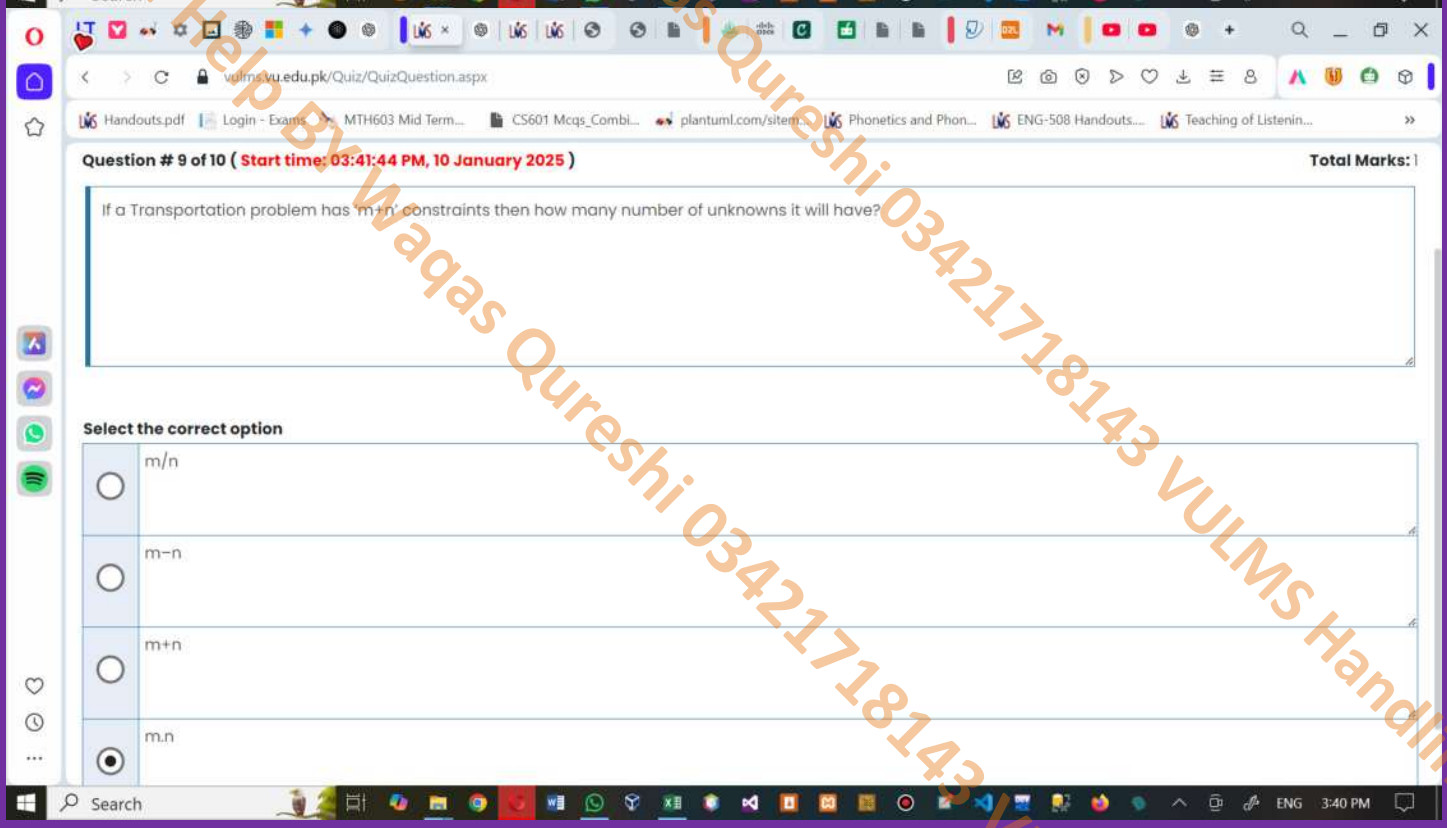
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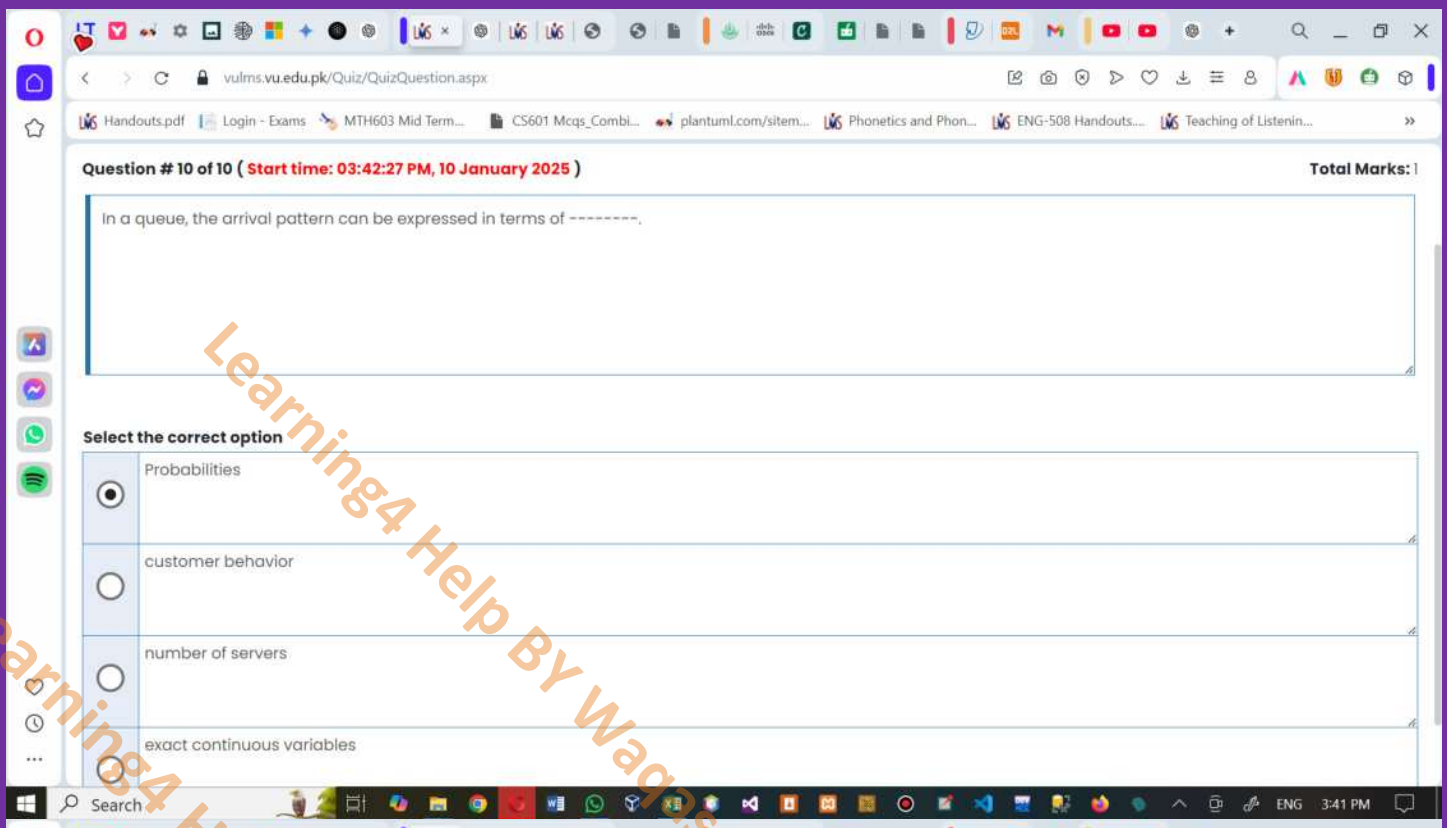


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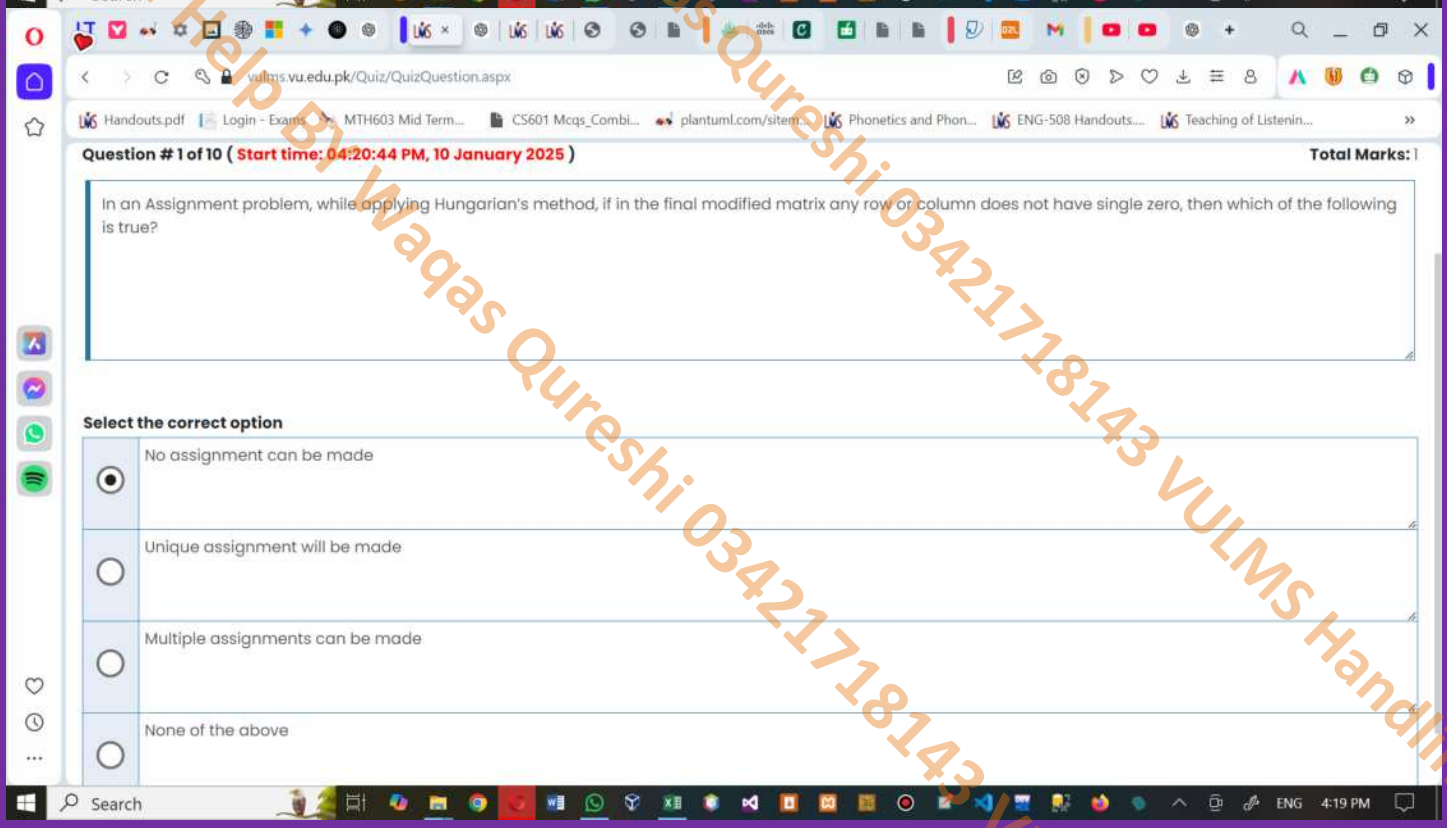


Question # 10 of 10 (Start time: 03:42:27 PM, 10 January 2025) Total Marks: 1

In a queue, the arrival pattern can be expressed in terms of --------.

Select the correct option

- Probabilities
- customer behavior
- number of servers
- exact continuous variables



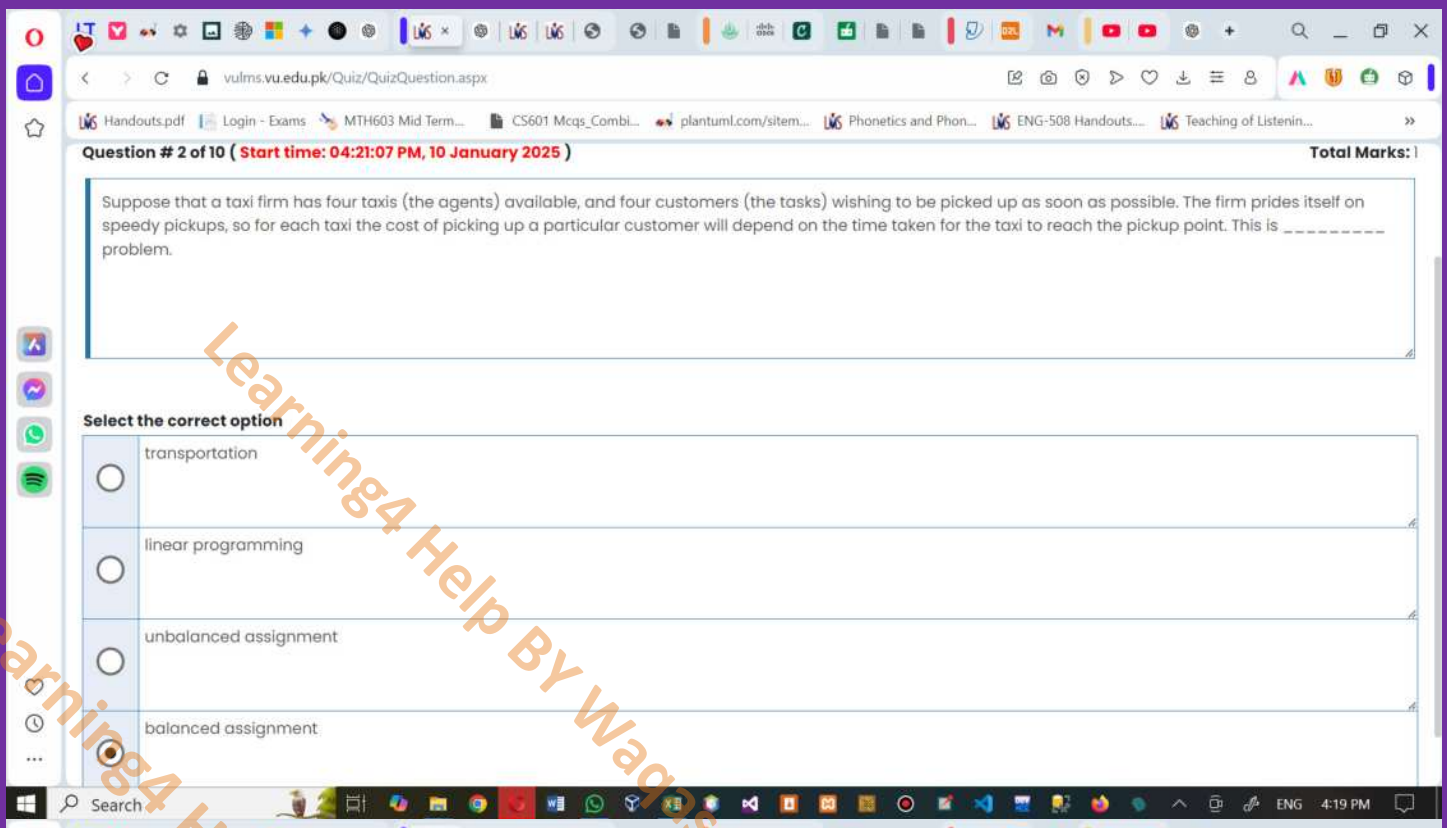
Question # 1 of 10 (Start time: 04:20:44 PM, 10 January 2025) Total Marks: 1

In an Assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?

Select the correct option

- No assignment can be made
- Unique assignment will be made
- Multiple assignments can be made
- None of the above

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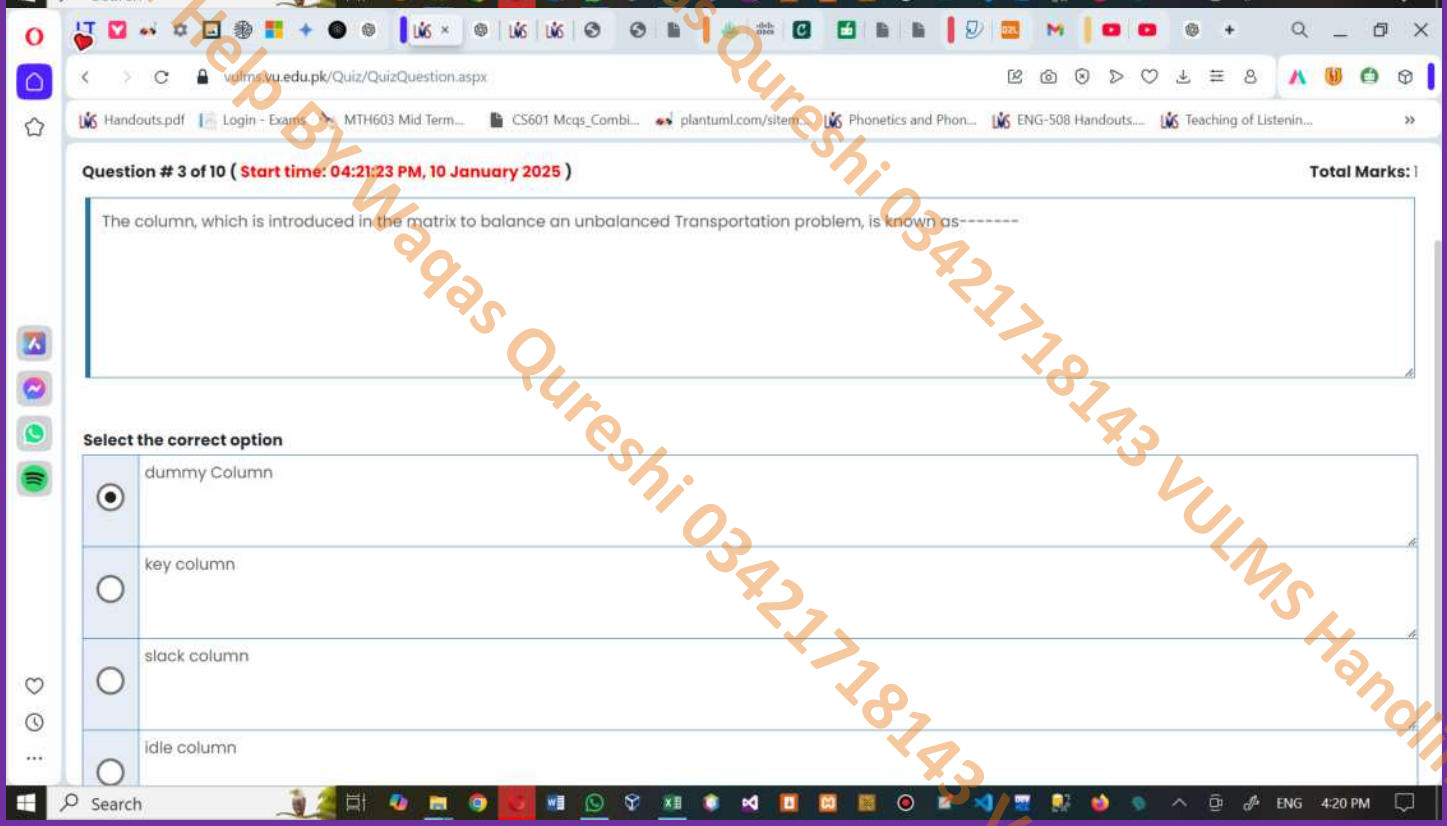


Question # 2 of 10 (Start time: 04:21:07 PM, 10 January 2025) Total Marks: 1

Suppose that a taxi firm has four taxis (the agents) available, and four customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the cost of picking up a particular customer will depend on the time taken for the taxi to reach the pickup point. This is _____ problem.

Select the correct option

- transportation
- linear programming
- unbalanced assignment
- balanced assignment



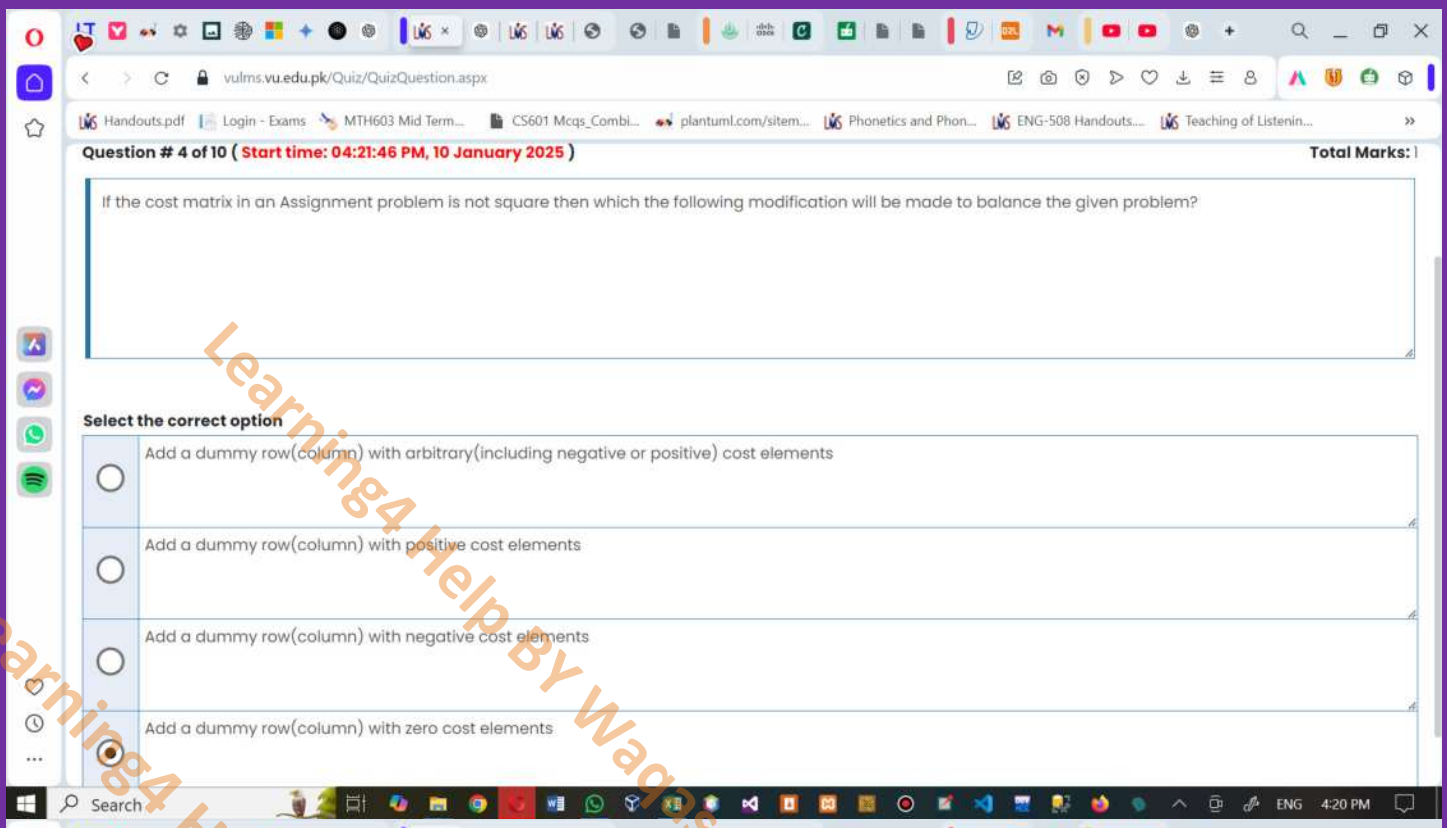
Question # 3 of 10 (Start time: 04:21:23 PM, 10 January 2025) Total Marks: 1

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

Select the correct option

- dummy Column
- key column
- slack column
- idle column

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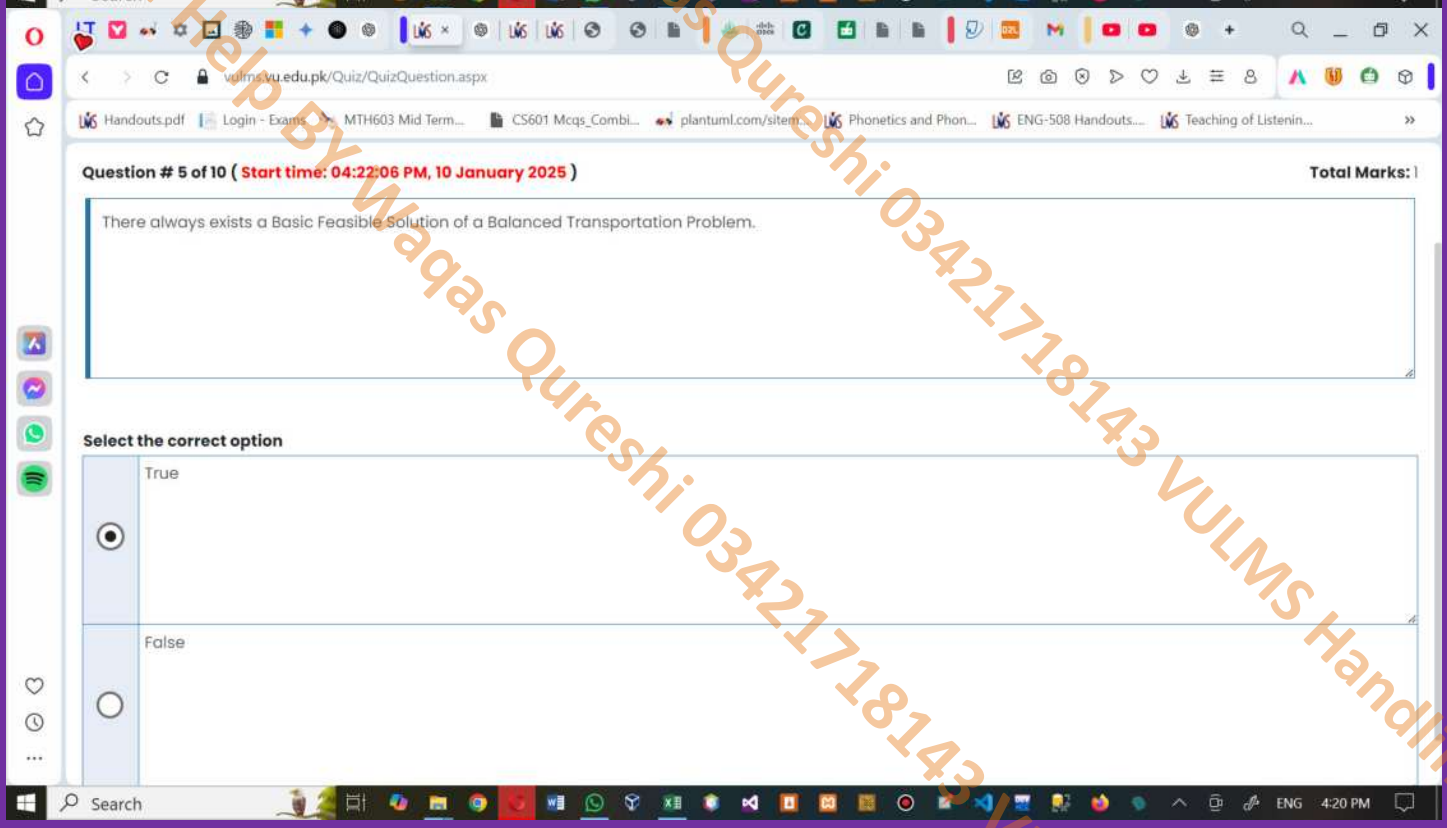


Question # 4 of 10 (Start time: 04:21:46 PM, 10 January 2025) Total Marks: 1

If the cost matrix in an Assignment problem is not square then which the following modification will be made to balance the given problem?

Select the correct option

- Add a dummy row(column) with arbitrary(including negative or positive) cost elements
- Add a dummy row(column) with positive cost elements
- Add a dummy row(column) with negative cost elements
- Add a dummy row(column) with zero cost elements



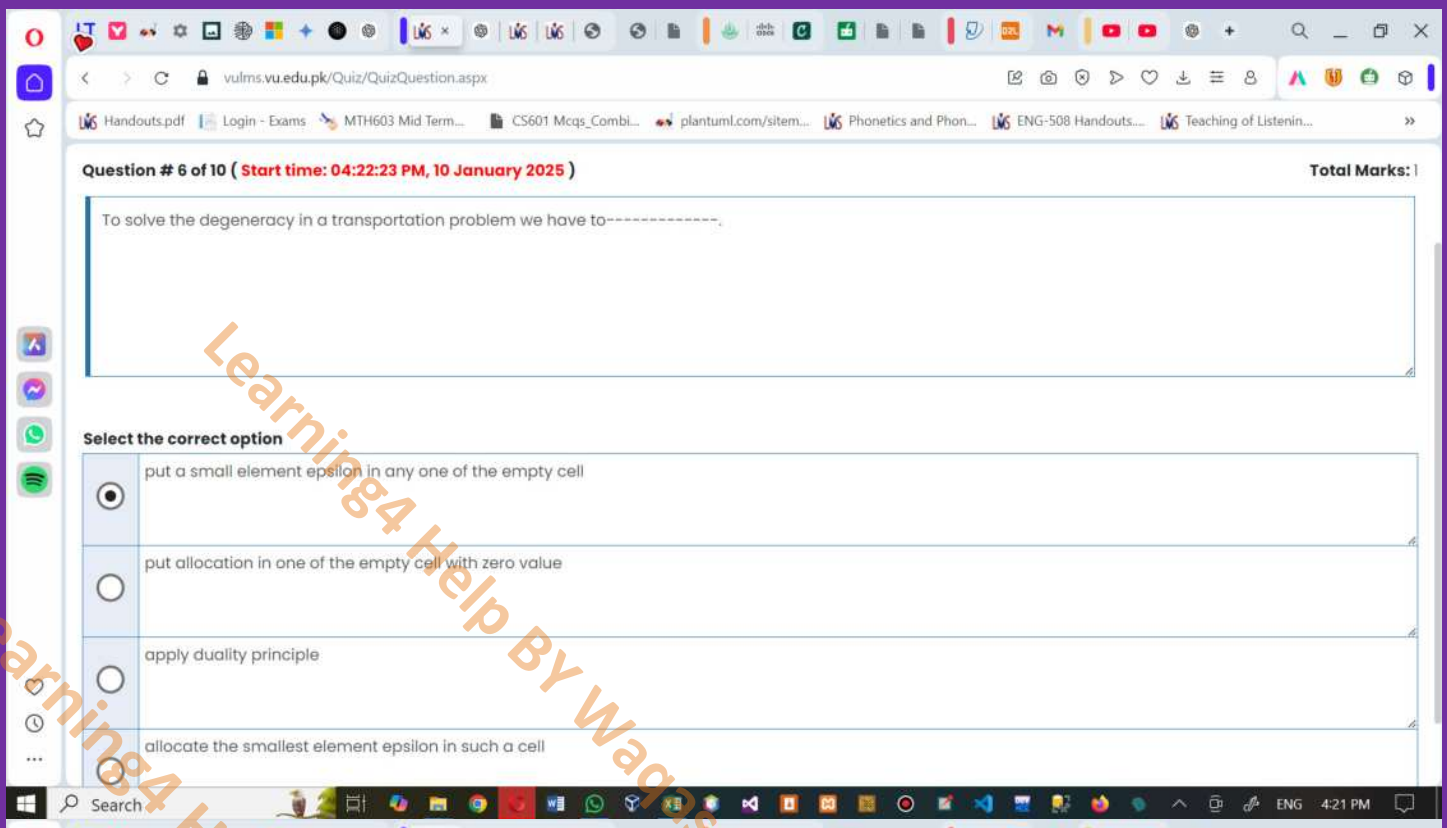
Question # 5 of 10 (Start time: 04:22:06 PM, 10 January 2025) Total Marks: 1

There always exists a Basic Feasible Solution of a Balanced Transportation Problem.

Select the correct option

- True
- False

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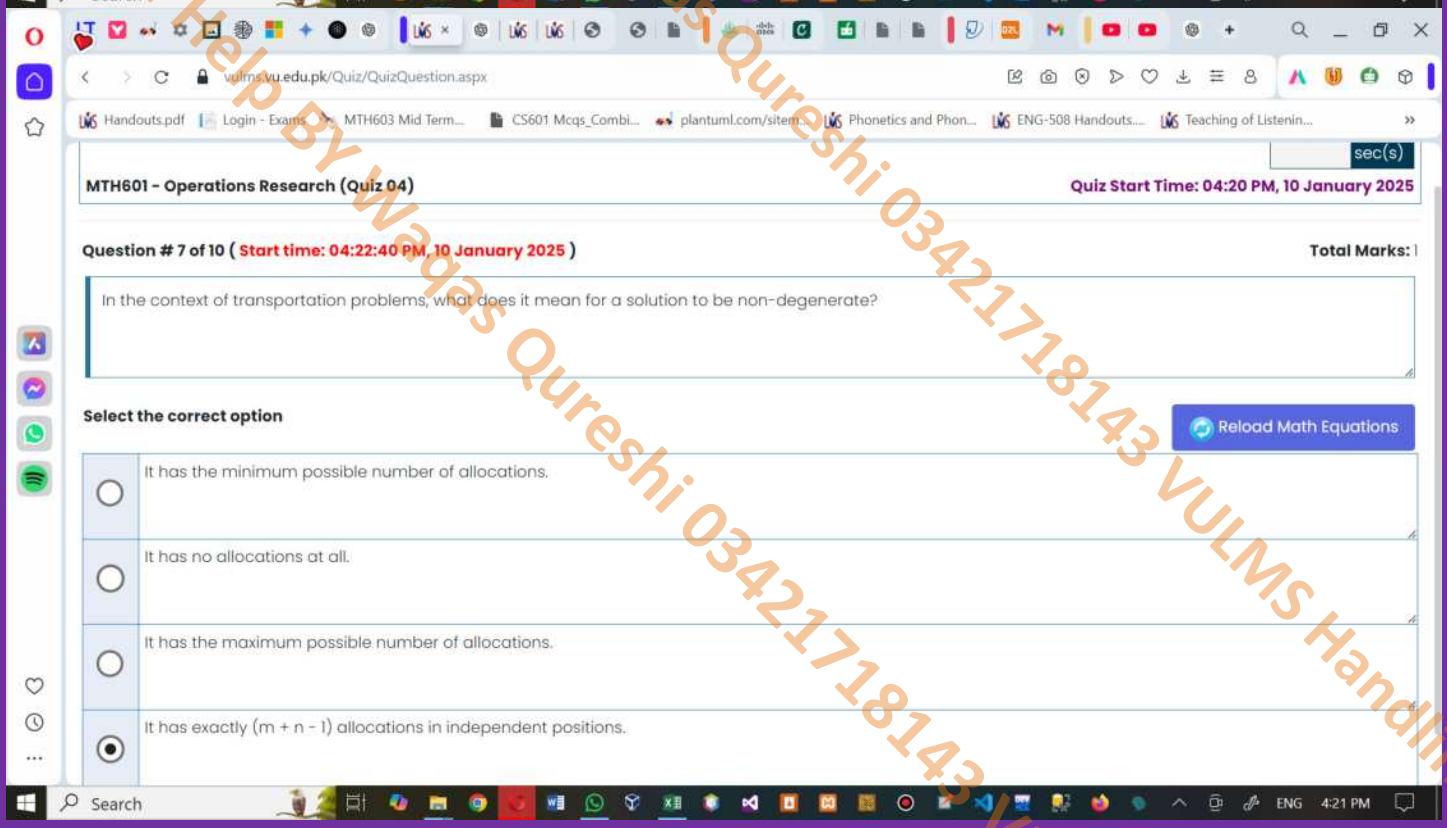


Question # 6 of 10 (Start time: 04:22:23 PM, 10 January 2025) Total Marks: 1

To solve the degeneracy in a transportation problem we have to----------.

Select the correct option

- put a small element epsilon in any one of the empty cell
- put allocation in one of the empty cell with zero value
- apply duality principle
- allocate the smallest element epsilon in such a cell



sec(s)

MTH601 - Operations Research (Quiz 04) Quiz Start Time: 04:20 PM, 10 January 2025

Question # 7 of 10 (Start time: 04:22:40 PM, 10 January 2025) Total Marks: 1

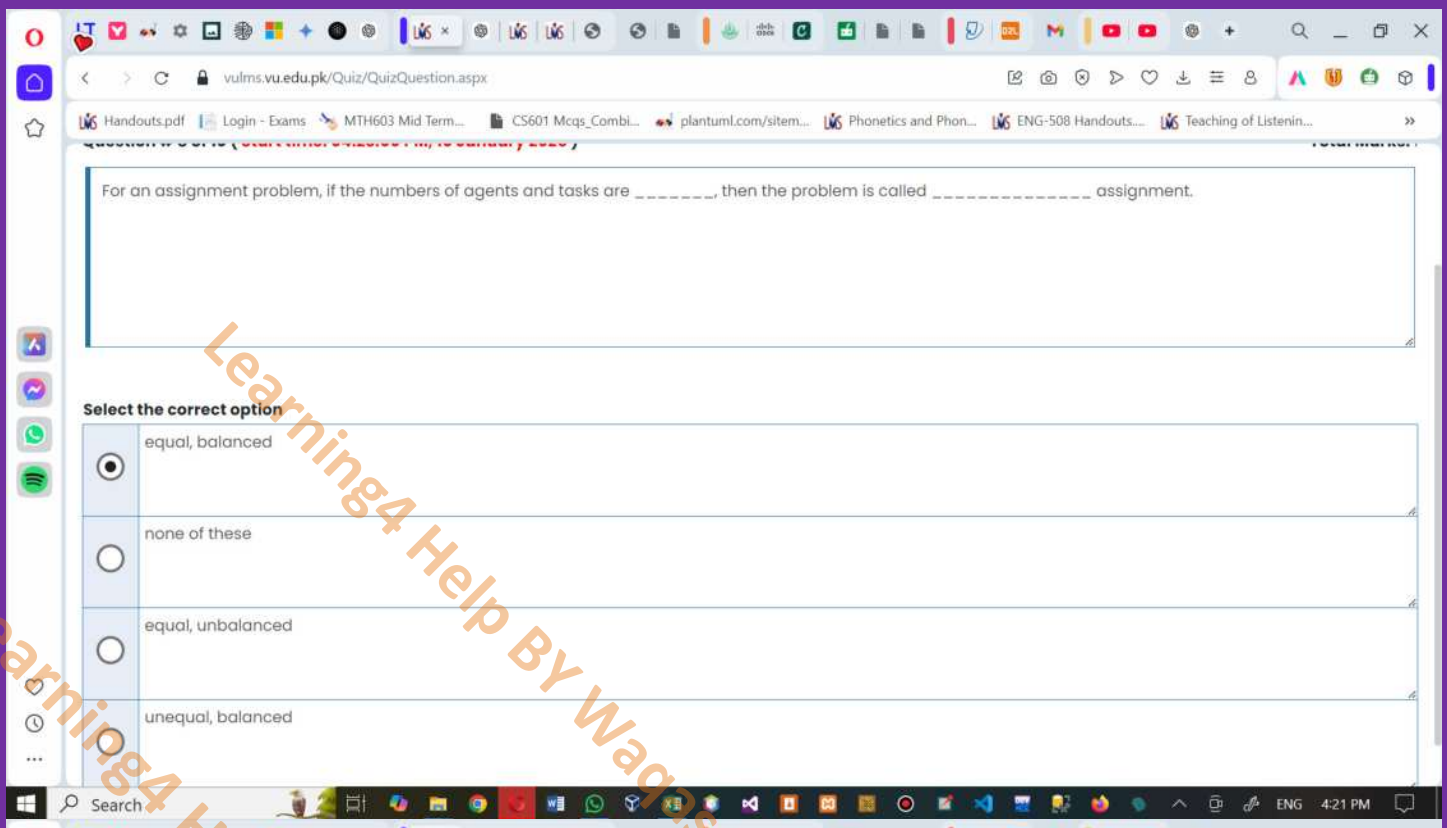
In the context of transportation problems, what does it mean for a solution to be non-degenerate?

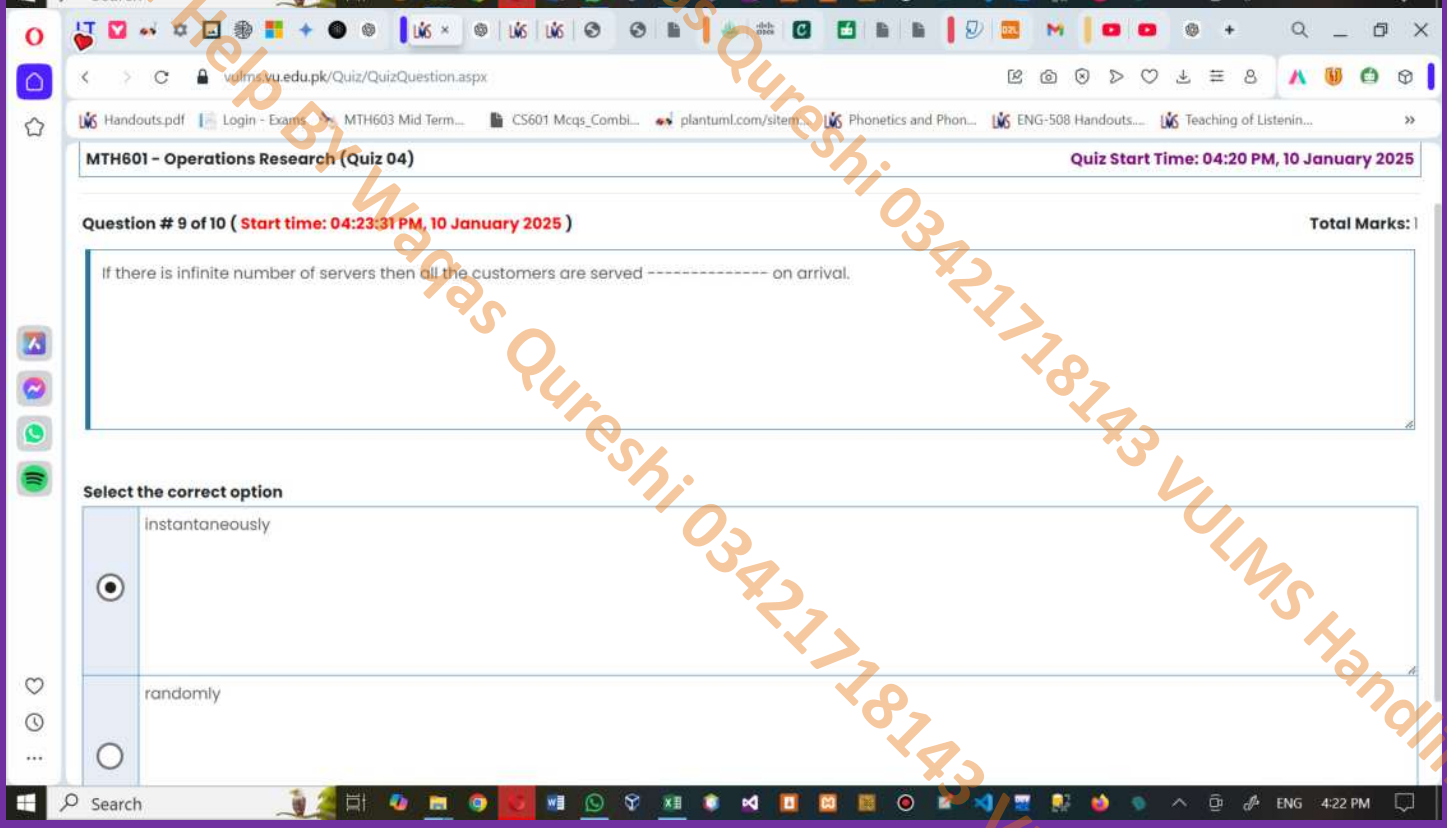
Select the correct option

Reload Math Equations

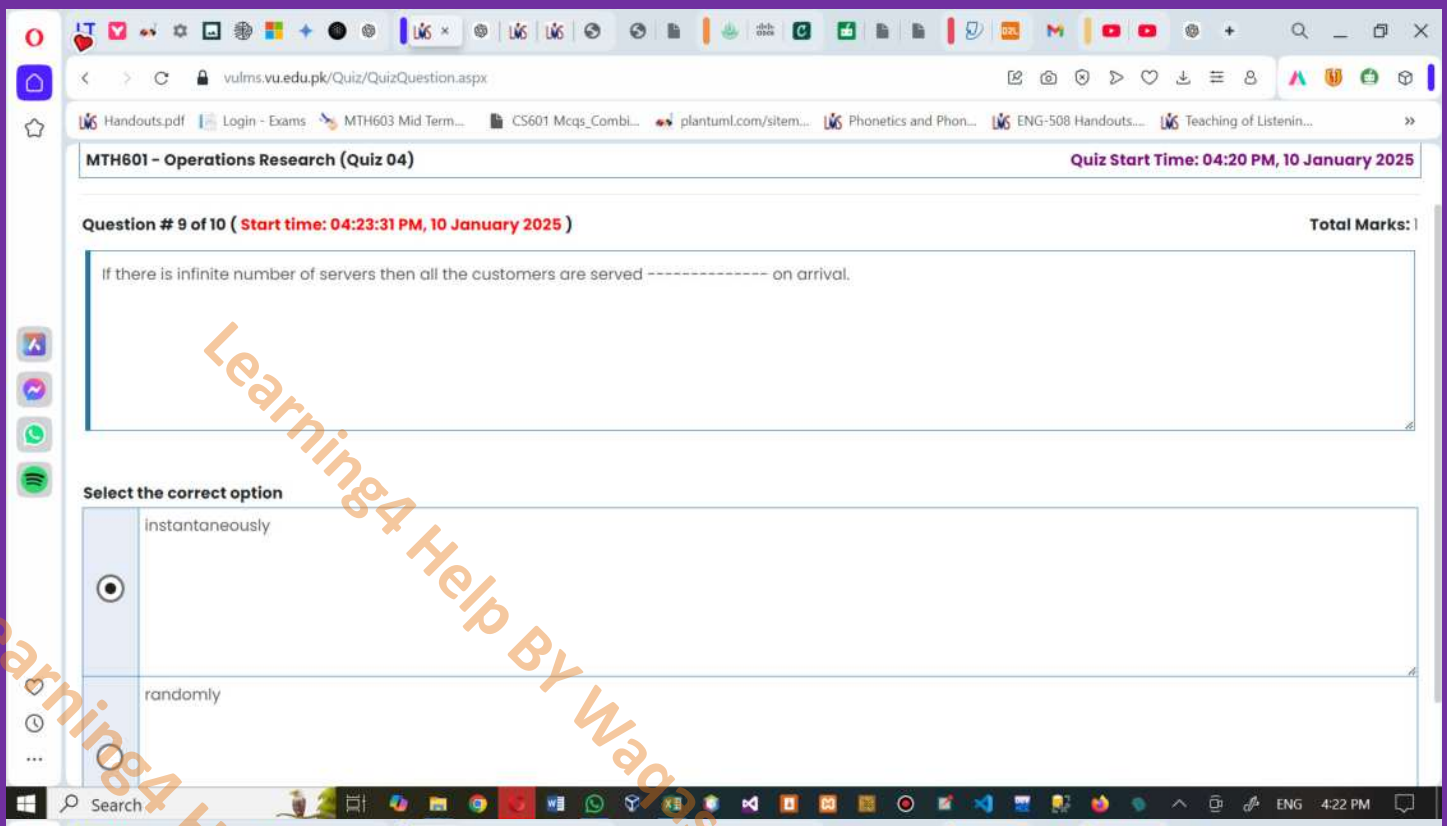
- It has the minimum possible number of allocations.
- It has no allocations at all.
- It has the maximum possible number of allocations.
- It has exactly (m + n - 1) allocations in independent positions.

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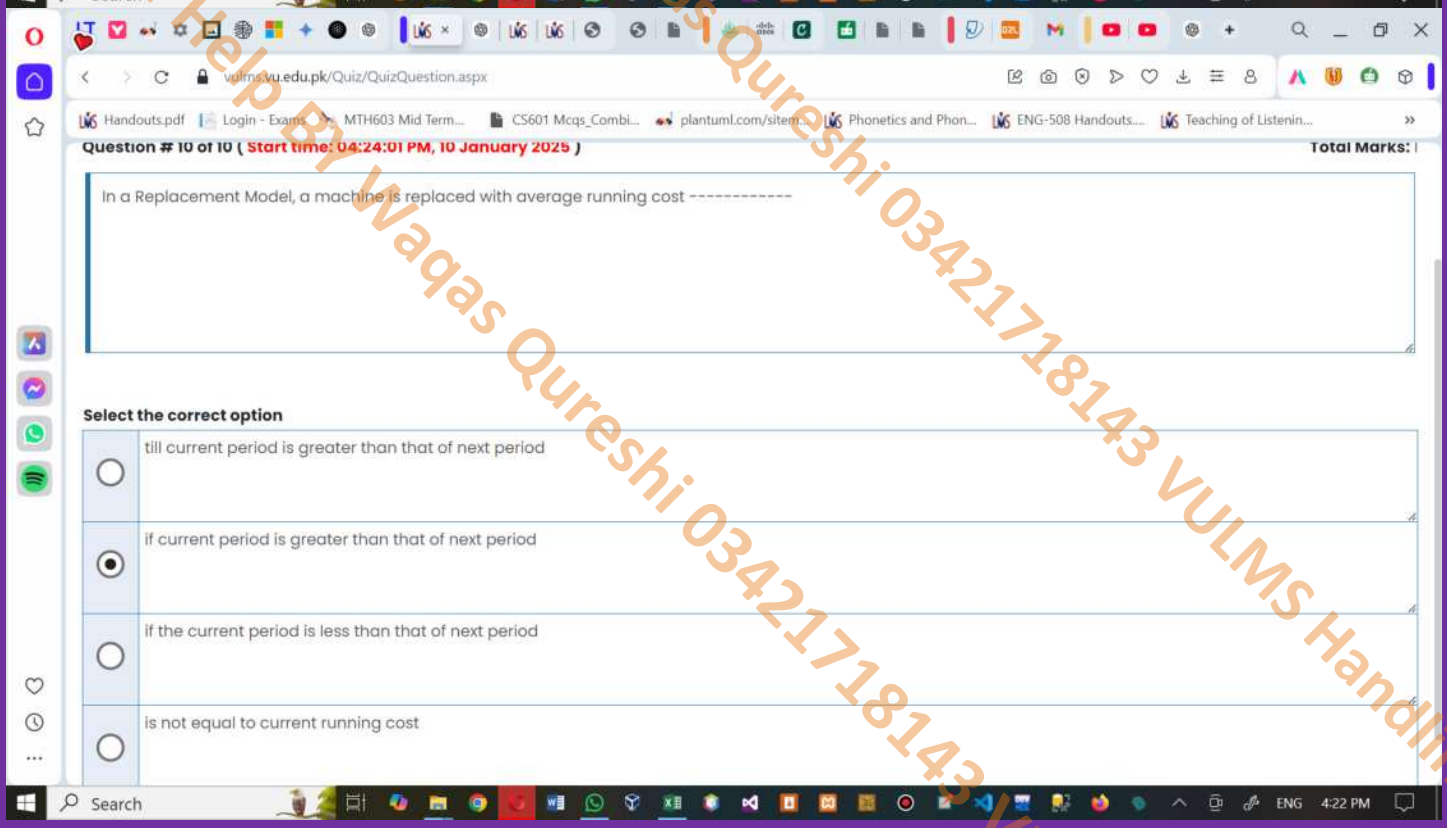
MTH601 - Operations Research (Quiz 04) Quiz Start Time: 04:20 PM, 10 January 2025

Question # 9 of 10 (Start time: 04:23:31 PM, 10 January 2025) Total Marks: 1

If there is infinite number of servers then all the customers are served ---------- on arrival.

Select the correct option

- instantaneously
- randomly



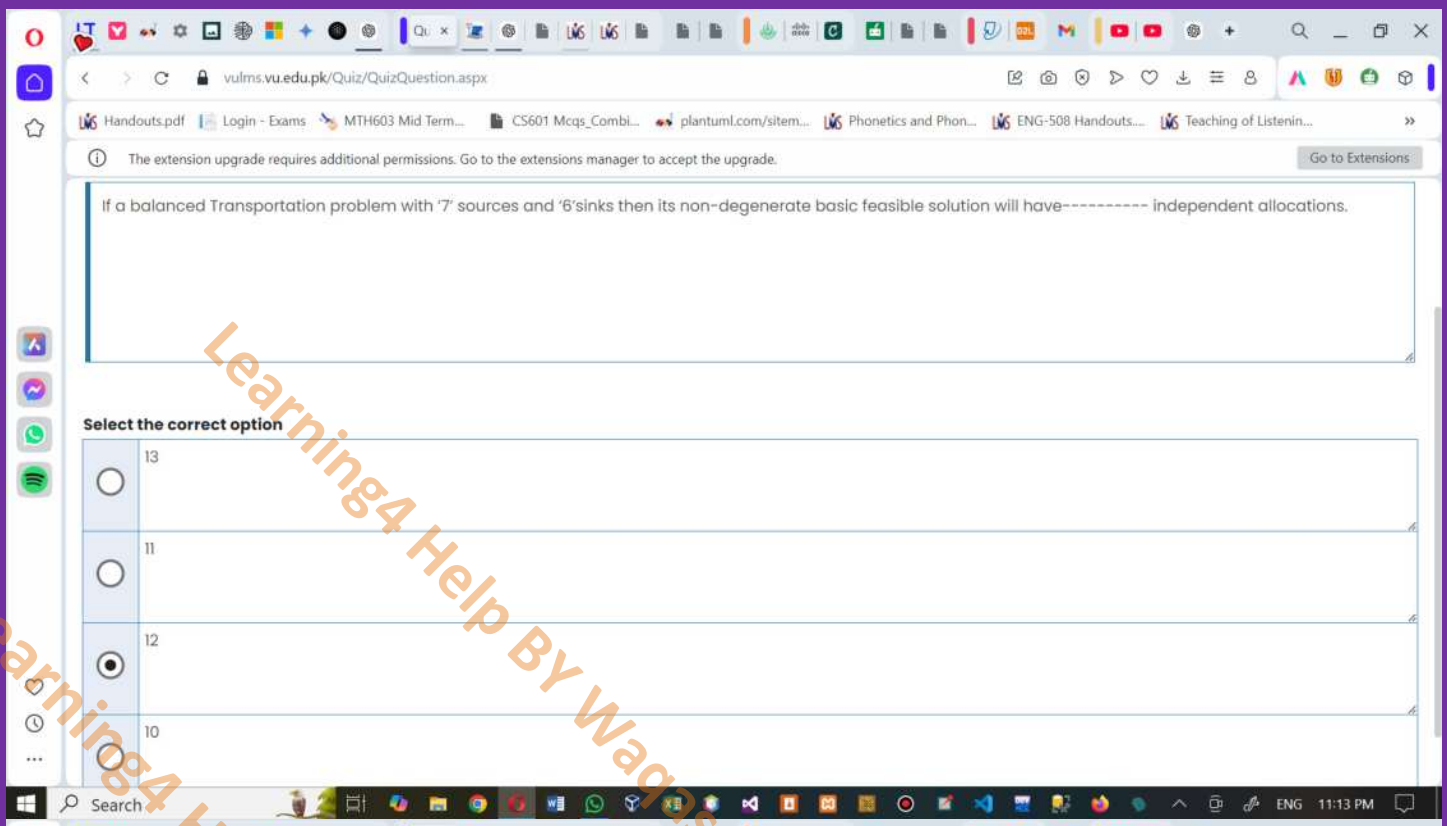
Question # 10 of 10 (Start time: 04:24:01 PM, 10 January 2025) Total Marks: 1

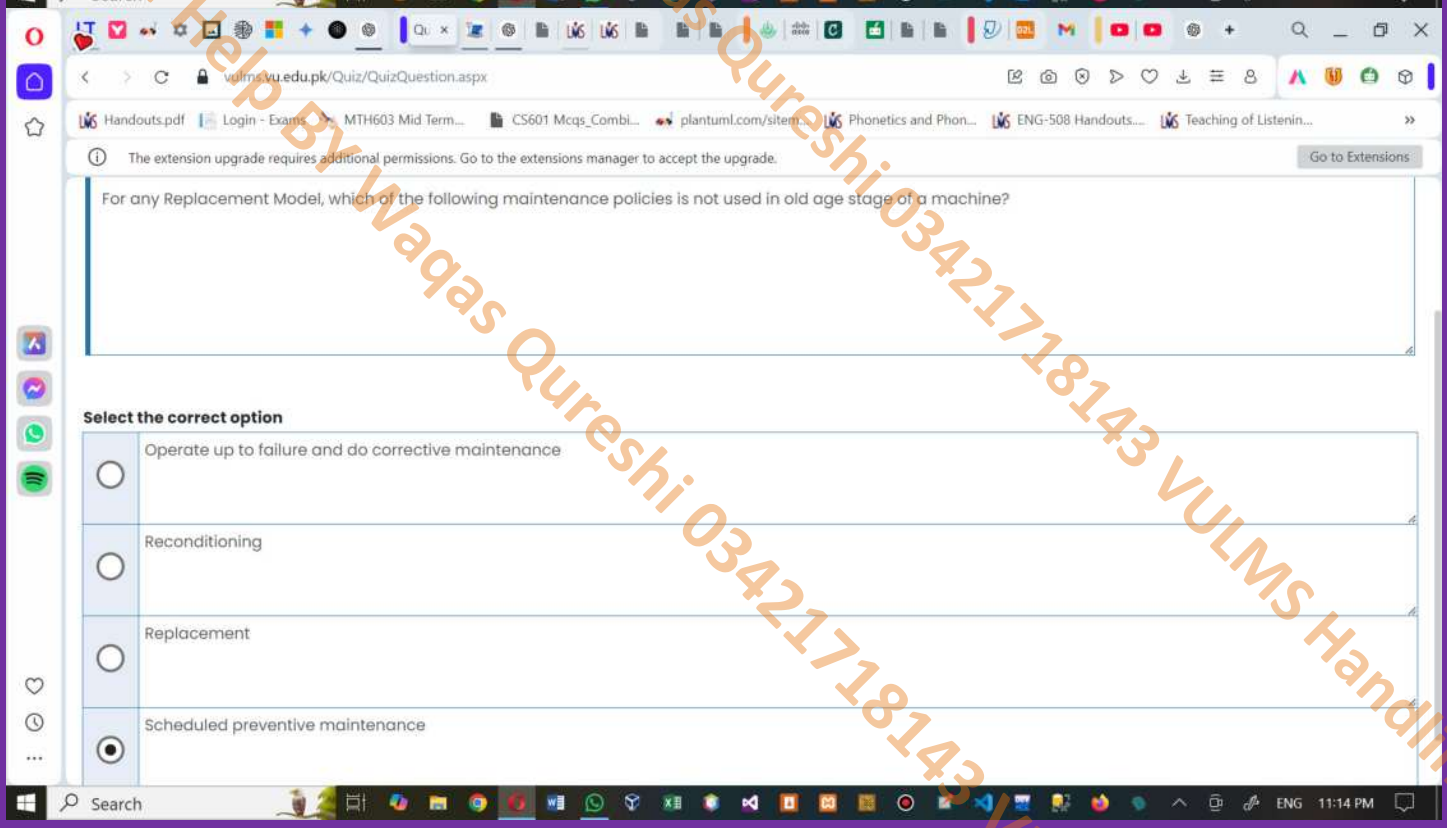
In a Replacement Model, a machine is replaced with average running cost ----------

Select the correct option

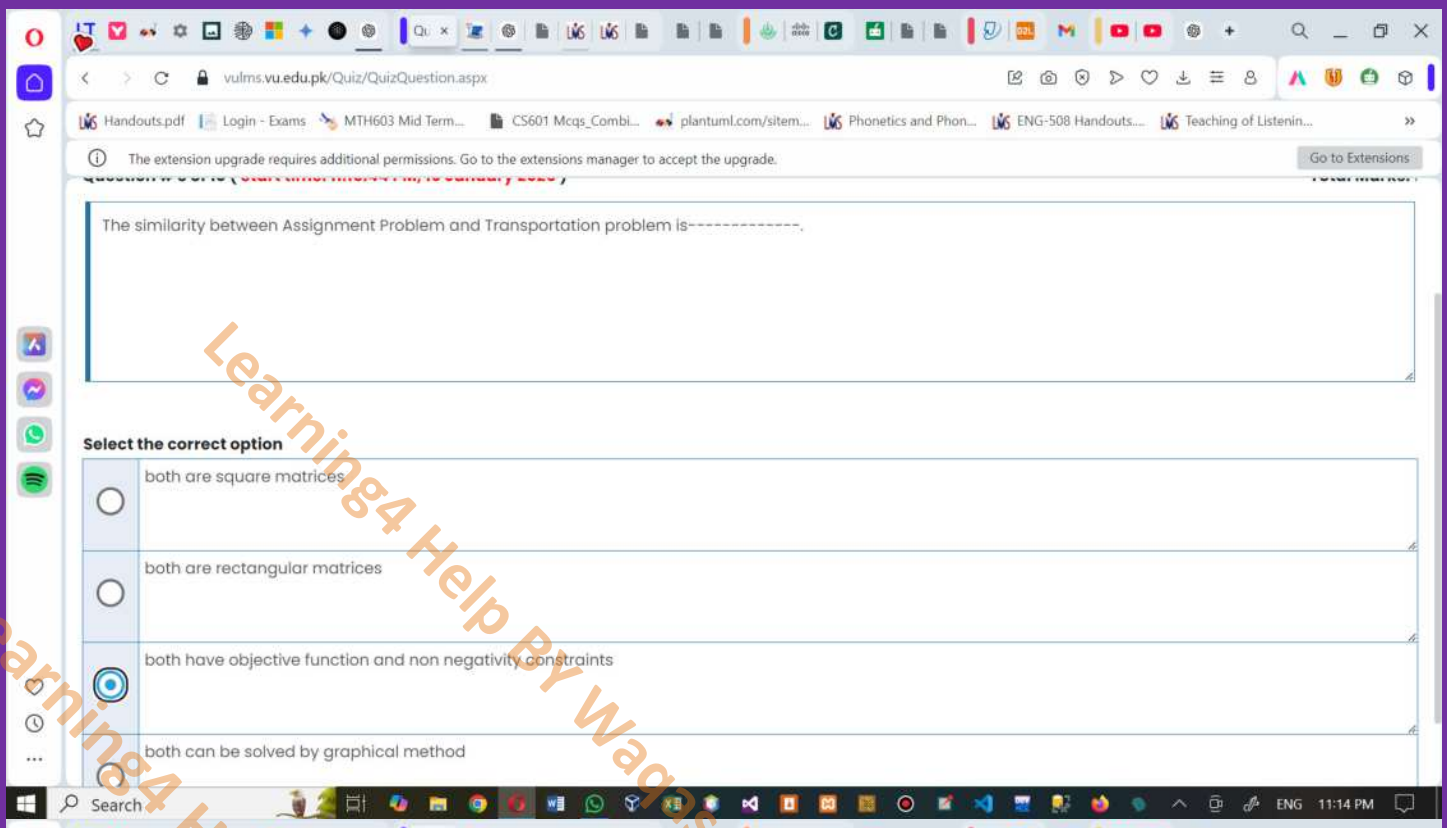
- till current period is greater than that of next period
- if current period is greater than that of next period
- if the current period is less than that of next period
- is not equal to current running cost

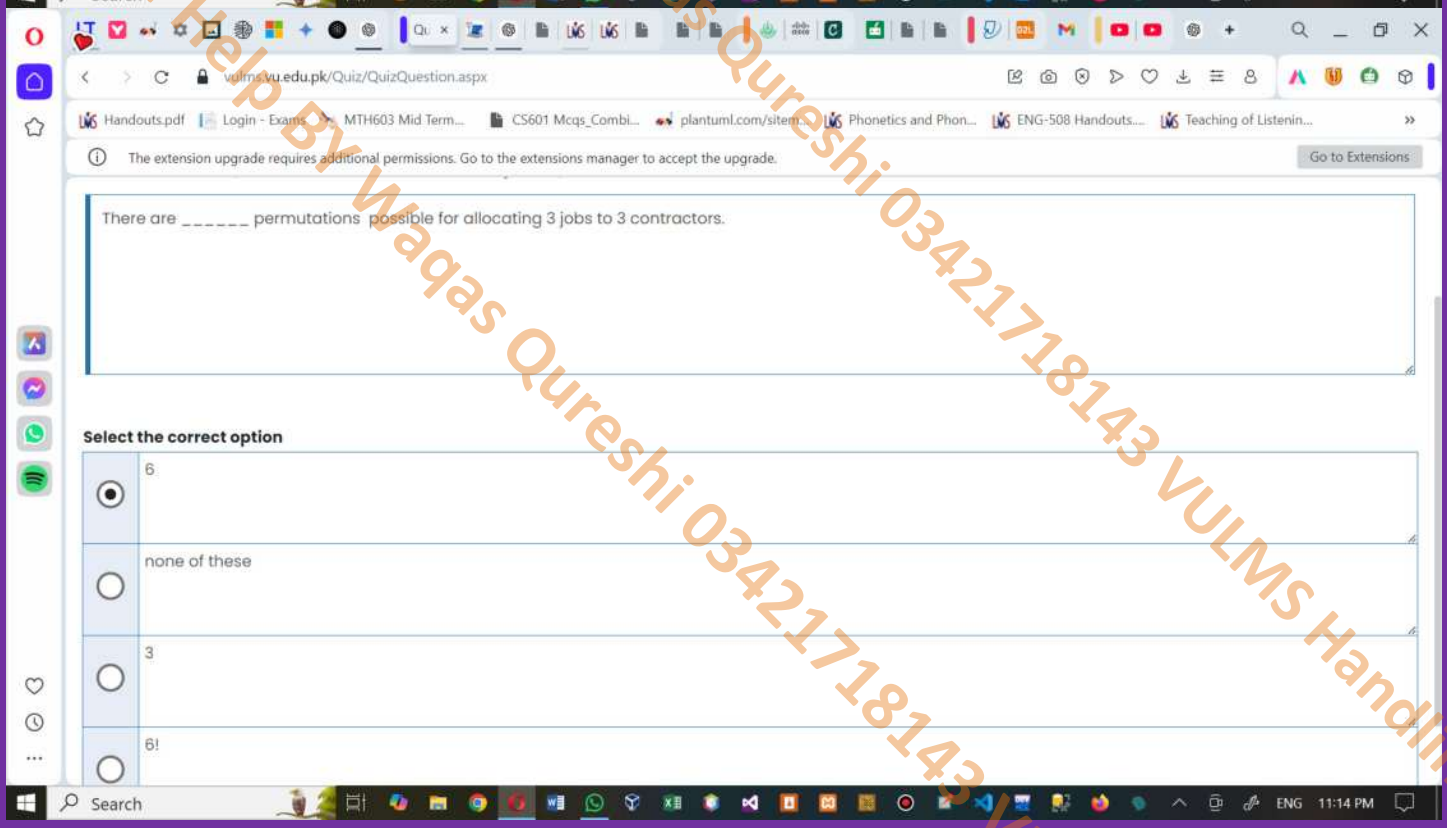
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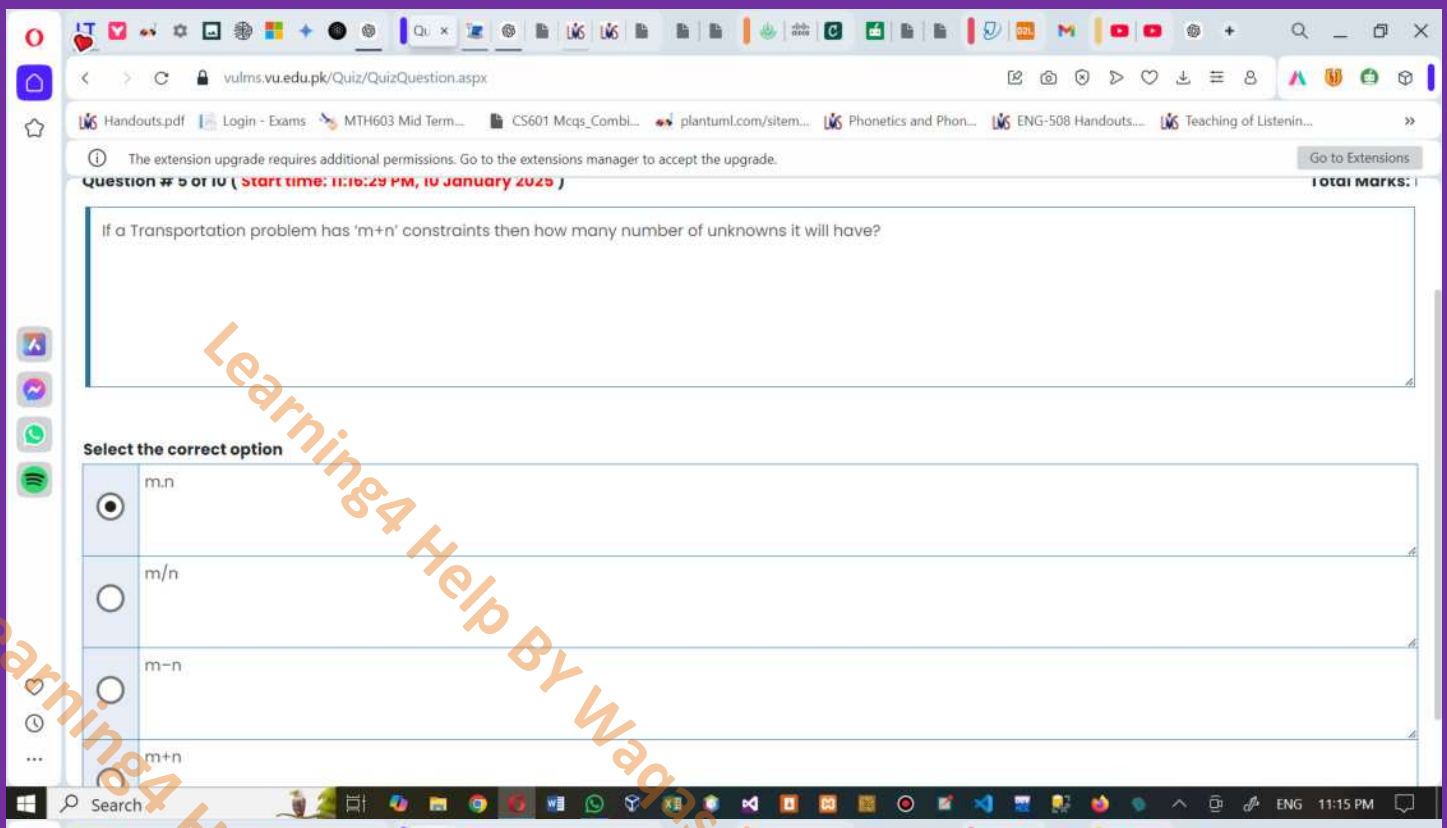


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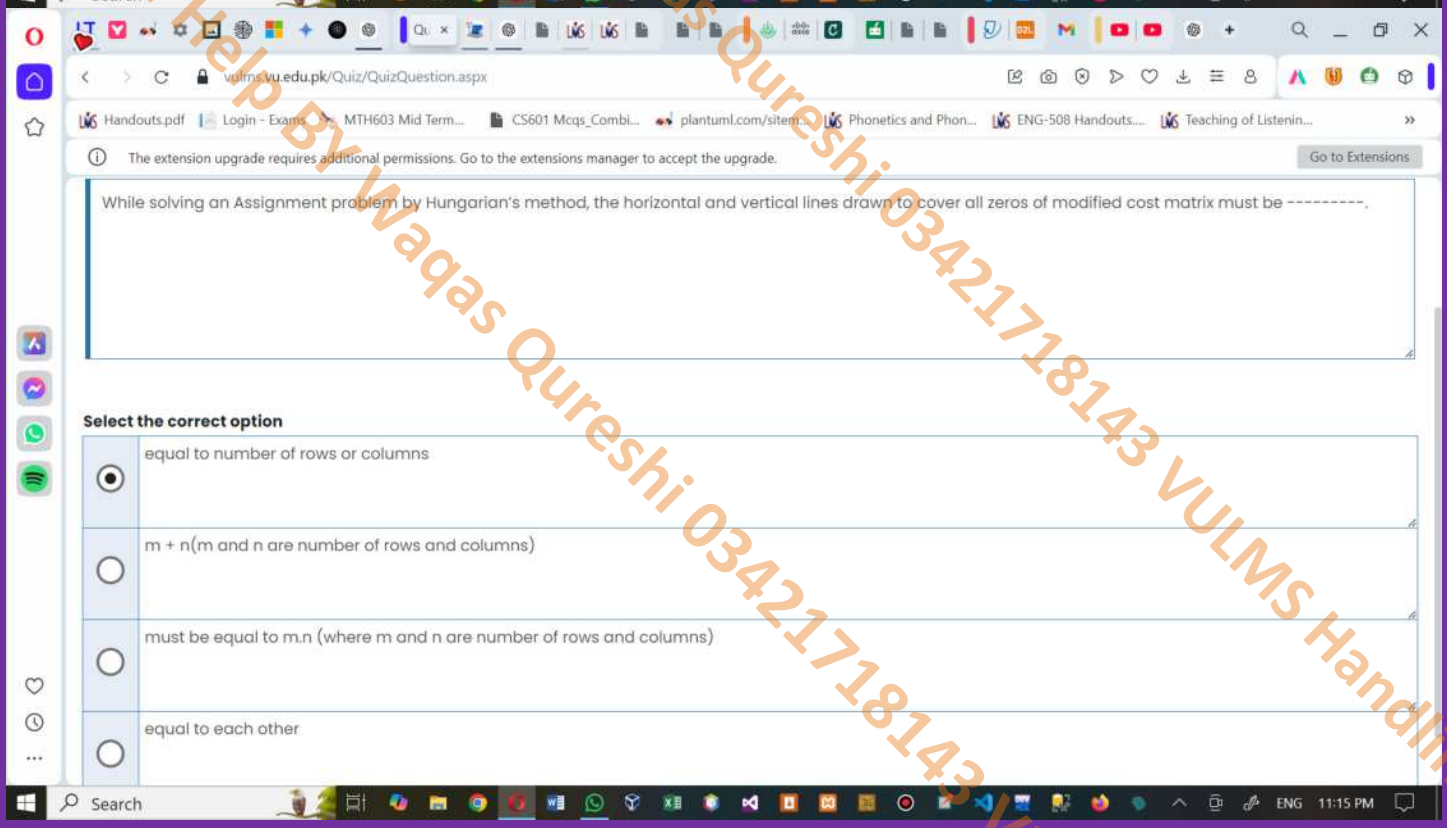
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If a Transportation problem has 'm+n' constraints then how many number of unknowns it will have?

Select the correct option

- m.n
- m/n
- m−n
- m+n

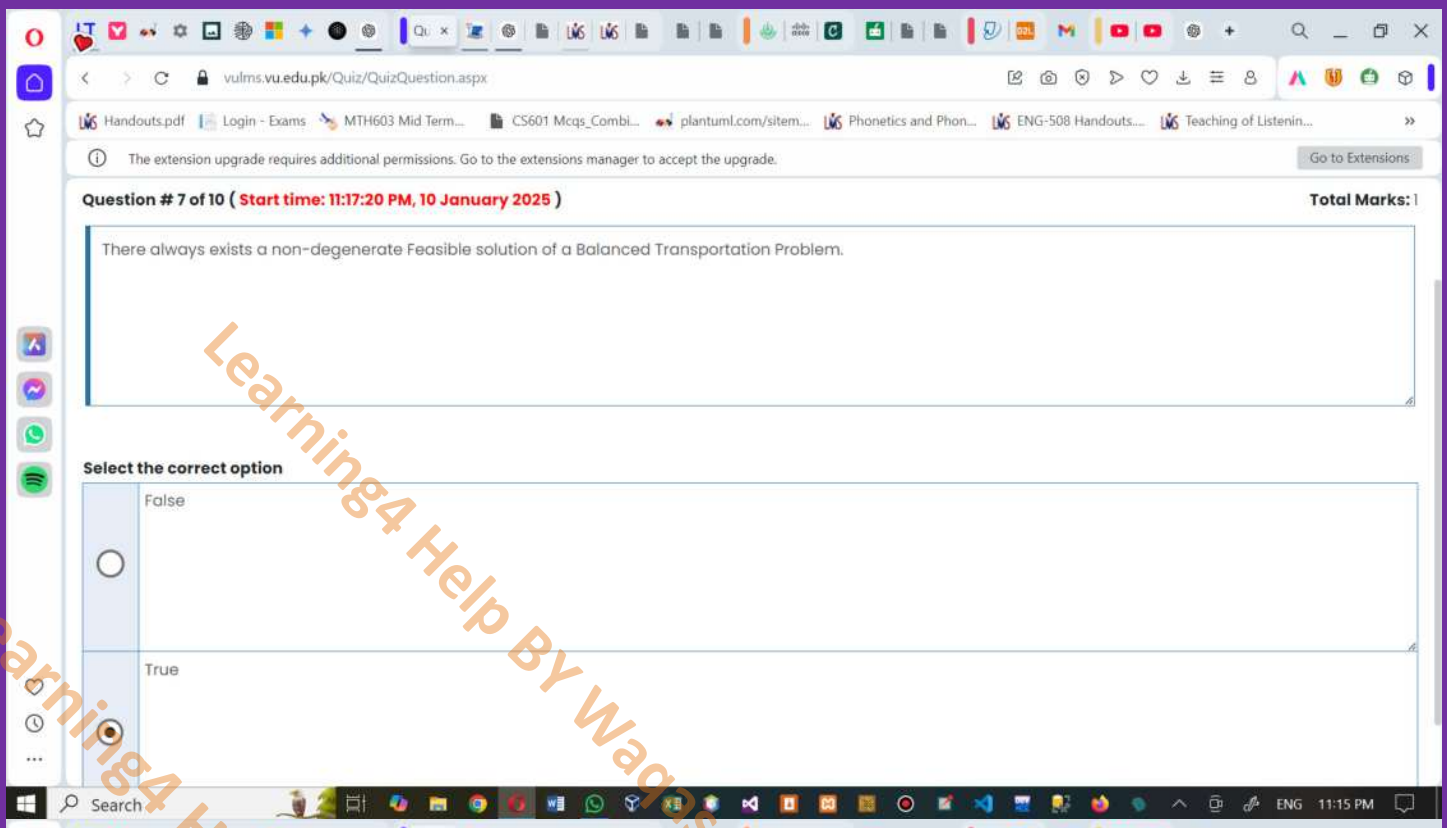


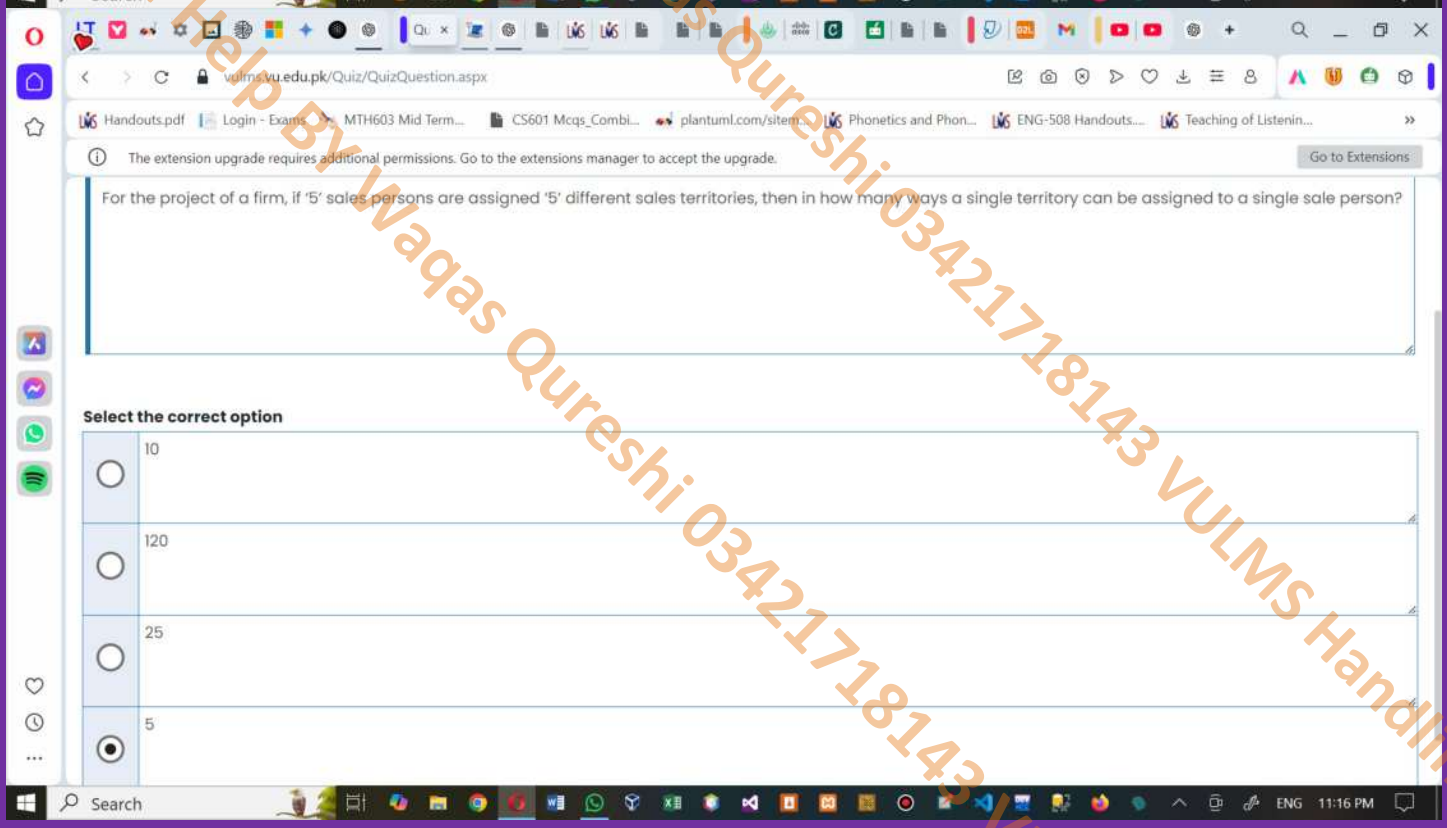
While solving an Assignment problem by Hungarian's method, the horizontal and vertical lines drawn to cover all zeros of modified cost matrix must be ---------.

Select the correct option

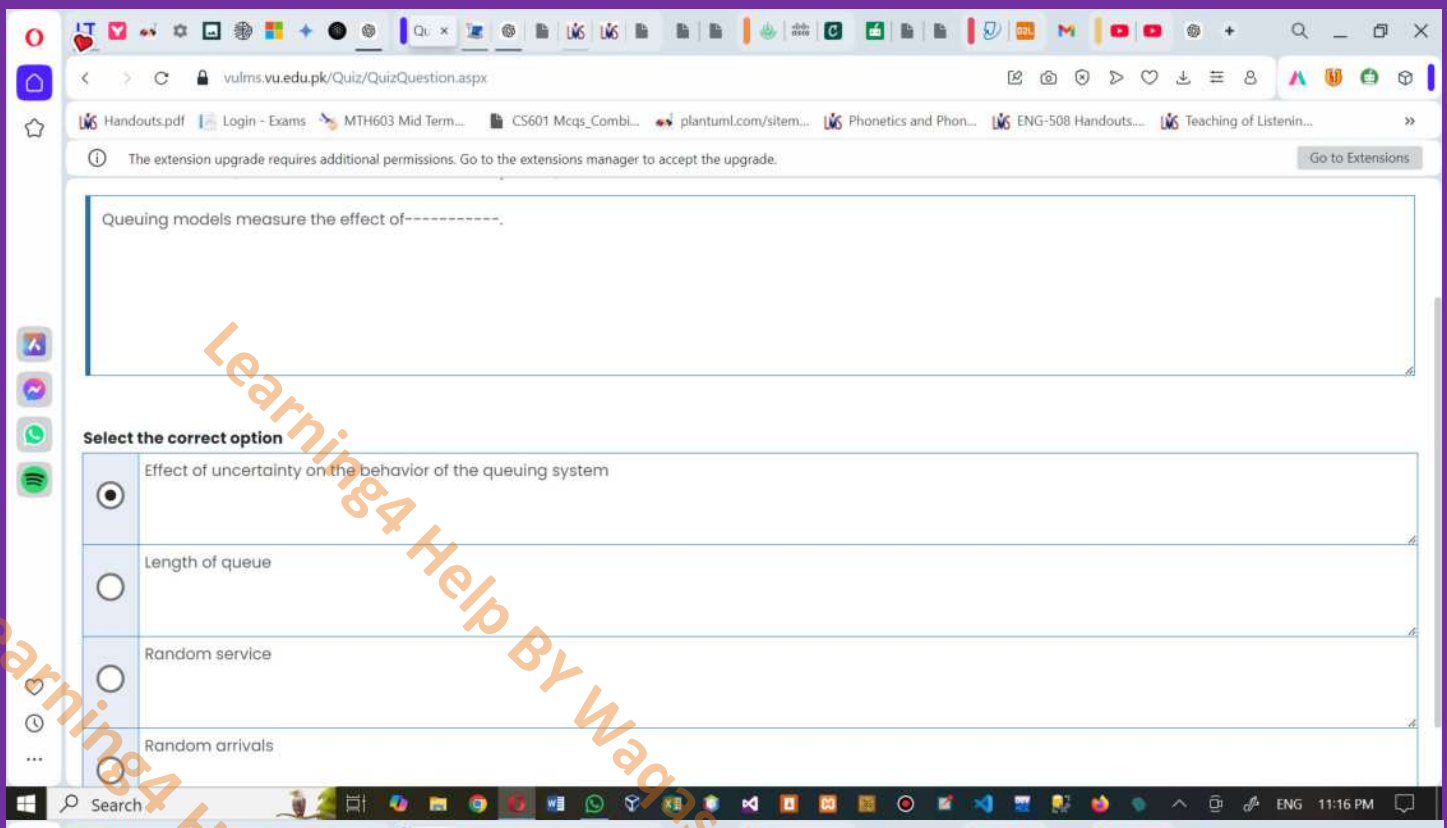
- equal to number of rows or columns
- m + n(m and n are number of rows and columns)
- must be equal to m.n (where m and n are number of rows and columns)
- equal to each other

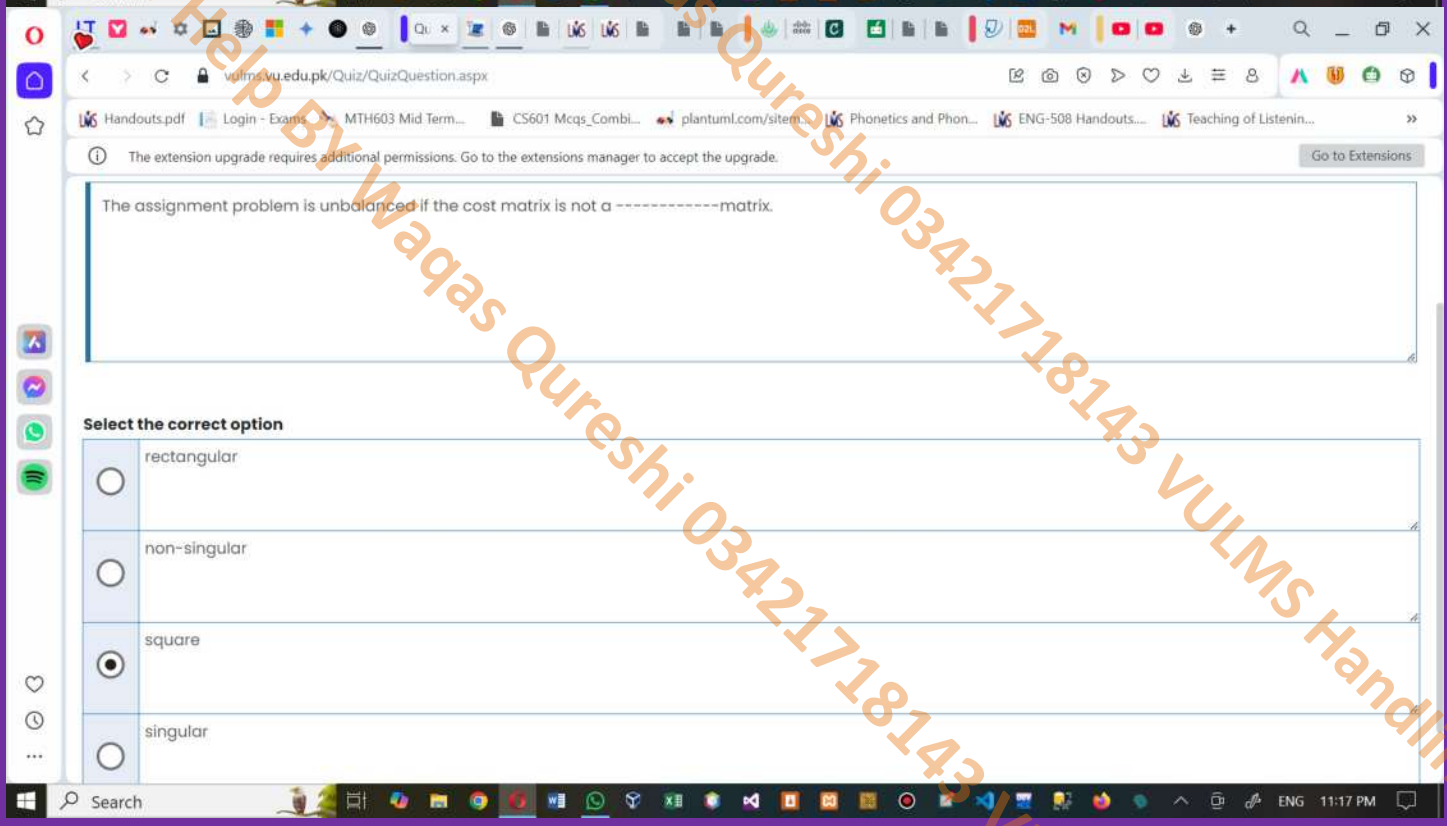
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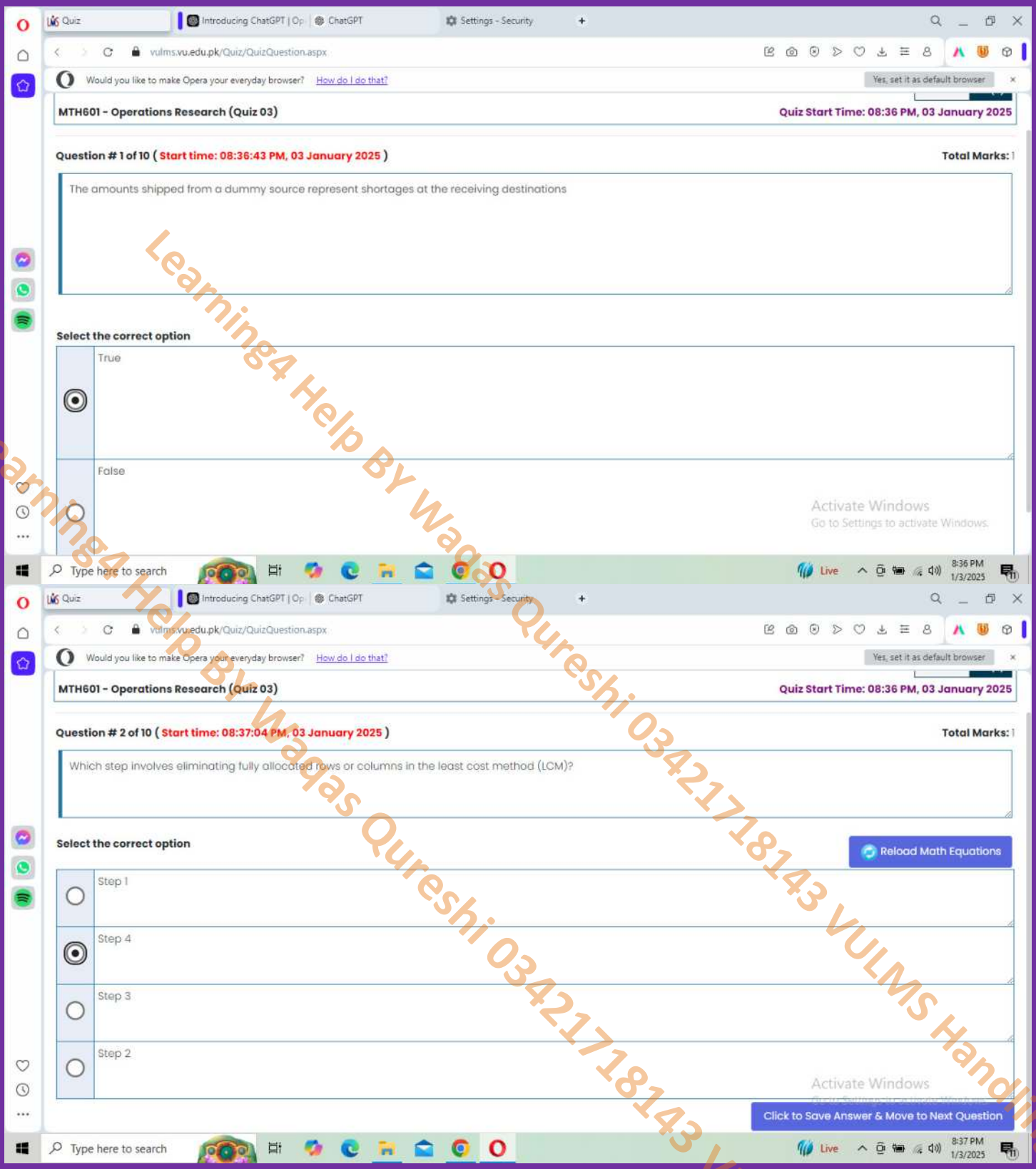


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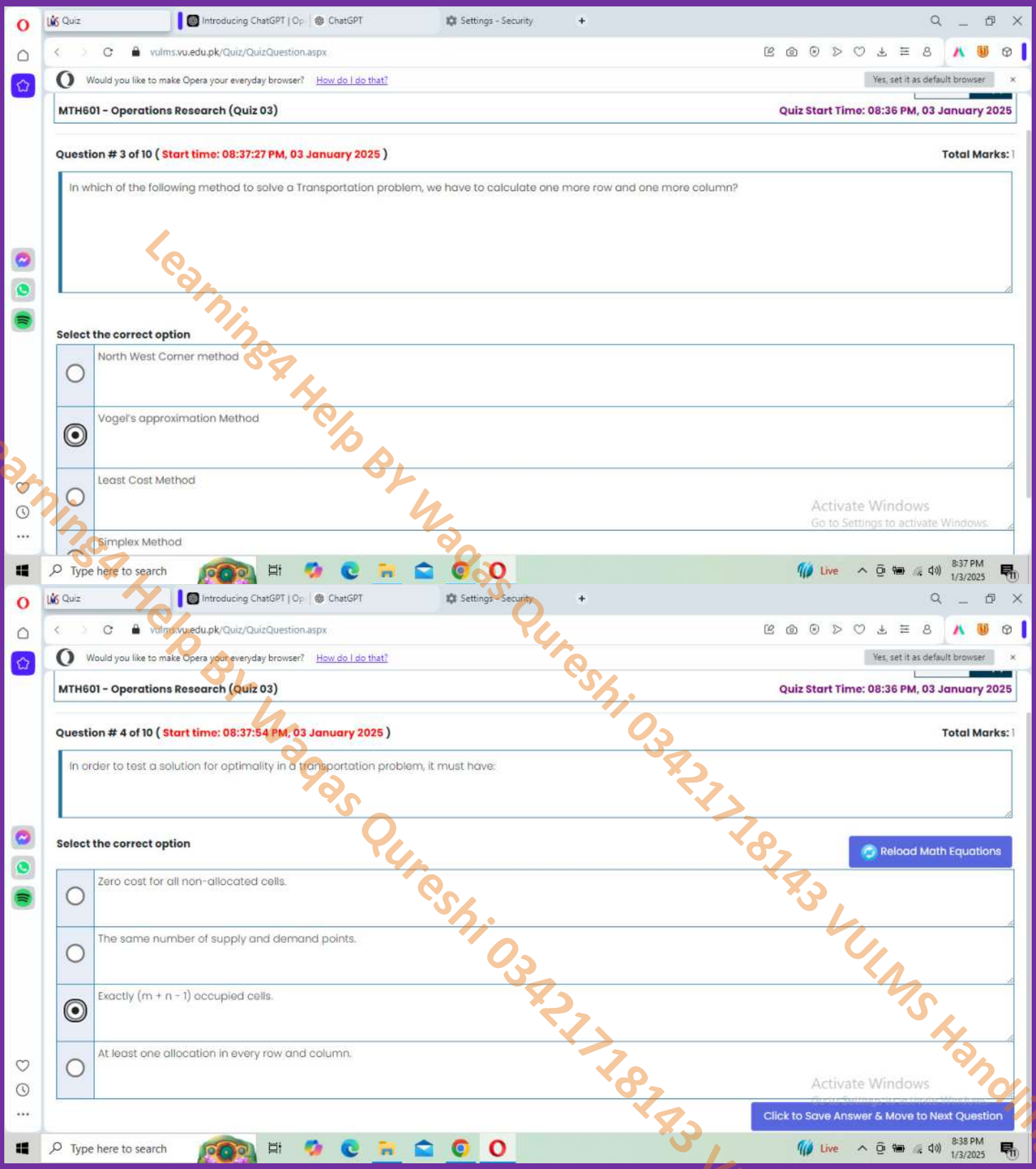




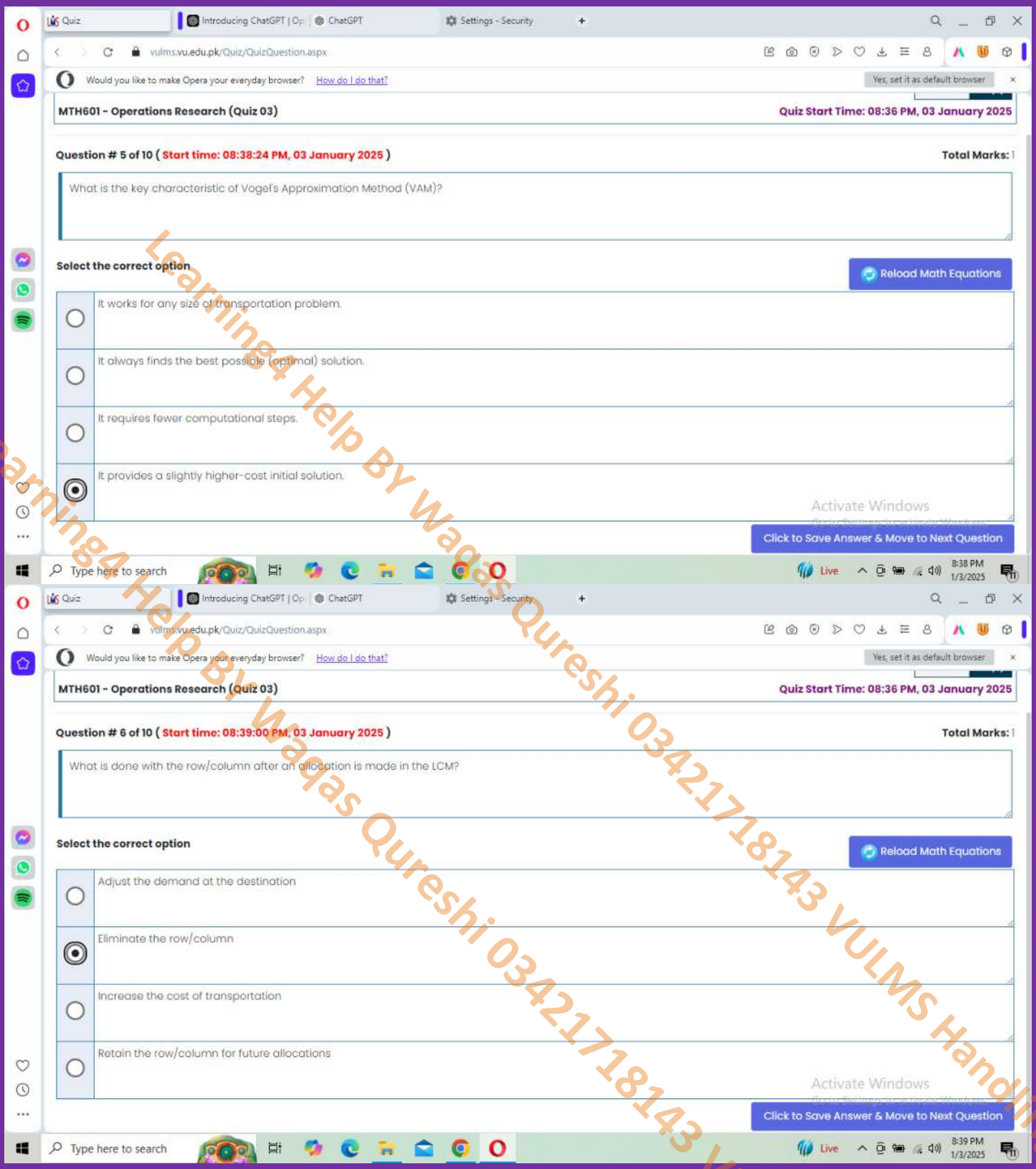
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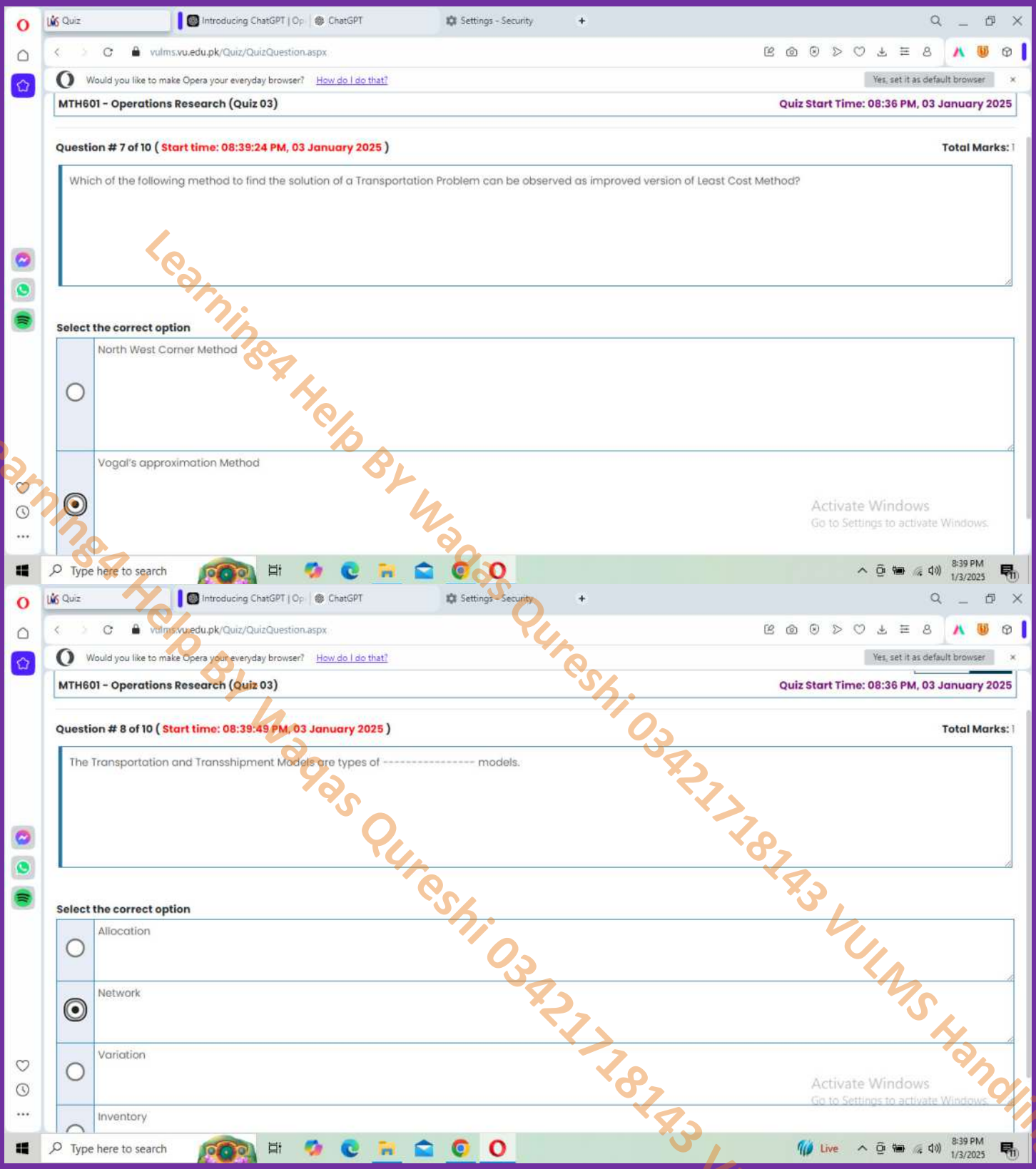
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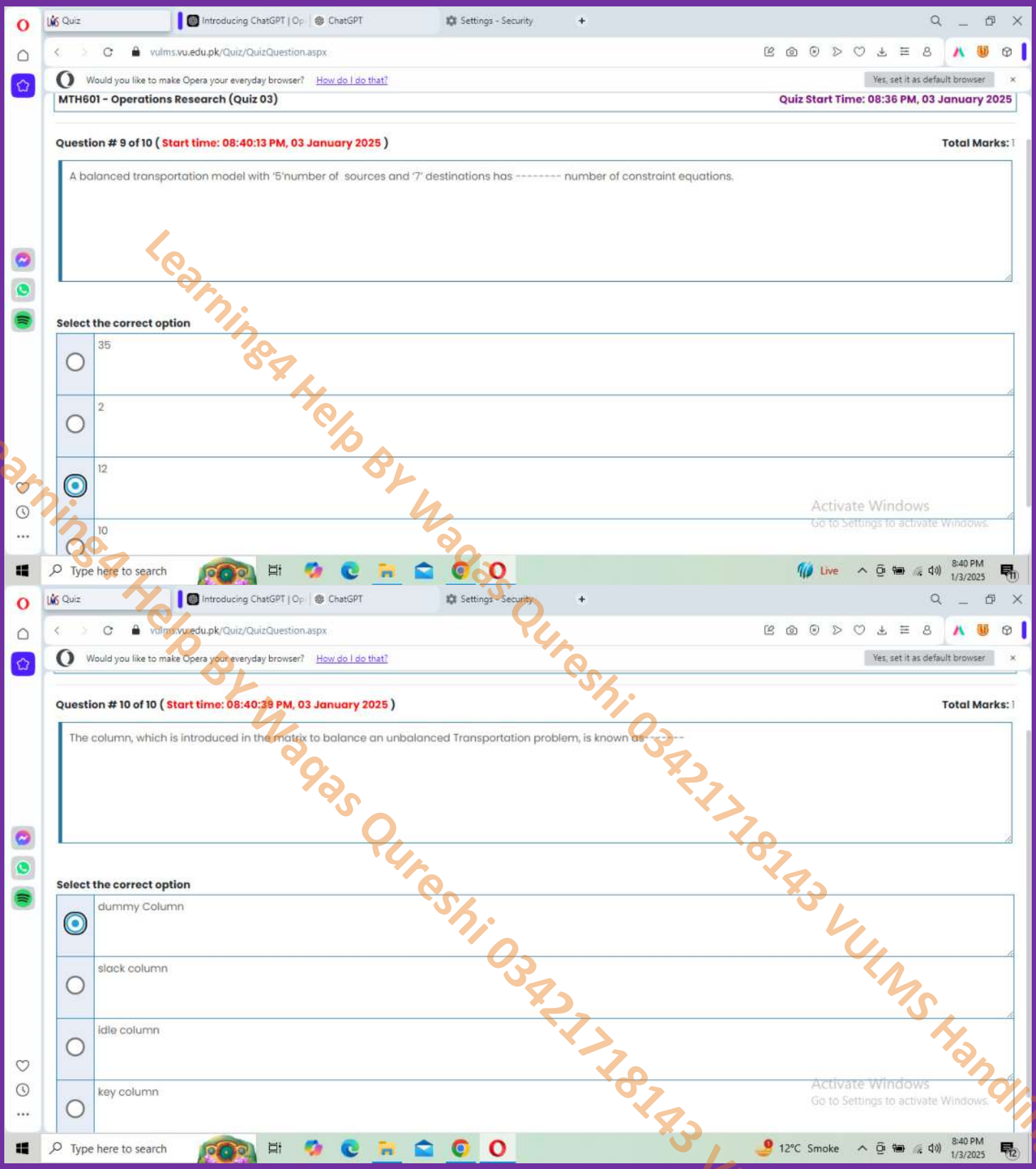
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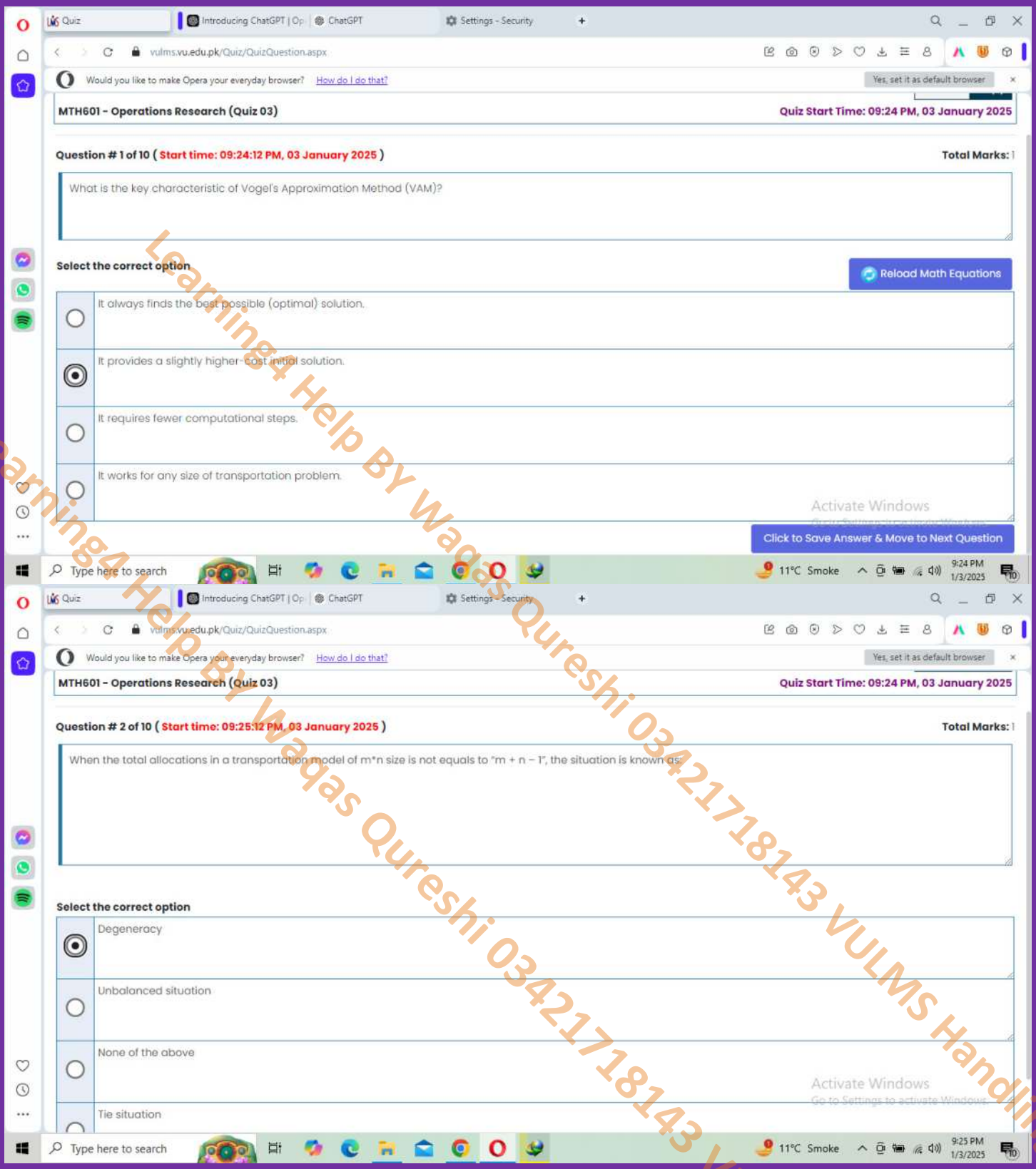
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MTH601 - Operations Research (Quiz 03)

Quiz Start Time: 09:24 PM, 03 January 2025

Question # 1 of 10 (Start time: 09:24:12 PM, 03 January 2025)

Total Marks: 1

What is the key characteristic of Vogel's Approximation Method (VAM)?

Select the correct option

Reload Math Equations

- It always finds the best possible (optimal) solution.
- It provides a slightly higher cost initial solution.
- It requires fewer computational steps.
- It works for any size of transportation problem.

Activate Windows

Click to Save Answer & Move to Next Question

Type here to search

11°C Smoke

9:24 PM 1/3/2025

MTH601 - Operations Research (Quiz 03)

Quiz Start Time: 09:24 PM, 03 January 2025

Question # 2 of 10 (Start time: 09:25:12 PM, 03 January 2025)

Total Marks: 1

When the total allocations in a transportation model of m*n size is not equals to "m + n – 1", the situation is known as:

Select the correct option

- Degeneracy
- Unbalanced situation
- None of the above
- Tie situation

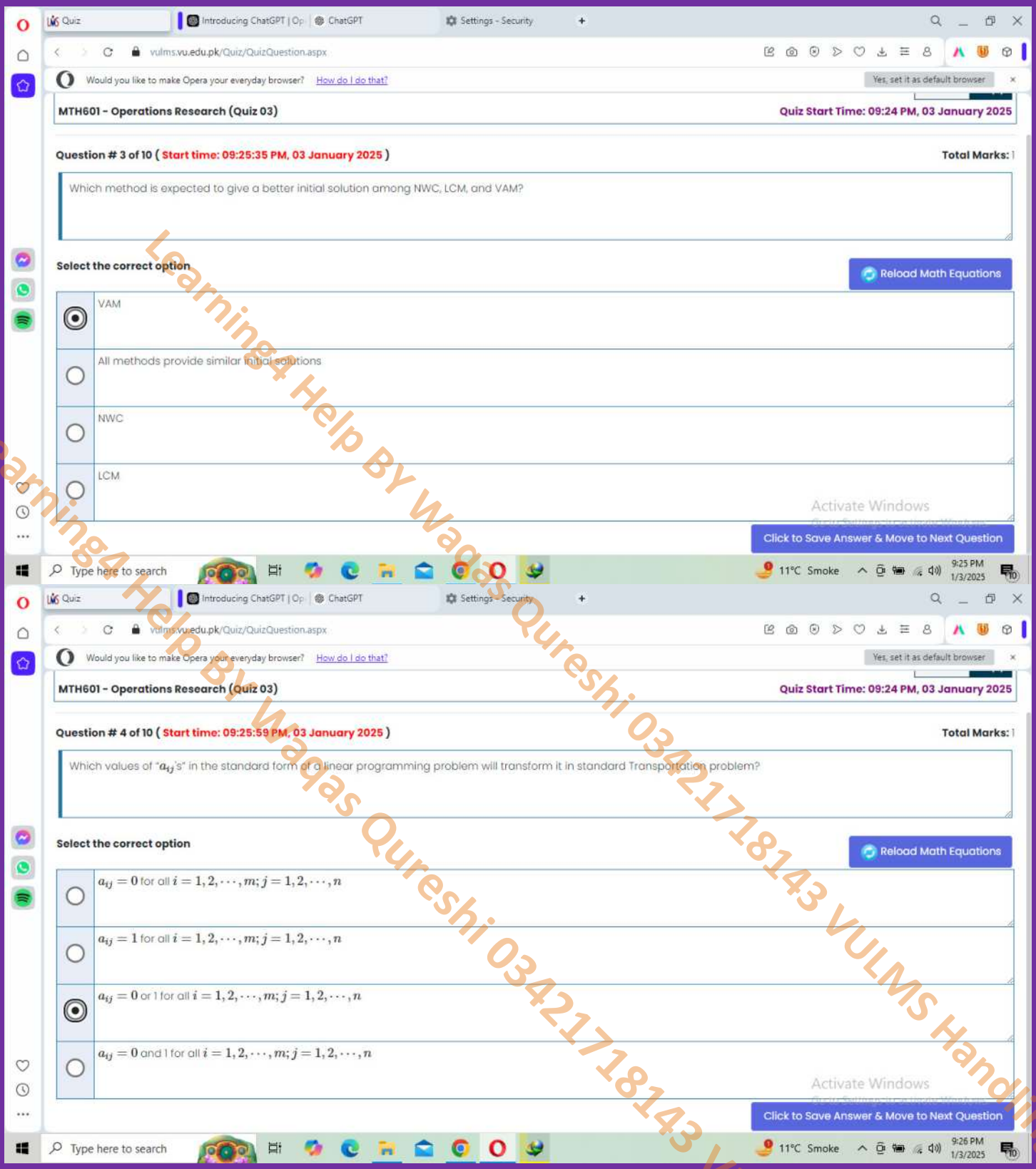
Activate Windows

Type here to search

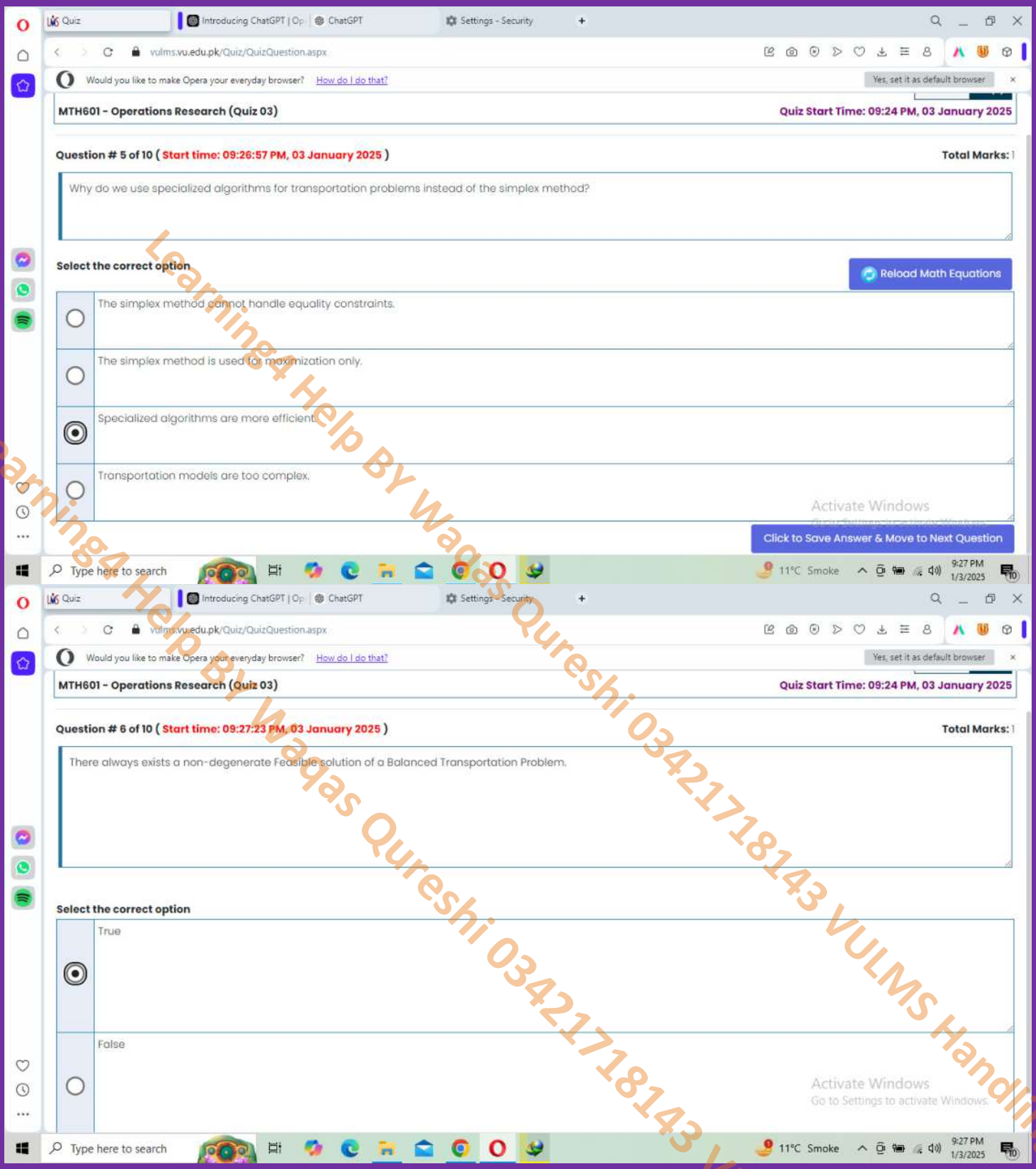
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9:25 PM 1/3/2025

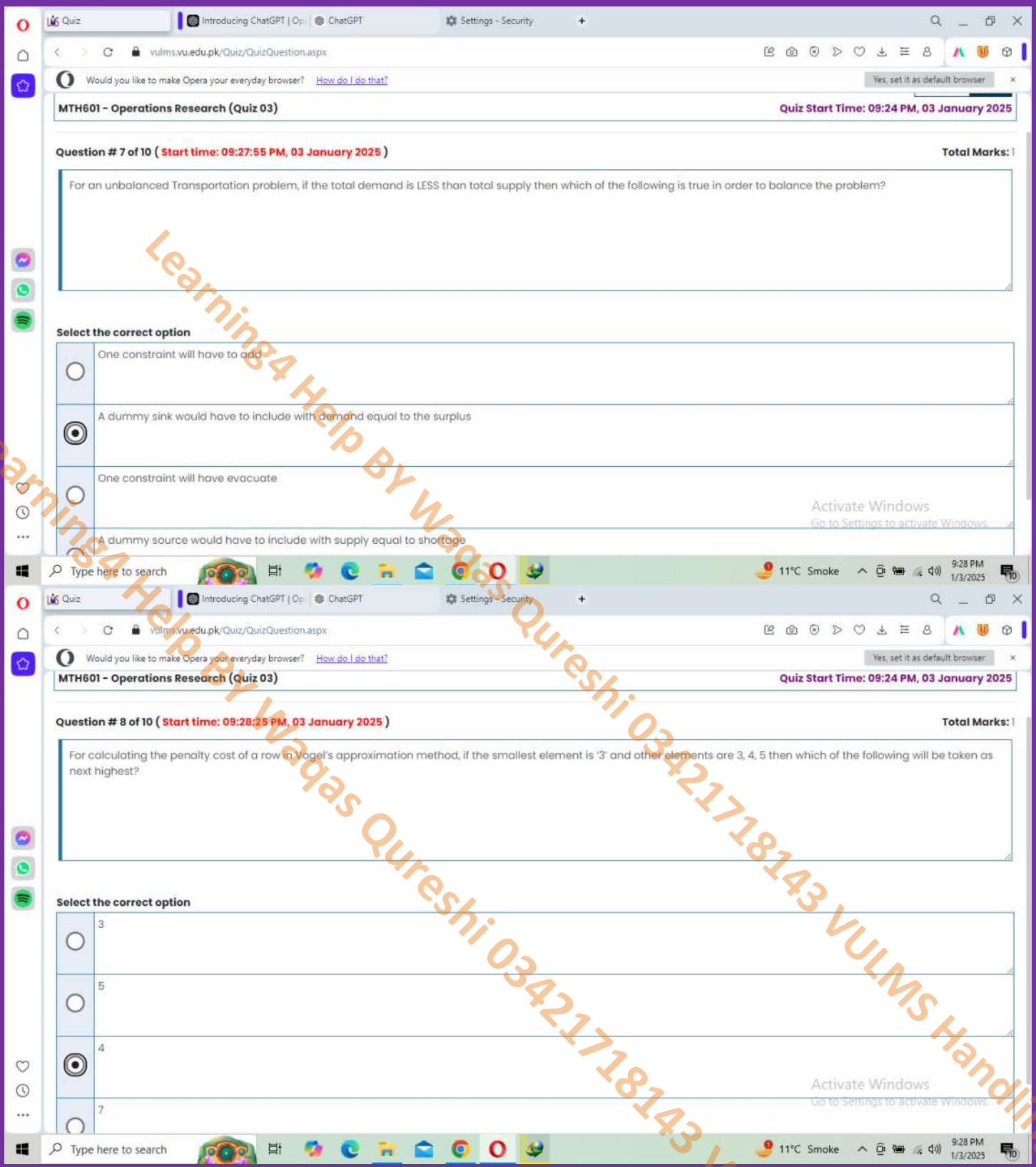
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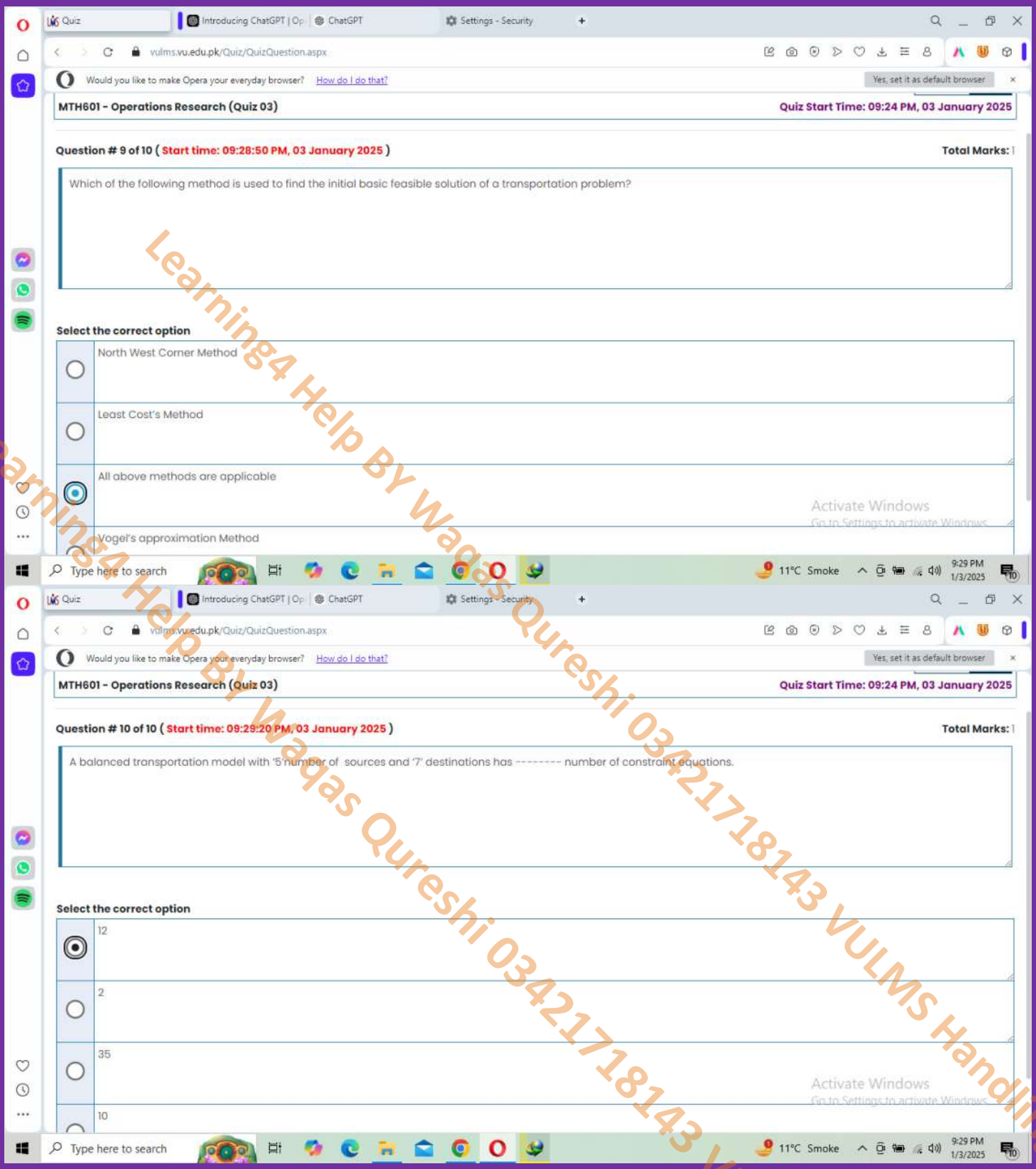
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